

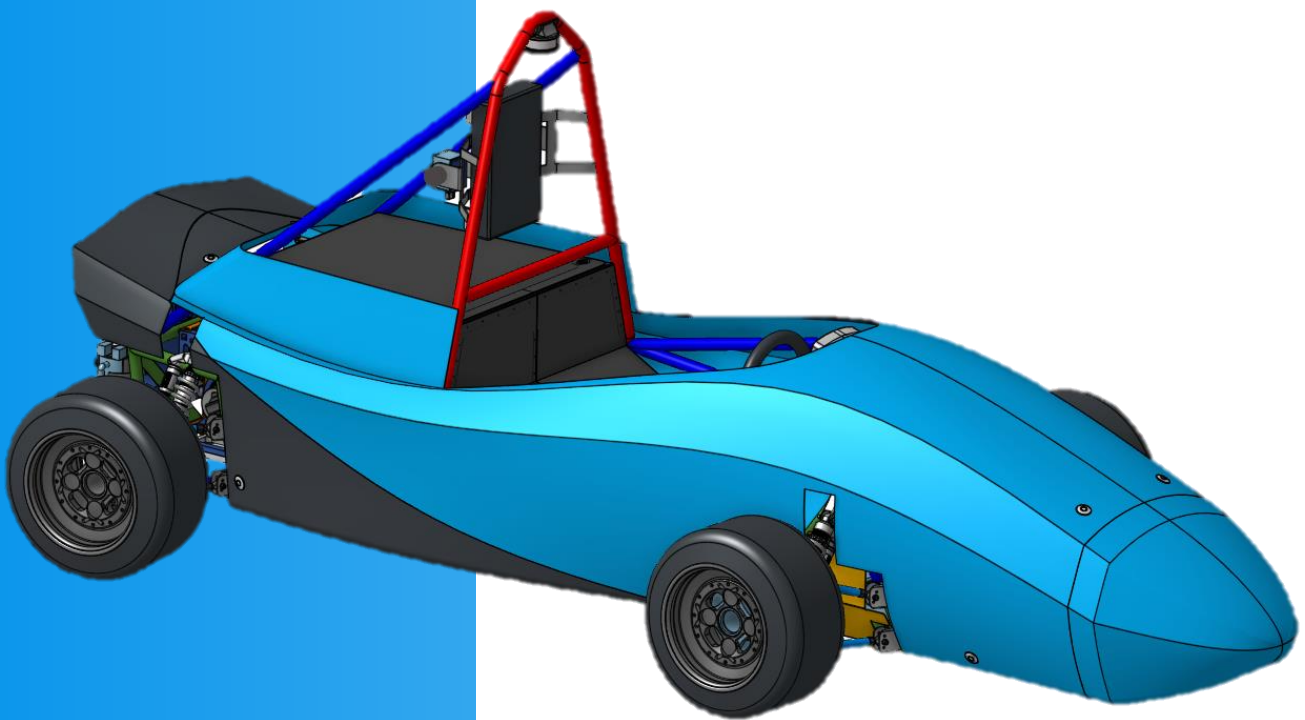


Mechanical Lead (SEM 1)

Chief Engineer (SEM 2)

FYP REPORT

NU23



Jye Hollier
2023



Summary of Contribution

As 2023 Mechanical Lead and Chief Engineer, I contributed:

- NU23 Mechanical Design
 - Chassis
 - Powertrain
 - Braking system
 - Fit and finish
- EV.Three.5 modifications
 - Modifications to final drive for implementation of spool
 - Adjustments to suspension to lower ride height
 - Motor controller adjustments
- Mechanical Design Assistance
 - Provided with all mechanical designs on NU23
 - NU24 initial design
- Management of manufacturing
 - Consultation with MEWS
 - Sourcing of materials
 - Ordering from suppliers
 - On-campus manufacturing
- Management of SMSP Track days
 - Development of test plans and running of experiments
- Design of test and exhibition tracks
 - Car park 1 test track
 - SMSP endurance track
 - Varley Group demonstration track
 - Engineering carpark Open Day demonstration track
- Application of Livery and Sponsorship Stickers to NU23
- Management of Dynamic Events at FSAE Competition
- Creation of Design Report Submission
 - Co-authored with Team Leader
- Concept, Integration and Management Presentation in Design Event
- Test Driver
 - Completed majority of test driving for EV.Three.5 and NU23
 - Driver training throughout year
 - Endurance event co-driver

Declaration

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Acknowledgements

I would like to acknowledge all past NU Racing members for their contribution in getting the team to where it was when we began. Without continuous, hard work teams like NU Racing struggle to succeed or grow and we were fortunate to have a solid base to start from in 2023.

I would also like to thank Malcolm Sidney for his continuous support of the team with technical assistance from design work through to manufacturing. This guidance has helped the team during some of the toughest times of the year.

I would like to thank Dr Alex Gregg for his unwavering dedication to the NU Racing team. His ability to lift energy and inspire the team to achieve through challenges is simply unmatched.

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My greatest appreciation goes to the team of 2023. While we got a little unlucky on the day, I truly hope everyone feels a great sense of pride from the achievements we made throughout the year and at competition. To walk away with three trophies, doubling the total number received in the Team's 20-year history, is an amazing achievement considering our size and the challenges we had to work through. 2023 was a big year with some significant changes made in the path for future NU Racing teams. I hope that everyone developed skills and friendships which will last well beyond the time we spent in TA.

I would also like to acknowledge the new leaders who have stepped up for 2024. Already working hard in the summer break to get the best start possible, I am excited to see what you will achieve with the 2024 team.

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Introduction

This report outlines works completed for the year 2023 in The University of Newcastle's Formula-SAE team NU Racing. This details the contribution to Mechanical Systems particularly in semester 1 as Mechanical Lead and overall changes implemented to the Vehicle in semester 2 as Chief Engineer.

2023 was a year of growth for NU Racing, with a focus on generating a more sustainable design ethos and adjusting timelines to ensure repeatable success at competition. It was also a year to realign subsystems to create a better performing vehicle for competition, while providing an opportunity to adjust the teams structure and timelines to better utilise team knowledge and increase preparation time for events.

Formula SAE is a global competition providing a platform for university students to create teams to design, build and compete in open-wheeled formula-style race vehicles. NU Racing has been a part of this competition since 2003 with a history of varied success. Now participating in the EV class, EV.Three was NU Racing's second competition EV and competed at the F-SAE Australasian competition at Winton Raceway in Victoria in December 2022. This car placed best in NU Racing's history, with 4th overall in the EV class.

This report begins with an overview of the 2022 competition performance of EV.Three, with a focus on its design strengths and weaknesses, and expanding on its limiting factors. With a focus on dynamic results, it will then explain how these were used to develop goals for 2023 and the key technical changes to be made to achieve these goals with the 2023 competition vehicle.

Three sections explain the Authors work in the chassis, powertrain, and braking system providing system reviews, 2023 designs, 2023 manufacturing, 2024 initial designs and 2024 recommendations as applicable.

Finally, it ends with notes on recommendations for 2024 and beyond providing overarching design goals and plans for execution. This aims to provide a basis for ongoing team development to continue on the path to creating dominant race vehicles.

Initial Vehicle Design

Review of EV.Three Vehicle and Competition Performance

EV.Three was NU Racing's third electric vehicle chassis and second competition entry in the EV class. It was undoubtedly successful, placing 4th in the overall EV class at the 2022 Formula-SAE Australasia competition. EV.Three was the culmination of three years of design work and manufacturing, with the 2022 team finalising design and manufacture for testing and competing. With a strong focus on reliability instead of performance, this proved successful at competition with consistent average event scoring leading to an impressive placement overall.



Figure 1 - EV.Three (NU Racing, 2024)

Whilst this was a great result for the team, the level of dynamic performance displayed by EV.Three would not be sufficient for success in the Australasian competition in future years. This result had to be taken in context of a year where general performance of all teams in dynamic events was lower than traditionally seen, likely due to cancellations of the 2020 and 2021 competitions. To highlight key areas where changes could provide significant improvements, analysis of dynamic event scores was completed to rank priorities for point gains. Skidpad and Autocross results were considered sound with some room for improvement however would not be a focus of large design changes. 2019 and 2022 results, paired with 2023 event scoring goals and total available points can be seen on the following page.

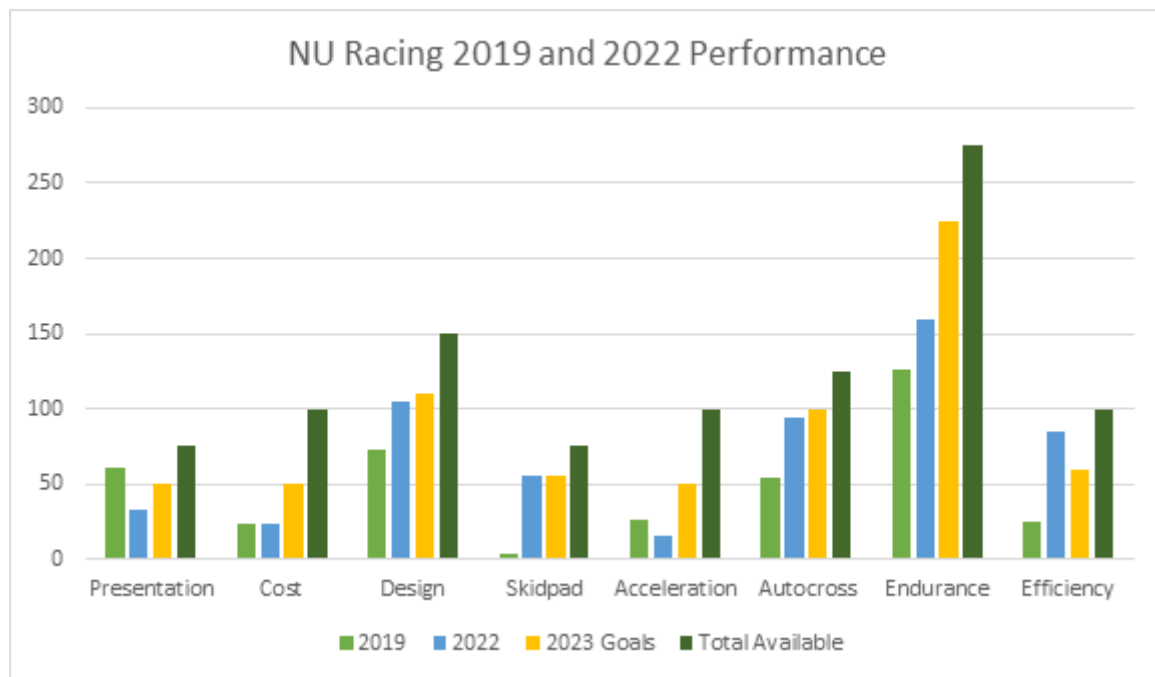


Figure 2 - Past NU Racing Event Scores and 2023 Goals (McLean, 2024)

The Endurance Event was clearly the biggest performance limitation in EV. Three for dynamic points at the competition. This was due to battery temperatures rising quickly at competitive pace which required the team to drive slowly for this event. High battery temperatures were also observed in the autocross event and all track days prior to competition. This had a severe impact on test time and this aspect alone provided great incentive to try and reduce these temperatures.

Whilst not having as many points available, the acceleration result at the 2022 competition was also poor with points scored nearing half that of 2019. This was despite a completely new drivetrain package with two advanced motors capable of 70 kW each. Personal experience driving EV. Three showed the vehicle never had any level of acceleration performance expected for a competitive F-SAE vehicle. This would be a key focus of dynamic improvement in 2023.

Another important issue with the 2022 drivetrain was that despite mechanically having the ability to provide torque vectoring, this was never implemented electrically and so the hardware provided no real improvement in performance. Whilst the independent rear-drive improved low speed manoeuvrability compared to a spool, skid pad results were not exceptional. Without torque vectoring implemented this mechanical arrangement in fact lead to traction issues and poor acceleration out of corners. As each wheel received the same torque but was free to rotate at any speed, the inside wheel would frequently break traction for a considerable distance exiting a corner. With equal torque split between wheels, the additional grip of the outside tyre was not utilised reducing acceleration, while the limited grip of the inside tyre was exceeded, breaking traction, and reducing the tractive force available from this wheel.

Contributing significantly to acceleration results and general performance was weight. At 302 kg, EV. Three was the heaviest EV vehicle at the 2022 competition. In a competition with a power limit but no minimum weight, reducing weight only helps with dynamic performance. Subsystems which appeared to be excessively heavy were the powertrain (including tractive system and cooling) and aerodynamics package. The aerodynamics package had no conclusive real-world validation and based on driving experience made little to no difference in vehicle feel at the speeds the car was driven at.

The steering system was also poorly designed at a fundamental level, with incorrect chassis attachment methods leaving unintended degrees of freedom which were missed through several years of design work and reviews. Whilst a fix was implemented for the 2022 year it was not effective with the steering wheel still able to be moved on its mounts left/right with a strong pair of hands. The system also was not compliant at the 2022 competition due to a lack of guarding. In addition, the assembly with universal joints was difficult to manufacture and experienced binding when driving.



Figure 3 - EV.Three Steering Wheel Movement Due to Chassis Mounting

Overall, there appeared to be significant discrepancies in subsystem requirements and capabilities across the vehicle, resulting in some systems being oversized with significant weight penalties, while others were undersized significantly limiting performance. Some systems also appeared to not work as intended in the real-world driving conditions of an FSAE vehicle. In particular:

- The accumulator power output (both peak and continuous) was significantly lower than what the powertrain was designed around
- The heavy aerodynamics package and cooling systems were not truly validated as appropriate
- The mechanical design of the drivetrain also did not align with the electrical systems driving it, leading to an increase in complexity which hindered performance

Another aspect of EV.Three was that due to its extended development time and the unique circumstances surrounding the 2020 and 2021 competitions, the chassis was technically not compliant at the 2022 competition. Whilst exceptions were granted due to many teams being in similar situations, at minimum significant chassis modifications would be required to create a compliant vehicle for 2023. Some work had been carried out detailing compliance issues and potential solutions, however no planning or decisions had been made by the previous team for a chassis moving into 2023. As a result, this was a significant issue which would require immediate rectification by the new team.

Despite being designed with serviceability as a priority, many systems on the car were also difficult to install/remove. This was particularly true for the drivetrain and aero package which both had many fasteners and required specific knowledge for installation. The Central Electronic Node also required significant disassembly for removal.

Whilst there were many improvements to be made to performance, it must be noted that the reliability inherent in the EV.Three design was still the most important requirement for any FSAE vehicle. It does not matter how well a vehicle is optimised for an event at competition, if an issue occurs preventing the car from completing an event no points will be gained and this is far more detrimental to overall performance compared to any gains made by vehicle optimisation. This was a very important concept for the team to keep in mind, as system reliability rules above performance always.

In conclusion, the top-level issues hindering EV.Three performance were:

- High Accumulator Temperatures
- Reduced Powertrain Output
- Excessive Weight
- Mis-aligned system bottlenecks

Testing of EV.Three/EV.Three.5

In the early stages of 2023, extensive testing was completed on the EV.Three chassis to evaluate subsystems and confirm key areas for improvement in the 2023 car. Accumulator changes were not feasible in the short period of time allocated to this testing, so instead the main experiments focused on the effects of weight and the performance of systems which contributed to this significantly. In particular, the key experiments which shaped the overarching theme of NU23 were testing of the aerodynamics kit, cooling system and powertrain. These were completed on a small test track on campus to significantly reduce testing times. This track was designed with a focus on easy setup to provide consistency across testing days. This design along with other track maps for SMSP and demonstrations are located in the appendix.

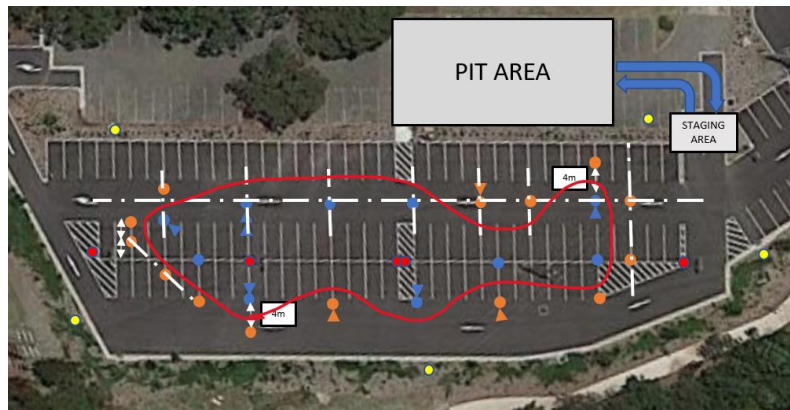


Figure 4 - Car Park 1 Testing Track

Lap and skid pad times were tested with the aero kit installed and uninstalled, as well as a with control test weights to reflect the weight of the aero kit with no drag/downforce. This showed no measurable gain in performance with the kit on compared to the control test, while there was a noticeable improvement in car feel from driver feedback with the weight removed.

The second main testing completed was to determine if the powertrain could feasibly be reduced to a single motor. It was known that the current motors were not being fully utilised to their potential torque output and so experiments sought to compare current performance to a single motor at full torque output. Whilst somewhat limited in gearing due to space restraints and sprocket availability, the coupling was modified with pieces to lock the halves together with a larger 60T driven sprocket added.

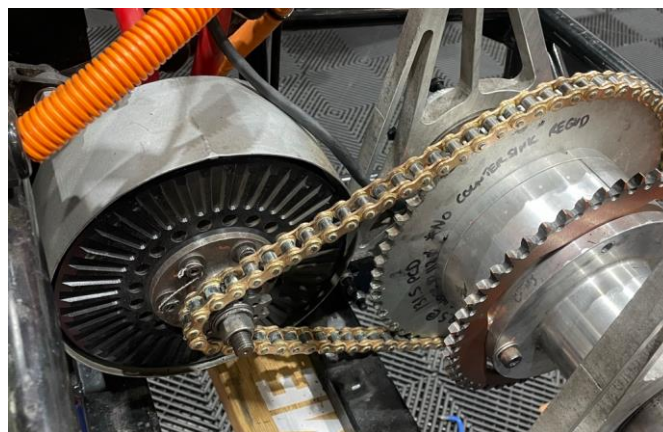


Figure 5 - EV.Three Final Drive Modifications

The improvements in wheel torque from changes to gearing and phase current from the motor, paired with weight savings from the removal of one motor and controller, showed promising performance and an increase in pace so the decision was made to develop this powertrain concept for NU23. It was noted that by finding smaller drive sprockets the wheel torque could be improved further with more gear reduction as the motor was still not being utilised in its full speed range or power output. Locking the wheels together also removed the traction issues experienced by the coupling arrangement as torque would always be naturally directed to the wheel with more grip.

The cooling system was also modified in tandem with the change to a single motor. Using only one radiator for cooling the motor and controller, as opposed to two in EV.Three, showed temperatures remained stable highlighting the excessive cooling capacity offered by the 2022 system. This radiator was also moved to the back of the vehicle removing the need for sidepods.

Together, the EV.Three chassis combined with the changes stated above was dubbed EV3.5, providing a platform for testing and driver training while design and manufacturing commenced on NU23. This testing confirmed the main changes for NU23 would be:

- Removal of aerodynamic devices
- Changing the dual motor powertrain to a single motor driving a spool
- Implementing a single rear-mounted radiator



Figure 6 - EV.Three.5

Whilst not tested, improvements to the accumulator to reduce cell temperatures were also highly desirable. With no low front wing (which for 2022 required an increase in ride height for clearance) it was also intended that the vehicle ride height would be lowered as much as possible without excessive scraping. This would lower the centre-of-gravity and improve dynamic performance.

NU23 Primary Goals

With the above factors setting up a general set of goals for improving the performance of NU23, there were also other team factors which would affect decisions for the vehicle's overarching design. These factors included:

- Team size and experience
- Time available
- Manufacturing resources
- Testing resources

EV.Three was the culmination of almost three years of design and development, while NU23 design started in late February with testing planned to commence mid-year. This realistically only allowed for 5 months of design and manufacturing. While some components could be transferred across, and a great deal of low-voltage systems copied, mechanically a significant amount of work was required. As a result, it was important to prioritise mechanical changes and understand flow on effects from changes in terms of new parts required to interface with these changes and the manufacturing load they would create.

The following details each subsystem and the changes anticipated. Many of these fixes were tested/designed as starter projects for new team members at the start of the year, with the prototypes for these designs explained above as tests in EV3.5.

Subsystem	Issues	Priority	Potential Fixes	Plan to Implement
Chassis	Non-compliant, new chassis not designed	High	Chassis Modified or New Chassis, conservative changes to minimise change	Required members modelled and ordered from VR3
Accumulator	Temperatures	High	Improved air cooling, increase number of cells	Consider chassis changes in to increase space for improved packaging
Powertrain	Excessive weight and complexity, traction issues, torque output, changes required to suit compliant chassis	High	Single motor/controller, spool final drive	Design NU23 powertrain with changes in mind
Steering	Not correctly restrained, no guarding for compliance	High	New design, implementation of gearbox	Incorporate mounting requirements into NU23 chassis
Aerodynamics	Wing performance did not justify weight, difficult to install/remove	High	Remove wings from NU23 design, develop body kit to provide sponsorship area	Design new single-piece body kit to suit NU23 chassis with extended side coverage.
Cooling	Excessive cooling capacity and weight	Medium	Implement rear mount for single radiator	Incorporate new mounting into NU23 chassis

Suspension	High Ride Height, must suit new chassis	Low	Adjusted coil over lengths to lower vehicle as no front wing requiring clearance	Adjust later in year, minimise changes by maintaining geometry where possible
Pedal Box / Braking	Pedal feel, not suitable for spool	Low	Revised pedal box design, revisions to plumbing to suit single rear caliper	NU23 chassis to suit EV. Three pedal box with potential replacement completed later in year. Plumbing changes as required for single caliper

Table 1 – Prioritised Subsystem Issues Requiring Change for NU23

With the above major subsystem improvements identified, careful consideration was made surrounding how these could be implemented whilst minimising the risks of change and manufacturing loads. With the required changes to the chassis and flow on effects to the powertrain, it was decided that to reduce manufacturing loads as much of the suspension would be carried over as possible, as it was an effective system with potential for more fine-tuning. To achieve this, all relative pickup locations would stay the same allowing for the reuse of a-arms, hubs, uprights, and steering arms.

Overall focus for NU23 designs would incorporate the following:

- Reduced manufacturing load
- Ability to change systems with revisions
- Reduced part count (and as a by-product reduced weight)
- Balance systems to better utilise hardware

It should be noted that whilst weight saving was an indirect goal it was not a primary focus of most designs. Conceptual designs were to be judged on functionality, reduced part count and manufacturability, with weight loss generally a by-product of simplified systems.

As the commonality between all subsystems, the chassis design was the most urgent design to finalise as this would have the most significant lead time for materials and construction. It would also be the basis for all subsystem designs. It would be designed with EV. Three suspension retained as stated above, while also allowing for the existing pedal box to be reused with potential for changes later in the year if resources allowed it. Ideally this approach would be used with other systems where possible, allowing for iterative development not constrained by bespoke mounting.

Chassis

The EV3 chassis was successful at the 2022 F-SAE competition. The culmination of design by Ryan Gorton and Perry McNulty saw the development of a chassis which provided much improved dynamic performance compared to EV.Two, and far better packaging compared to EV.One.

As detailed in Liam O'Neil's report, there were several compliance issues in the 2022 chassis which would require modifications if it was to be used as a second-year chassis in 2023. These were:

- Accumulator Container Mounting – A maximum of two accumulator mounts could be placed between triangulated chassis nodes however there were five.
- The top member of the tractive system side impact member did not reach the minimum height of 240mm.
- The Top height of the Rear Impact Structure did not reach the minimum height of 240mm.

A rules enquiry submitted in February asked for clarification on possible modifications to the chassis to make it compliant. The enquiry detailed possible fixes for the accumulator mounting through the relatively simple addition of three members, while suggesting modifications to the rear to incorporate increased heights for the SIS and RIS requiring significant changes to the chassis.

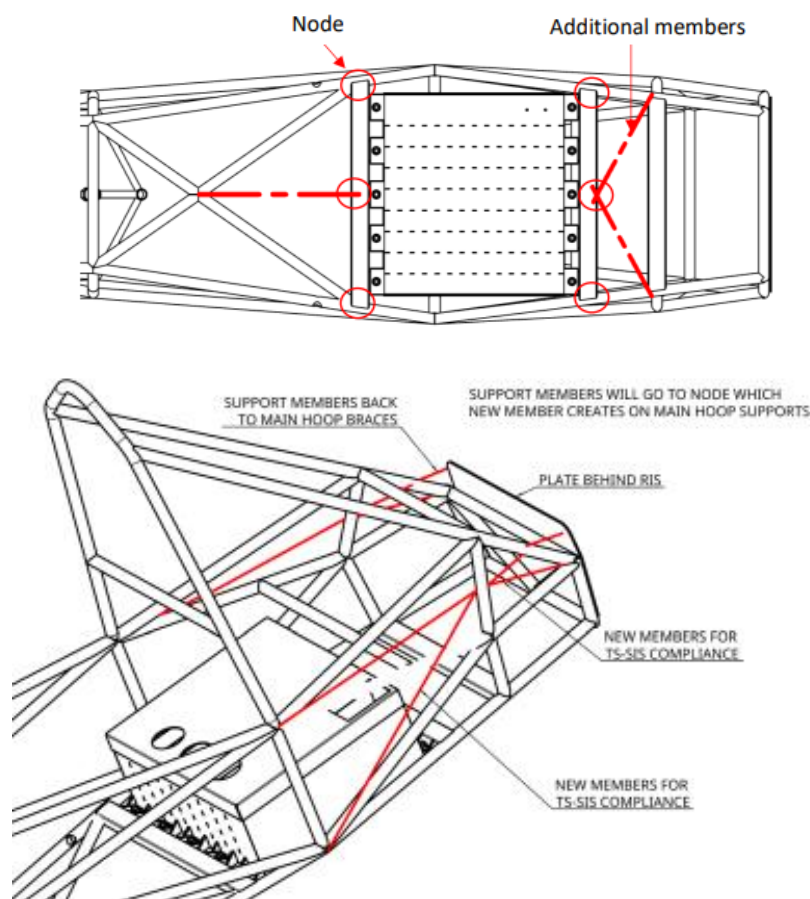


Figure 7 - Proposed Changes to Accumulator Mounting and SIS & RIS Structures (O'Neill, 2023)

The reply to this enquiry stated that whilst the modifications would be deemed compliant, it suggested that for improved design scoring it would be best to develop a new chassis for the 2023 competition. This response provided confirmation that the proposed designs would rectify compliance issues, and so the path to implement these changes needed to be chosen.

Options for Rectification:

After several discussions within Leadership with the assistance of 2022 FYP Students, three options were developed to implement chassis design changes for 2023. They were as follows:

- 1- Modify the existing chassis – this would be achieved by ordering accumulator mount braces and new members for the SIS, RIS and main hoop supports. The rear of the chassis would be cut apart for the installation of the new members mid-way through the year.
- 2- New rear half chassis – this would see a complete rear half chassis ordered and manufactured throughout the start of the year. When completed, all members of the existing chassis behind the roll-hoop would be cut off and the new rear welded on.
- 3- Complete new chassis – this would see a full new chassis designed and manufactured whilst still using the EV3 chassis. Half-way through the year all components required would be transferred across to the new complete chassis.

Other Issues:

Aside from the compliance issues which would require rectification, there were other issues with the 2022 chassis which were considered during this process. These included:

- Tight accumulator packaging – this had a significant effect on the possible routes for airflow and was believed to be a significant factor in the overheating issues experienced at 2022 track days and competition
- Poor steering system design – the use of rod-ends left the steering wheel with unintended degrees of freedom. This was somewhat addressed for the 2022 competition however had slop and some binding requiring a new solution.
- Driver ergonomics – whilst a relatively minor issues, there were complaints from drivers around the amount of available elbow room leading to increased fatigue in the Endurance event.

There were many factors to consider in deciding the best approach moving forward for the 2023 competition chassis. These focused mainly around difficulty and manufacturing timelines, as well as downtime from testing and flow-on effects to car systems and how these changes would affect manufacturing loads. The following table summarises key points for and against each option outlines above:

Option	Benefits	Downfalls
1 – Chassis Modifications	- Least amount of change - Least number of tubes required - Front suspension unchanged - Majority of suspension pickups remained	- Still likely required changes for steering - Large downtime while members welded in - No potential for changes to accumulator packaging
2 – New rear half Chassis	- Reduced downtime for welding - Potential for minor improvements to accumulator packaging however still limited by existing roll hoop	- Still likely required changes for steering - Downtime while halves welded together - Completely new rear suspension pickups
3 – Complete new chassis	- Can be completely manufactured while testing continues - Allows for more significant changes to accumulator packaging and can incorporate new steering design	- Requires most changes - Largest order from tube supplier with added complexity in notching - Largest manufacturing load - Would require all new suspension pickups

Table 2 - Chassis Change Implementation Options

It was decided that the best option moving forward would be to design and manufacture a completely new chassis for 2023. Whilst this would require the most design and manufacturing work, it would minimise testing downtime and removed risk with manufacturing and testing being completed in parallel. This would also allow for options to be investigated for packaging improvements to be implemented without constraints of the previous design.

To reduce the manufacturing burden associated with a complete chassis, the design was to reuse the entire suspension from EV3. As a result, relative suspension pickup locations would not be modified in each corner allowing the suspension to effectively be unbolted from the old chassis and installed into the new chassis immediately. The following details the design of the 2023 chassis and changes that were made compared to the 2022 chassis.

2023 Design

With the successful characteristics of the 2022 competition vehicle, there was an emphasis placed on minimising changes to the 2023 chassis where possible. This reduced the risk of developing new issues with compliance and would also allow some components to be transferred directly across.

As accumulator temperatures had a significant effect on vehicle performance not just in the endurance event but also at track days limiting testing, it was intended that with the flexibility provided of a completely new chassis, additional room would be provided for a larger accumulator with improved internal air flow.

The following goals were selected for the design:

- Ability to reuse existing suspension as much as possible
- Ability to reuse existing pedal box
- Maintain geometry in the front half of the vehicle as much as possible
- Alter rear to comply with Acc mounting, SIS and RIS requirements
- Increase space for improved accumulator cooling

To begin, the existing 2022 chassis was modelled in OnShape parametrically using the frame tool as a starting point to make alterations.

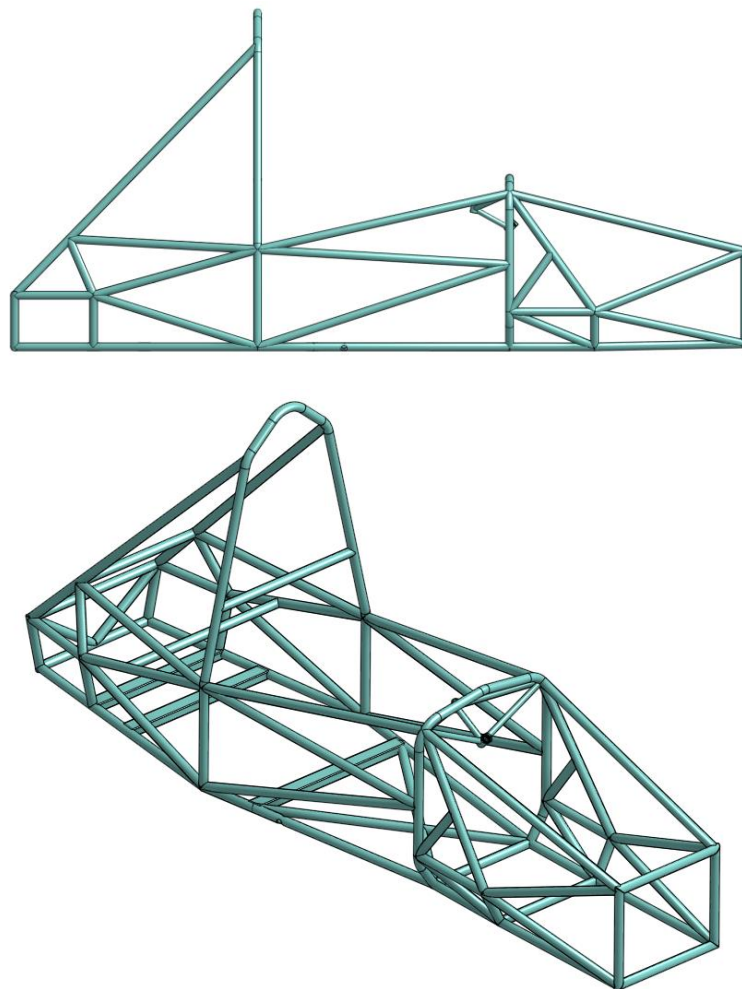


Figure 8 - EV.Three Chassis Model

At a bare minimum for compliance, The TS-SIS and RIS nodes had to be moved higher to 240mm. The flow on effects of these changes were considered initially to gain an understanding on how this would affect other subsystems, particularly the powertrain. At minimum, the following changes would be required:

- New Motor mounts
- New final-drive brackets
- New rear extension tubes

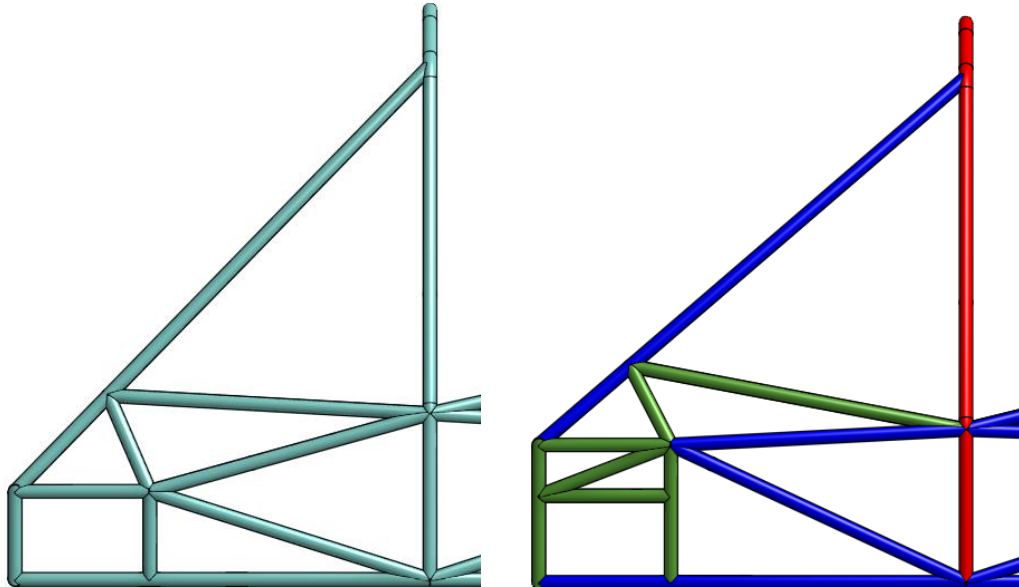


Figure 9 - Changes to TS-SIS and RIS

It was important to consider flow-on effects initially, as any other changes which had the same impacts they would not be an issue as these parts would have to be remade anyway. At this stage the powertrain configuration and layout had also not been decided, so packaging was to be left conservative with space requirements.

The addition of three members on the underside of the chassis corrected the compliance issues for accumulator mounting, and at this point the chassis was compliant, and any further changes weren't necessary for competition.

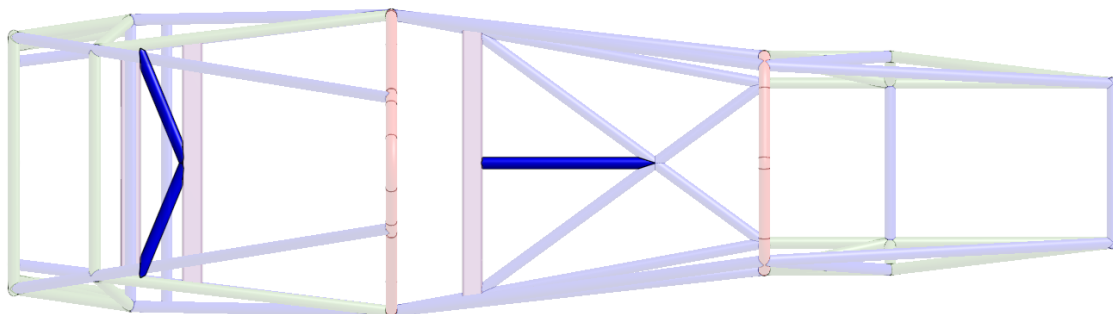


Figure 10 - Additional Members for Accumulator Mounting Compliance

Despite this, time was spent investigating potential changes for accumulator cooling and how the chassis could be modified slightly to allow for these changes to fit within its constraints. After many

discussions with the accumulator cooling engineer, it appeared the best option moving forward would be to rotate the segments 90 degrees and place push and pull fans on either side of the accumulator. The BMS was also to be moved to the top plate to improve packaging.

The overall width required in the chassis would provide enough room for a segment's length, the lined accumulator walls and two 38 mm axial fans. The accumulator length was determined by fitting four 120 mm axial fans plus two lined walls. This would allow for eight far wider and less restrictive segments or nine slightly wider segments, providing flexibility in cell arrangement should changes be desired later in the year.

Overall, in order to fit these changes, the accumulator opening had to be increased in length by 70 mm and width by 50 mm. This would have the flow-on effects of pushing out the rear frame members, requiring the rear suspension to move out wider increasing rear track, and back increasing wheelbase. The main hoop was also widened by 60 mm to keep the bottom frame members (and suspension nodes) on the same plane. This had the byproduct of increasing the cockpit width improving driver ergonomics.

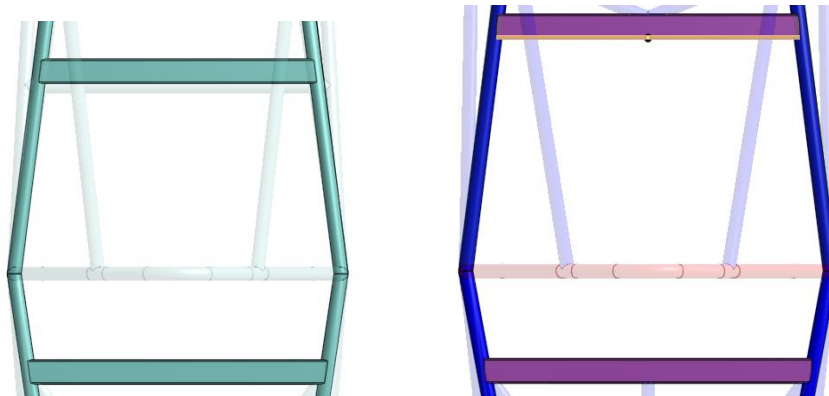


Figure 11 - Increased Accumulator Packaging in NU23

When considering powertrain packaging, there was approximately 30 mm of space which could be gained by moving the motor closer to the final drive. By decreasing the space available for the powertrain by 30 mm, the increase in wheelbase was reduced to 40 mm. Changes to suspension positioning in terms of width was far more difficult to alleviate due to nature of member layout. Whilst not ideal, an increase in rear track of 50 mm would only bring the rear wheels to equal track with the front wheels. Some testing was completed on EV.Three with rear wheel spacers to replicate this track change. With no noticeable change felt by drivers it was deemed acceptable.



Figure 12 - Reduced Powertrain Packaging Length in NU23

It was noted that these changes would have several effects on the manufacturing load for 2023. This included:

- A new accumulator would need to be produced as the EV3 accumulator would not fit the new chassis
- The CV housings in the powertrain would each need to move out 25 mm to keep the axle length the same so they could be reused.
- New firewalls, seat, controller mounting, and floor would be required.

From the front hoop forward the chassis remained identical to 2022, allowing the pedal box and front suspension to bolt up with no modifications required. The final chassis design can be seen below.

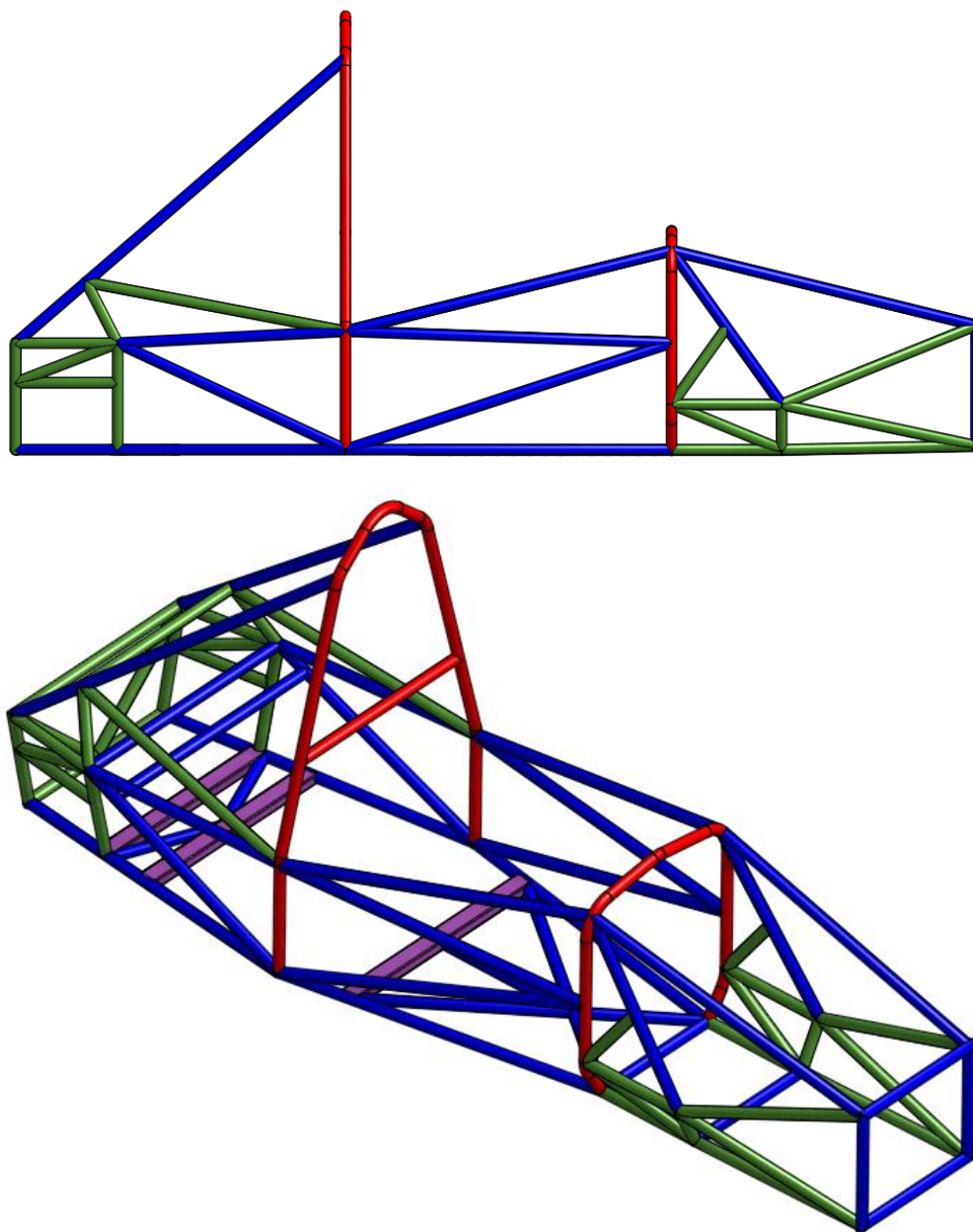


Figure 13 - Finalised NU23 Chassis Design (with optional members)

Optional members were added to provide a fourth tube for motor mounts to attach to as well as bracing for the rear suspension nodes. As the powertrain was not finalised and the requirement for suspension pickups being located on nodes not a requirement in the rules, these were to be notched last allowing them to be left out with no holes remaining in the chassis. When finalised in manufacturing these members were not included on the chassis.

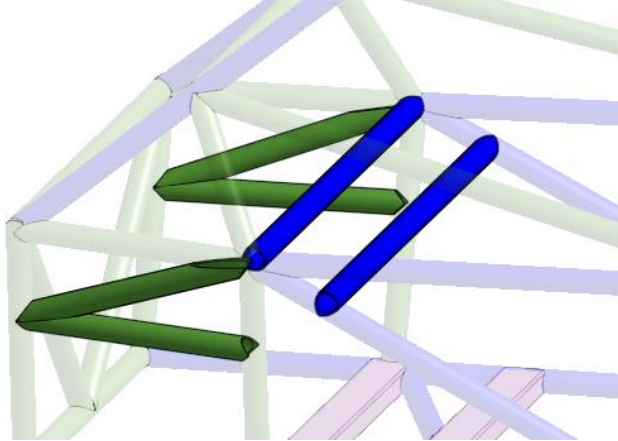


Figure 14 - Optional Chassis Members

Manufacturing

Notching

With the chassis design finalised, the tubes had to be prepared for manufacturing. For a spaceframe chassis, all members must be notched to fit together tightly at nodes allowing for a weld to be completed around all of the joints. For small, bespoke tube frames notching can be completed manually through a notching tool or even simply with a grinder.

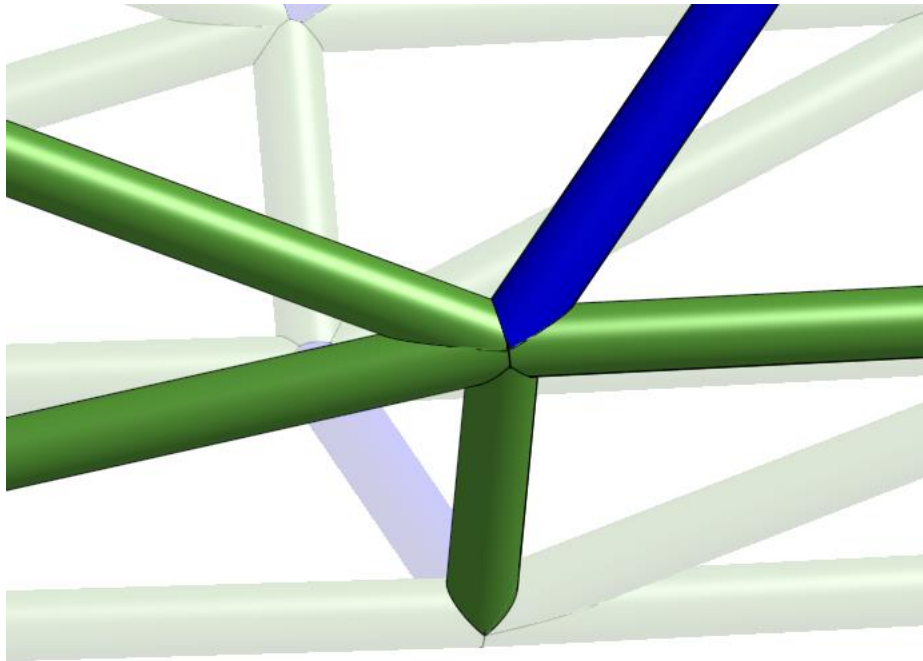


Figure 15 - Front Upper Suspension Node NU23

Due to the complicated and large nature of an FSAE-style spaceframe chassis, an alternative option is to have the notching completed by a computer-aided manufacturing process. One option is to have the tubes laser cut which provides a 2-dimensional cut around the tube in the required profile. While somewhat effective, this does not account for the angled nature of cuts generated in a model where tubes from several different planes may meet at a node.

For this reason, an alternative process is CNC notching with a cut created by a multi-axis tool path. This allows for near perfectly notched cuts with angled cuts reflecting the exact geometry of the modelled tubes. This is a unique process Canadian company VR3 specialises in, marketing to FSAE teams around the globe. VR3 was used by NU Racing for the 2022 chassis and feedback from the machinists who welded the tubes together was very positive. Whilst a more expensive option, the accuracy and reduced manufacturing time a VR3 tube kit would bring was deemed to be worth the cost. As a result, it was planned that after modelling the chassis, tubes would be prepared and ordered through VR3 as soon as possible.

To prepare the chassis for manufacturing, all notching must be completed in modelling software. Whilst a relatively straightforward concept, it is very important that the order in which the chassis is welded up is carefully considered. This will make assembly easier for the machinist, and prevent situations where tubes are not physically able to be added.

Prior to commencing the notching process in OnShape, several discussions were completed with the machinist from the MEWS workshop who was to weld the chassis. Several views of the chassis were printed out allowing for the ordering of assembly of these tubes to be discussed and planned out.

This order would in turn represent the order of hierarchy in which the frame members were to be notched in CAD. The finalised order can be seen in figure below. The ordering of notching was to be red, yellow, green, turquoise, blue, indigo then violet.

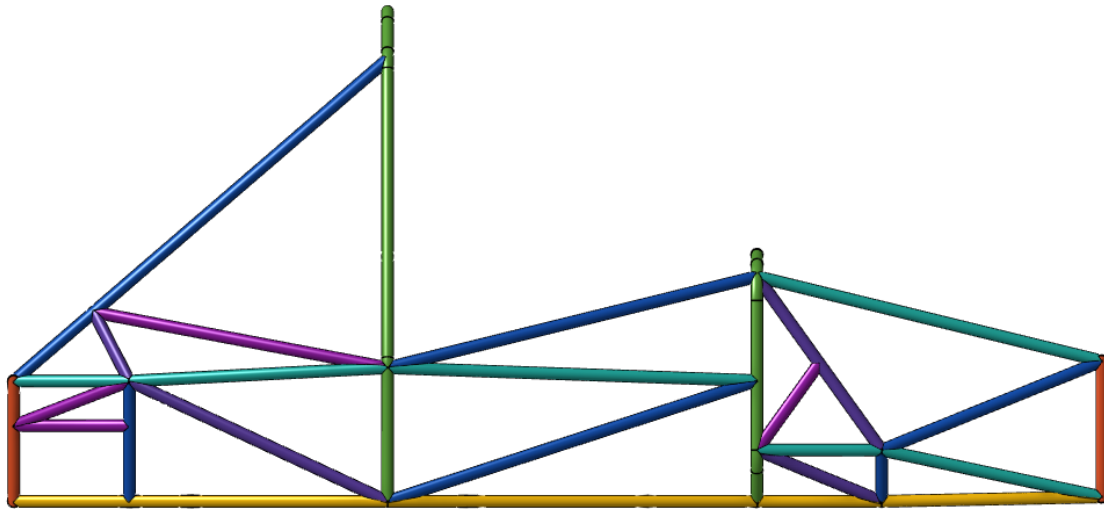


Figure 16 - Chassis Notching Order

Whilst this order will be somewhat consistent among chassis with similar geometries, it is still essential that machinists are brought into the design phase prior to the notching being completed to ensure all parties agree on the order of manufacturing. Tube notching in OnShape is somewhat time-consuming and so redoing notches due to changes from feedback is best avoided. For each node, two members must be chosen as a 'primary mitre' to create an unbroken surface at the node for all other tubes to be notched to. This primary mitre will be made between the most critical members or first members to be tacked at the node.

The overall process for modelling a frame OnShape is as follows:

1. Sketch all geometry by placing points at each node of the chassis.
2. Using the frame tool and setting the correct tube diameter and wall thickness based on SES requirements and design choices, create members by selecting the nodes the member runs between.
 - An important note in this step is, where appropriate, use the frame tool to generate multiple members at once in one operation. This should be used for tubes which form the primary mitre at a node which are of the same wall thickness, and for the front bulkhead and RIS where members create an enclosed hoop. Generating these members in one operation automatically generates the initial mitre required for the nodes in these areas.
3. Complete the initial mitres for all remaining nodes.
 - This will be required where the primary members to be mitred are not the same wall thickness or the members cannot be made in one operation for another reason. This process is detailed further below.
4. Use the frame trim tool to trim all remaining members in the manufacturing order required. This must be done at all nodes and so an ordered approach is required. Sometimes trimming may have to be completed over several operations at a node for the trims to work.

It is a relatively straightforward process, however as stated above some nodes will require additional steps due to limitations in the OnShape frame toolset. For the 2023 chassis, this was required at the lower nodes of the front hoop, as well as the upper rear nodes of the TS-SIS. At these nodes, the two members which form the primary mitre joint for all other members to be notched to have different wall thickness, and so could not be made in one operation.

Unfortunately, the OnShape framing toolset does not provide an easy way to form a mitre joint between members in this scenario. Instead, the best solution found for these nodes was to generate a plane representing the mitre cut and trimming each member to this surface. In addition, as the members are cut square when generated, they must be extended prior to trimming to prevent a hole forming on the other side of the join. Once extended and trimmed, this creates a seamless joint with no holes allowing all other members to be trimmed to this surface. This process can be seen in the following images.

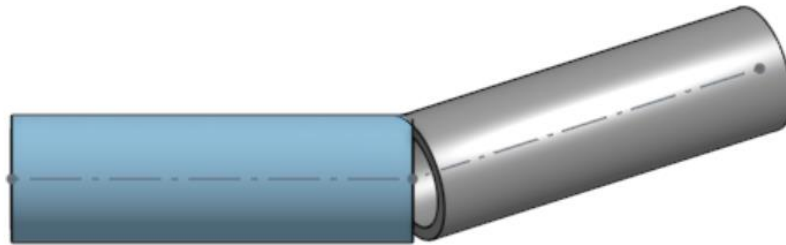


Figure 17 - Mitre Joint Prior to Trimming with Frames to Node

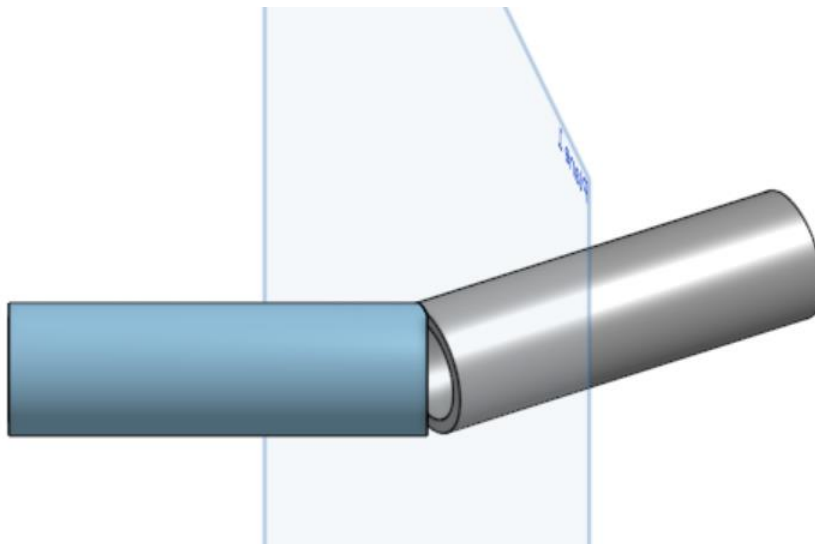


Figure 18 - Trimmed to Mitre Plane Resulting in Gap

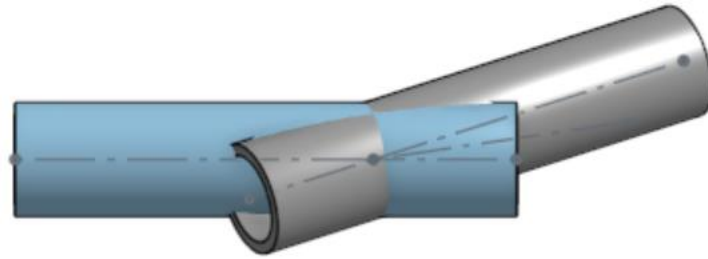


Figure 19 - Tubes Extended for Corrected Mitre

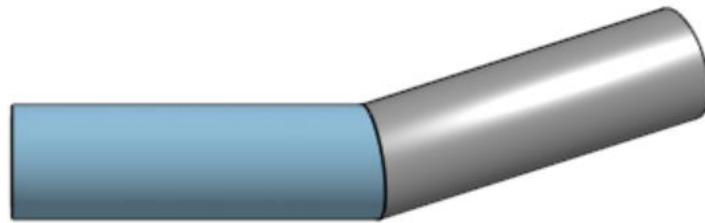


Figure 20 - Finalised Corrected Mitre

Ordering process

Once all members are notched, the assembly can be sent to VR3 for review. Due to the number of notches and joins present it is unlikely they will all be correct first go so there may be some changes required by VR3. Once all members are notched correctly, individual part files can then be exported for manufacturing. It is critical that all tubes are checked for wall thickness and that all documentation required is consistent.

A balloon diagram of all members is also required. For 2023 this was generated by importing the parts studio into an assembly and then creating a balloon diagram. The tabulated list of tubes and sizes was done manually.

With all documentation for the chassis complete, the tubes were approved for manufacturing and production commenced at VR3. While being manufactured, drawings were prepared for chassis welding in MEWS.

Drawings for manufacture

While there are many additional items to be welded to the chassis such as suspension tabs, steering components, harness attachment points, radiator supports, firewall brackets, body kit tabs, pedal box and accumulator, this section focusses on manufacturing of the bare tube chassis. Utilising CNC tube notching greatly simplifies the welding process as the accuracy of the notching means the tubes essentially assemble themselves. Drawings were required for general node heights and locations; however, jigs were not necessary to locate members as had been used previously.

Located in the appendices, drawings were supplied of a bottom view locating all floor members, dimensioned views of all hoops, and a side view showing heights of members at nodes. It should be noted that dimensions provided did not reference the centre point of nodes at this is not easily measurable. Instead, dimensions were generally provided to the underside of the of the member at the node.

Welding

With these drawings provided, general assembly was relatively straightforward with no significant issues to overcome. The chassis was officially ordered on the 3/04/2023 and arrived on the 1/05/2023. Most chassis tube welding was completed within three weeks, while the remainder of suspension pickups were finished on 8/6/23 providing a rolling chassis. Overall, chassis construction took five weeks, and this is a good conservative figure for timelines. It should be noted that an additional month was allocated to brackets, firewall and steering to finalise a rolling and steering chassis. These items have room to be streamlined in future years.



Figure 21 - Chassis Welding with Suspension Pickups Completed 8/06/2023

With the chassis complete in line with the tight timelines outlined for the team it was considered very successful. Providing a rolling chassis in early July led to the first drive at the end of July, the earliest any team has ever begun testing in NU Racing history.



Figure 22 - Rolling Chassis with Majority of Mechanical Systems Fitted 7/07/2023

Powertrain – Performance

Electric motors are ideal in their ability to apply essentially 100 % torque from 0 RPM. They do not stall like a traditional internal combustion engine and can also run at high speeds. When designing an electric FSAE powertrain, there are several factors which must be considered to ensure appropriate vehicle performance and system efficiency from an electric drivetrain.

The fundamental goal of a powertrain is to make the car accelerate. This is achieved by applying a torque to the driven wheels. The higher the torque, the faster the car will accelerate. Therefore, when designing a powertrain maximising wheel torque is the overall goal (Note of course that there are friction limits which place a limit on reasonable wheel torque required).

When looking at top-level design, in an electric drivetrain it can be assumed that the system is 100% efficient. Whilst not exactly true, it drastically simplifies how various aspects of the powertrain, from electrical input to mechanical output, are linked and affect each other.

There are four fundamental energy transformations that occur in an electrical FSAE car powertrain design, all of which must be understood. These occur in the accumulator, motor controller, motor, and gearbox.

Accumulator

The accumulator provides high voltage DC power for the tractive system, converting it from chemical potential energy stored in cells (For EV.Three/NU23 this was in the form of Sony VTC6 18650 cells). Cells are arranged in series and parallel to create the desired pack voltage and capacity (For EV.Three this was 6P and 96S providing 18Ah and 403.2V when fully charged). The current capability of the accumulator is also directly linked to the capacity – determined by multiplying the capacity in Ah by the discharge rating of the cells (c). As the VTC6 cells are rated to 10C for short periods, for EV.Three this resulted in a peak supply voltage of approximately 400V and a peak current supply of 180 amps.

As a pack is discharged the voltage will reduce. In addition, when cells experience high discharge rates approaching discharge limits, significant voltage sag will be observed. As a result, the EV.Three accumulator could only output approximately 62kW when fully charged as at 180A (which translates to 30A per individual 18650) pack voltage drops to approximately 345V. This is not the 72kW found by multiplying peak voltage by peak current. This will also drop very quickly as the battery voltage settles as seen in the first few 0.25Ah of the graph below. Discharge curves for most cells are available online and are a very useful tool to approximate cell output under peak load conditions to appropriately spec an accumulator. The following graph shows discharge curves for the VTC6 cells at different discharge rates.

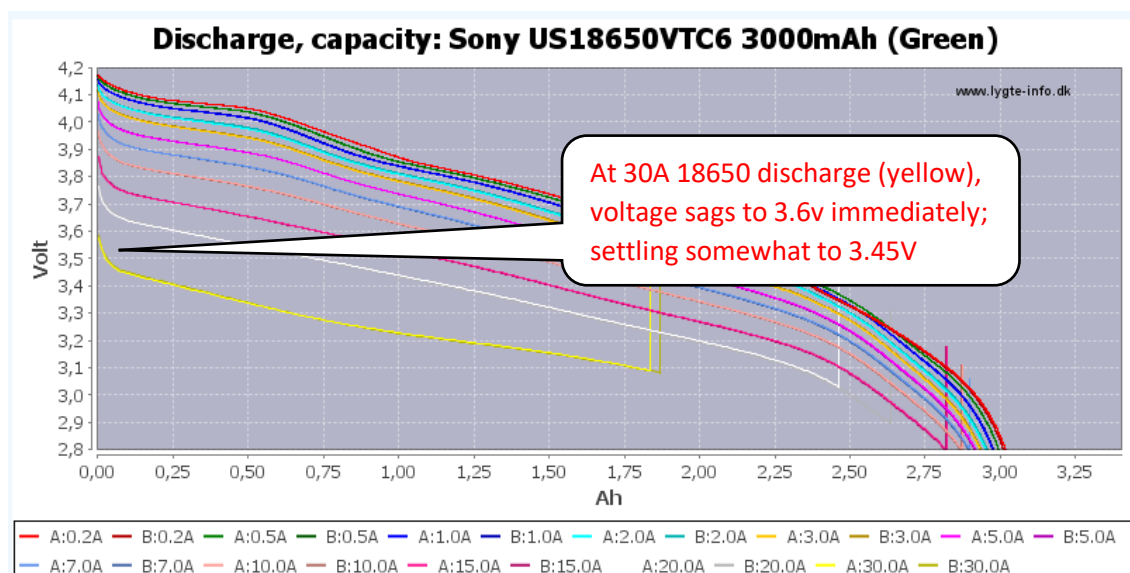


Figure 23 - VTC6 18650 Cell Discharge Voltage Curves at Various Current Levels (Test of Sony US18650VTC6 3000mAh (Green), n.d.)

Higher current through cells produces more heat and as power loss is a squared relationship, significantly more heat is generated at high accumulator loads. To model heat generation it is important to use RMS current and not average current for an endurance run as this more correctly reflects the squared relationship. Despite what intuition may suggest, for a given power load and accumulator energy capacity, a higher accumulator voltage with lower current output will not reduce heat generation.

Heat generation is only directly related to the total number of cells in the accumulator as a larger number of cells reduces individual cell load for a given power output. Generally, a lower battery voltage will reduce motor power output and therefore reduce accumulator load reducing temps, however this is not a direct relationship and vehicle performance is affected.

It is desirable to maximise accumulator voltage and minimise cell current for a given power output to increase system efficiency however this does not affect cell temperatures. Additional cells in parallel will reduce loads on individual cells for a given current output. An accumulator should be designed such that it does not exceed maximum motor voltage whilst also being able to provide the required voltage at full power draw.

The accumulator has a BMS which senses current draw and will fault if accumulator current exceeds the rated amount for approximately 10 seconds. Whilst a good last measure for cell protection, this will not prevent current spikes from occurring which can damage cells.

Motor controller

The motor controller takes the Direct Current (DC) supply from the accumulator and inverts it to a 3-phase Alternating Current (AC) output as required by the motor. Whilst there is a large amount of feedback and tuning which must occur in the motor controller, as well as complicated switching electronics, for a top-level understanding it is best to think about the DC power coming in and AC power going out. With efficiency assumed to be 100% these electrical powers are equal.

The motor controller switches the DC supply and converts this into a vary voltage 3-phase AC output to power the motor. This is achieved through the DC source being switched by Insulated-gate bipolar transistors (IGBTs) at very high rates. Pulse Width Modulation (PWM) control allows the controller to vary its Root Mean Square (RMS) output voltage as determined by the torque request and motor speed to provide the required phase current.

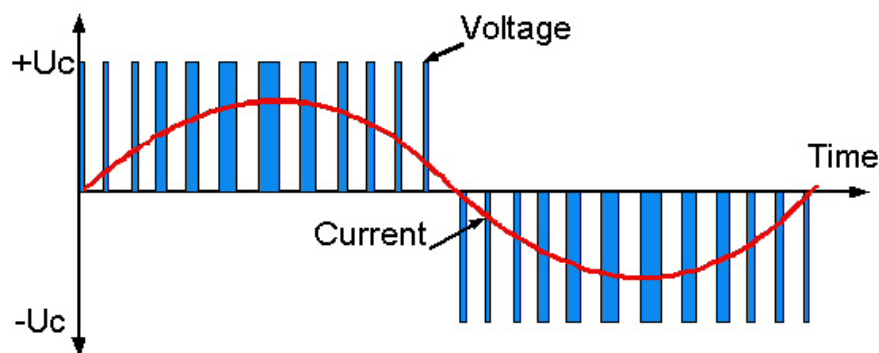


Figure 24 - Use of PWM to Create Alternating Current from Fixed DC Supply (PWM Inverter Circuit, n.d.)

The above graph is a simplification of the controller output however depicts how the controller can vary phase voltage to create an AC phase current. The magnitude of this alternating wave can be varied by adjusting the maximum duty-cycle at the wave's peaks and troughs. To change from positive to negative voltage the phase cable is either switched to TS + or TS -. Each phase output has two IGBT's to switch TS + and TS - resulting in a total of 6. Importantly, the inverter only switches the supply voltage and cannot provide a peak phase voltage larger than the battery voltage source (the limit is in fact lower due to switching losses).

The most significant setting for torque output in the motor controller is peak and continuous phase current (Irms). Peak refers to short term loads (up to 2 minutes for the Emrax 188) while continuous refers to loads which continue over long periods of time. These limits are tied to the current capabilities of the coils in the motor, and so are set based on the motor parameters in the datasheet. Due to the short periods of acceleration in FSAE events (approximately maxing out at 5 seconds) under all driving conditions peak output should be expected unless the controller has de-rated (usually due to excessive motor or controller temperatures)

When providing peak torque output, the motor controller outputs the required AC phase voltage to produce the desired AC phase current. For torque outputs other than 100%, this is proportionally reduced (i.e. a 50 % torque request corresponds to 50 % of peak RMS current). A key point to understand is that whilst there are relationships between phase current and accumulator current, a phase current limit does not provide a fixed DC current limit for the supply (accumulator), it corresponds to a supply (accumulator) current you are likely to see for a given RPM. This will increase with RPM as explained in the motor section.

Motors

Electric motors take the 3-phase AC power from the motor controller and turn this into rotational mechanical energy. Again, assuming 100% efficiency, the mechanical power output of the motor is equal to the electrical power transferred on the phase wires from the motor controller, which is equal to the DC power supplied by the accumulator.



Figure 25 - Emrax 188 Motor (Emrax 188, n.d.)

The torque output of an electric motor is proportional to the phase current in the windings. More current results in more torque, and this is a linear relationship as magnetic forces generated are proportional to current. Phase current and motor torque output are linked by a constant which can be found in the motors data sheet (for an Emrax 188 this is 0.6 Nm/1 Arms).

If mechanical power output is considered, power is proportional to torque and angular velocity. As electric motors have essentially a constant peak torque output (due to a constant peak phase current), power output under peak load is proportional to motor RPM. This can be seen in the red line shown on the below graph for the Emrax 188 High Voltage (HV) motor.

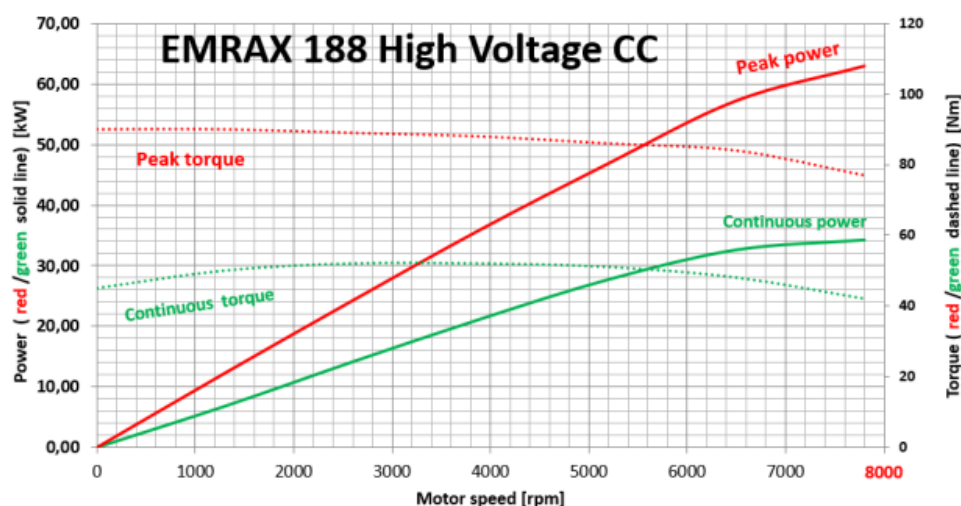


Figure 26 - Emrax 188 HV Dynamometer Test Data (Emrax 188, n.d.)

Motor mechanical output is equal to the electrical power being transferred in the phase wires. Therefore, electrical power also increases with RPM. Peak phase current is held constant, and so this electrical power increase relates to changes in RMS phase voltage.

As a motor spins faster, back-electromagnetic force (back EMF) is induced in the motor coils by the rotating magnet assembly. This back EMF increases in proportion to motor rotational speed. Therefore, to achieve a constant phase current under peak loads, the motor controller must supply a phase voltage which increases with RPM. This relationship can also be found in the data sheet (For an Emrax 188 HV $V_{rms} = 0.0384 * RPM$)

This increasing voltage is how the electrical power provided to the motor increases with RPM to match the increase in mechanical power output. The following table summarises the above. Note this is for increasing RPM at a constant motor torque (e.g., 100% torque request for an acceleration run).

Mechanical output	Phase wires	DC battery supply wires
$P = T * \omega$	$P = V_{rms} * I_{rms}$	$P = I * V$
Constant torque	Constant RMS current	Decreasing voltage due to sag
Power increases with RPM	Increasing RMS voltage	Increasing current with RPM

Table 3 - Summary of Energy Transfer During Acceleration Run

Motor specific load

Motor specific load is an important aspect of motor selection which must be factored in to ensure performance is as expected, particularly at higher RPM. RMS phase voltage must increase with RPM and as stated in the controller section, this voltage is limited by battery voltage as explained in the controller section. It must also be considered that as loads on the battery increase and the pack is drained, its voltage reduces.

A value known as specific load speed relates motor RPM to supply voltage. For an Emrax 188, this value is 16 RPM/1 VDC at full load, and 19 RPM/1 VDC at no load. This can be used to calculate the maximum motor speed by estimating pack voltage at the associated load and multiplying it by the specific load value. As battery voltage depends on power draw and maximum motor RPM depends on battery voltage, this must be trialled several times to approximate the useable speed range of the motor for the battery supply used. Motor power can be approximated by a peak power graph, such as the one in the motor section, to estimate power draw from the accumulator. An example of this calculation is also provided by Emrax in the motor documentation.

When combining all these factors, the net result is that motor performance at high RPM is heavily dictated by battery voltage performance at large loads, so it is critical that a cell is chosen which does not sag excessively.

For the EV.Three.5 drivetrain, assuming the motor reached peak power at 7000 RPM, power output would reach 60 kW. When fully charged at a 60 kW load the EV.Three accumulator voltage would sag to 345 VDC. $345 * 16$ equates to an expected motor maximum full-load speed of 5,500 RPM. At this point, motor torque output will have dropped significantly. Looking at the motor dyno graph, at 5500 RPM the motor can only output approximately 50kW, and so the initial power draw assumption is incorrect and must be trialled again at a lower power output (with RPM >5500 but <7000).

Testing with EV.Three.5 shows 5500 RPM is close to the full-load maximum speed as in the Motec data following the RPM curve has flattened off by the end of the run despite full throttle being applied the entire time. A significant reduction in torque is noticed from approximately 4000 RPM which can be seen in the steep reduction in accumulator current.

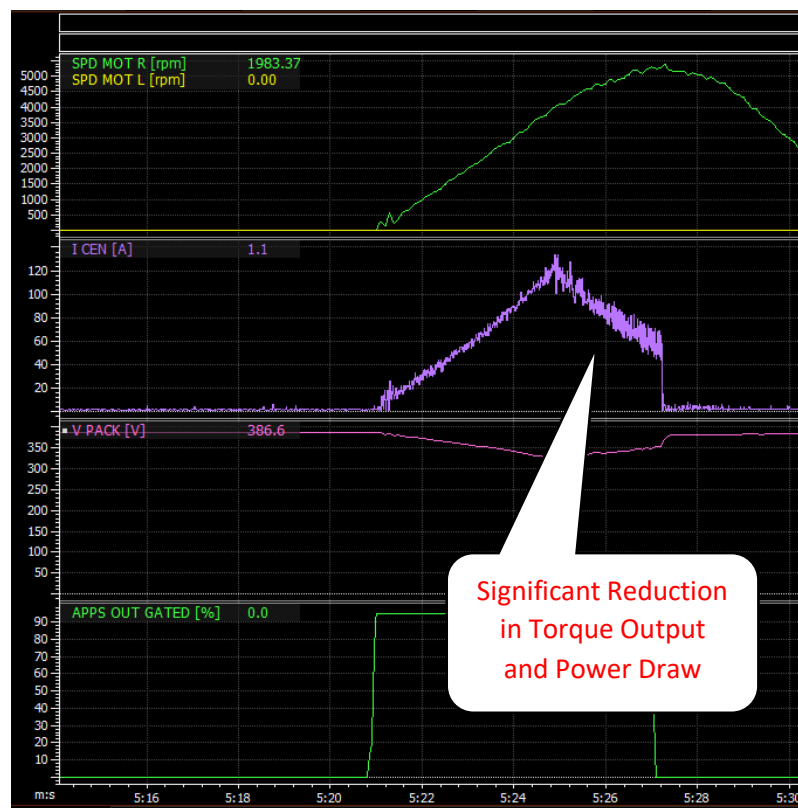


Figure 27 – Motec Data of Acceleration Run of EV.Three.5

In this image, 100% throttle is applied for the whole run, as shown by the bottom green line. This requests the motor controller to output 100% peak phase current to the motor (200 Arms) to produce peak torque output. Initially this is occurring correctly as the battery power draw increases linearly in line with the dyno graph provided by Emrax. The ICEN (accumulator DC current) gradient actually increases as accumulator current compensates for voltage sag.

Critically, at approximately 4000 RPM, power output reduces significantly as the motor passes its full-load base speed. This is because the controller has maximised its output V_{rms} (i.e. the duty cycle is maximised). This remains maximised, however as RPM continues to increase, resulting in a back-EMF increase, the effective coil voltage (phase voltage – back EMF) drops reducing phase current and resulting in a steep reduction in motor torque and power output. The RPM at which this critical change occurs at is directly related to supply voltage, and this is why a fully charged accumulator can result in better acceleration performance.

The car continues to accelerate beyond this critical RPM as phase voltage is still higher than the back EMF, however the effective coil voltage is not sufficient to result in the 200 Irms of phase current required for the maximum torque output requested. The car will continue accelerating (at a decreasing rate) until the motor torque drops to a point that it equals the cars resisting forces.

This significant reduction in power output greatly reduces acceleration performance. There is, however, a method to increase power output as these higher RPM's, known as field weakening.

Field weakening

The basic idea of field weakening is that through adjusting the phase of the drive current, the motor can be operated from a battery voltage that is less than the back-EMF of the motor. Functionally this control method uses some phase current to create a field which cancels out some of the magnetic field generating the back-EMF.

Whilst this does not allow the motor to continue producing peak torque, it does allow the power output to flatline in a constant power region, as opposed to dropping dramatically as seen in the Motec data. This does not increase peak torque or increase peak acceleration; however, it does reduce the rate at which torque decreases after the base speed. This is more extensively discussed in Josh Dawsons report.

The following two diagrams illustrate how field weakening improves motor torque and power output at high RPM.

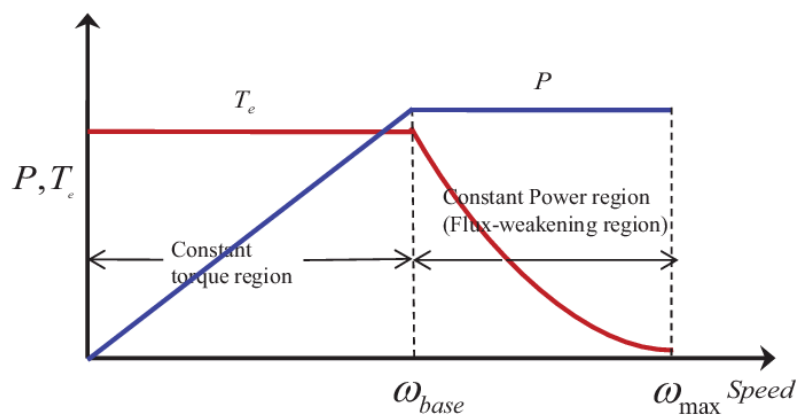


Figure 28 - Power and Torque with Field Weakening Enabled

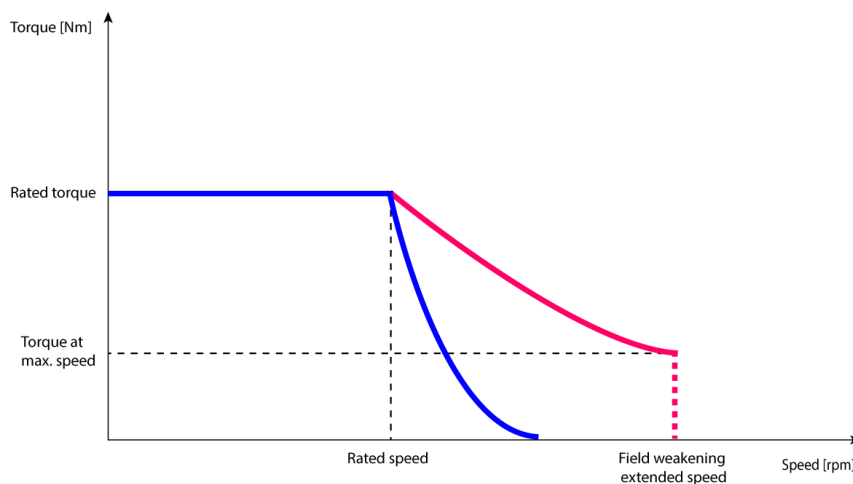


Figure 29 - Torque Output with and without Field Weakening

Downsides to field weakening are that the motor becomes less efficient as some phase current is not used for torque output. Whilst not recommended for constant running, it is acceptable for FSAE applications where the speeds at which this is required only occur for very short periods of time, and maximising motor output allowing for reduced motor size is desirable.

Gearbox

The final step is the gearing which takes the mechanical power from the motor and puts it at appropriate rotational speeds for the wheels. Again, power in is assumed equal to power out, so as $\text{torque in} \times \text{rotational speed in} = \text{torque out} \times \text{rotational speed out}$.

Maximising wheel torque is the objective of the powertrain, and so ideally the gearing is changed to the point where wheel torque is at the limit of grip. Unfortunately, there is still the limit of maximum motor RPM. Therefore, gearing should be maximised such that the motor hits its maximum RPM effectively at the end of an acceleration run, whilst factoring in the effects of voltage sag and maximum speed under load.

By doing this, in a single speed powertrain motor RPM is maximised for any given vehicle speed, which maximises motor power output and therefore acceleration. Note however that maximum motor RPM in service is not simply the motors maximum speed rating as explained in the motor section. This complicates the optimisation of gear ratios.

Limits to consider

As discussed above there are many limits which must be considered in powertrain design. These include:

- Peak motor phase current
- Battery current capability
- Motor maximum RPM

There are also many others to consider such as:

- Drivetrain mechanical strength (mounts, chains, belts, gears)
- Motor controller limits
- Cabling limits
- Competition power limit (80 kW)
- Motor maximum voltage
- Motor specific load speed

Phase current and motor RPM are controlled by the motor controller. If the parameters are set up correctly as per the motor documentation, these should never be exceeded.

The drivetrain should be mechanically and electrically designed to withstand peak motor output loads. This should ensure cabling and mechanical parts are not a limiting factor. The accumulator maximum voltage also cannot exceed the rated voltage of the motor.

Battery current limiting

Battery protection and competition power limiting are more difficult to achieve, mainly due to some motor controllers not having the ability to directly limit supply current in settings (this was the case when using the Unitek Bamocar D3).

There are 2 methods which have been explored by NU Racing to provide sufficient limiting to ensure battery current and competition power limits are not exceeded.

This is required when the motor can output more power than the battery can provide. This was the case for the dual 188 setup in EV.Three's drivetrain, although was not in EV.Three.5/NU23 with a single 188 drivetrain.

Power limiting with phase current

For the 2022 competition, power limiting was achieved by limiting phase current. Whilst this was somewhat effective, it is not the ideal method. This reduces motor torque output across its entire RPM range, when really it may only be drawing too much power at the top of its RPM range (as power is proportional to RPM). This means that torque output is lost where it does not need to be.

The second issue with this method is that as stated previously, it is not actually a power or current limit for the battery. This relies on maximum motor RPM not exceeding that which was seen when the limit was set up. If left to continue acceleration past 75 m, this setup would see the motor RPM continue to increase which would increase power output and therefore increase accumulator draw, surpassing the 180 A limit. This was observed at the 2022 competition.

By limiting this way, with constant torque output, the same performance can be achieved from less or smaller motors with more gearing. For example:

Comp had two motors limited to 160 Irms (approx. 80 Nm) geared at 3.84:1. This provided a wheel torque of approximately 307.2 Nm per wheel.

Two smaller motors which can handle 100 Irms (approx. 50 Nm) geared at 6:1 will provide 300 Nm per wheel.

One motor at 200 Irms (approx. 100Nm) geared at 6:1 provides a total torque applied of 600 Nm or 300 Nm per wheel.

Whilst all these scenarios use different phase currents, the differing gear ratios result in different motor RPMs. This results in essentially equal power output and accumulator draw. Gearing, phase current and motor configuration can all be adjusted with this method, but overall, you will end up with the same wheel torque as you are limited by power output at peak RPM in acceleration.

Power limiting with torque based on RPM

The alternate way to limit power is by having a motor torque output limit which varies based on RPM and accumulator voltage. As power output is proportional to torque * RPM, in drivetrain configurations with high motor power output, torque requests sent to the controller can be reduced as speed increases. This allows the motor to produce more torque at lower speeds (ideally at the limit of grip) whilst being limited at higher RPM. This maximises initial acceleration and allows maximum power to be reached earlier, without exceeding comp or accumulator power limits.

A simple way to implement this would be to continuously compare expected motor power output to instantaneous maximum accumulator power output (as maximum power output will decrease with accumulator voltage; only maximum accumulator output current is constant). By multiplying the motors speed by the requested torque output this can be compared to the instantaneous accumulator voltage multiplied by its maximum discharge current factoring in expected losses. If the accumulator output is exceeded, the torque request must be reduced.

This technique was explored in early 2023, however as the team moved to a single motor the accumulator draw limit was unlikely to ever be reached and so maximum phase current could always be applied. This method takes advantage of the full torque output of the motors.

2022 Review

The EV.Three drivetrain used two Emrax 188 HV motors gears at 14:50. At competition peak phase current was set at 130 Arms providing a total wheel torque of approximately 460 Nm. This provided disappointing acceleration performance however was reasonably successful in Autocross. Due to not implementing a proper method of battery current limiting there were several instances where DC current exceeded 200 A, well above the 180 A rating of the 6P Energus modules.

Initial Testing

Initial single motor testing with EVThree.5 had the controller peak phase current increased to the Emrax 188's peak rating of 200 Irms. Paired with the installation of a 60 Tooth (60T) driven sprocket and 12T drive sprocket, this increased peak wheel torque to 500 Nm providing a noticeable improvement in acceleration. This was promising with options for even smaller drive sprockets to be investigated.

2023 Design

For NU23, the same peak phase current and driven sprockets were to be used as EV.Three.5, however 10 and 11 tooth drive sprockets were found for testing. This further increased wheel torque to 545 and 600 Nm respectively. Whilst providing impressive off-line acceleration, as the motor was spinning faster, voltage limitations reduced output towards the end of an acceleration run. To try and improve higher speed performance whilst keeping this level of peak wheel torque, two changes were considered. Firstly, increasing the accumulator's voltage could provide improved performance and increase the motors base speed, while secondly, field weakening could be implemented to improve performance above the base speed.

Tractive System Voltage

Due to the increased chassis size, there was room for an additional segment to be added in the accumulator. This would provide an additional 12 Energus modules (which were available as spares) in series increasing maximum tractive system voltage from 403.2 to 453.6V. This would require a new accumulator container and alterations to the BMS wiring, however these needed to be redone from the initial NU23 accumulator due to manufacturing defects in the mounts and changes to the voltage sense wire routing. It would also require new narrower segment housings and slight revisions to the precharge and discharge circuits. Whilst a significant workload, many components affected required revisions prior to competition anyway. Generally, simply adding more cells in series is not ideal as this also affects the energy capacity of the accumulator. It was believed, however, that this would also improve accumulator temperatures by lowering individual cell loads and provide a more reliable cell arrangement for future years. The increased capacity would also be beneficial to endurance performance.

Concerns for this major tractive system change mainly stemmed around the motors voltage rating as the original datasheet explicitly stated a maximum battery voltage of 400 VDC. This was the original factor driving tractive system voltage for EV.Three. This did not appear consistent with the dynamometer data nor more recent datasheets provided by Emrax and so an enquiry was submitted to the company. The response was surprising, with confirmation that "all our motors will be fine up to 850 VDC. You only have to make sure that you limit maximum RPM in order to not exceed max speeds in case of low load conditions and higher voltages. Previously defined »nominal voltage« was voltage required to achieve continuous performance of the motor, nowadays so called »max. recommended voltage« is voltage required to achieve peak performance of the motor". Upon receiving this information and ensuring that all systems which interfaced with the tractive system would be suitable for operating at this higher voltage, the decision was made to increase system to 450 V with the addition of a ninth segment in the accumulator.

Field Weakening

Field weakening testing was also completed mid-way through the year to provide some data on whether this would be a feasible strategy to improve high speed performance for improved top-end acceleration. Whilst presenting risks due to the team's minimal knowledge on this complex control method, it was somewhat averted as the team had a spare motor should any issues occur. A car park test day was completed with field weakening trialled at different levels with promising results. With field weakening enabled even at low amounts (10 %), maximum motor RPM (and therefore maximum speed) at the end of a 75 m acceleration run increased by approximately 10% (~6000 to ~6500 RPM).

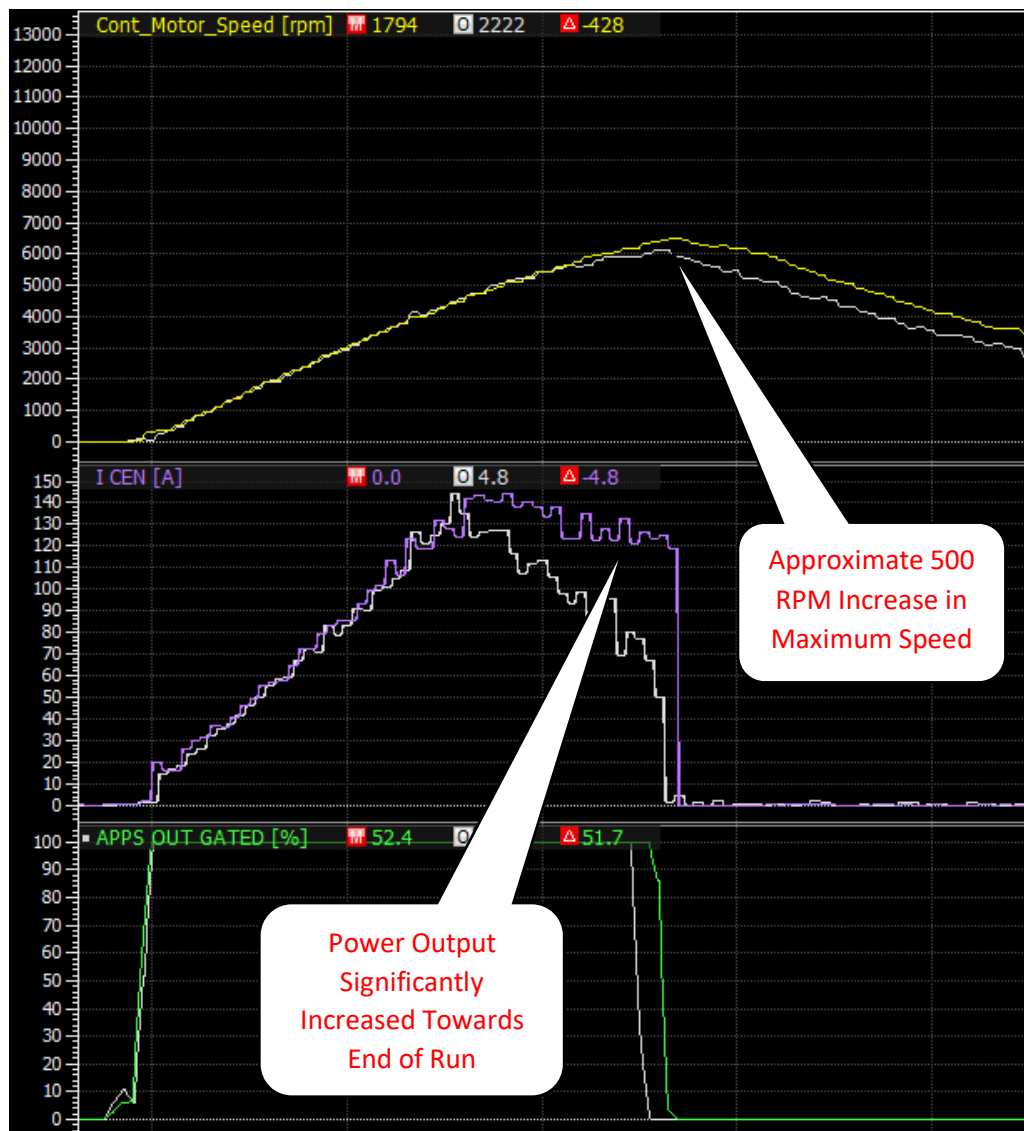


Figure 30 - Field Weakening Compared to No Field Weakening on 75 m Acceleration

Whilst unfortunately beacon timing was not functioning for this testing, the improvement in acceleration in the last 25 m of a run was noticeable. With these promising results it appeared that acceleration performance would be satisfactory for competition. After increasing to 40% field weakening the controller faulted due to current oscillation and so at this point it was decided to stop testing. The plan was to reduce the amount back to 10-20% for further testing at SMSP to validate this option.

Testing Incident

Unfortunately, an incident occurred in the workshop in the evening after this testing. A spin test of the motor led to the controller locking onto full output despite the throttle being lifted. The motor spun at very high speeds before smoke started being emitted. Within just 2 seconds the dash E-stop was activated, however by this time it was too late. The motor sustained obvious critical damage with grinding apparent as it was rotated by hand. This was a shock to the team, especially after the performance improvements observed in the earlier testing.

It should be noted that prior to this event several incidents had occurred during test days where the controller did not respond to a reduction in torque request and maintained full torque output after

the driver lifted throttle at the end of a straight. A significant safety issue, this occurred on the first test day of NU23 with its new LV systems and repeated a handful of times intermittently spread across many track days. The issue was frustratingly random with no clear cause, however seemed repeatable reappearing within a short time of a first random event. Improvements in CAN communication with the controller confirmed that it was receiving a torque request of zero, yet the motor would continue to be powered by the controller. More information surrounding this can be found in Jacob Bush's report. No clear cause for this was ever found.



Figure 31 – Motec Data of Catastrophic Motor Damage

Upon reviewing the data from the motor incident, it made little sense. The motor RPM data peaked at 6498 RPM, which is well below the maximum speed rating of 8000 RPM for a few seconds. The phase current and overall power output were also well within the motor's limits. The unknowns surrounding this event were a concern as it was believed that the controller settings changed were done correctly and within safe margins. When reaching out to Emrax, they also agreed that a motor failure should not have occurred in these operating conditions.

Upon extensive reviews of the logged data and controller documentation, a small detail in one setting provided a very likely explanation for the events which occurred. It is believed that this incident was caused by three main factors, based on work completed by both the 2022 and 2023 teams. The key issue was the parameter N-Lim. This is a parameter which sets the motor speed limit as a percentage of N-100% (which is entered as an RPM value; set at 6500 for EV.Three and kept the

same for NU23). N-Lim is a speed controller setting, which overrides torque control and places the controller in a speed control mode when reached (and while a torque request is still present which is greater than the speed controller torque output). It was found that N-Lim was set to 100% in 2022 and not reviewed in 2023.

Somewhat confusingly, however not uncommon in industry after discussion with a professional, an N-Lim of 100% in Unitek software does not set the speed limit at 100% of the N-100% RPM value as might be expected. Instead, it turns off “cruise control” which is Unitek’s term for speed limiting. As a result, there was no speed limit ever applied in the controller settings across 2022 and 2023. This did not make sense as the RPM data did not exceed 6500 RPM, however through further reading it was found that N-100% also sets the ceiling from which CAN RPM data is scaled. As a result, if the motor ever did rotate at a speed higher than 6500 RPM it would only be reported at 6500.

The second contributor was turning on field weakening. This, as explained earlier, would allow the controller to operate the motor at speeds far higher than its base speed. It is unknown what the potential speed ceiling would be with 40% field weakening enabled and only the inertial load of the powertrain on stands.

The third contributor was the controller continuing to output power despite the throttle being released. As stated above, it is still unknown why this occurred, and it was by freak chance that it happened whilst high field weakening was enabled, and the car was on stands.

These three factors combined resulted in a controller at full output with no speed limit set and significant field-weakening enabled. It is believed that the motor spun at incredibly high-speed causing significant mechanical stress resulting in internal damage. This theory was consistent with findings when the motor was pulled apart – the magnets on the outer rotating assembly had begun to break up from the intense centripetal forces placed on them. The actual speed the motor reached is unknown with the data only conclusively saying it was greater than 6500 RPM. In reality it must have been significantly higher than 8000 RPM for damage to occur in such a small period of time.

It is believed that having the speed limit inactive never previously caused an issue even when spinning the motors under no load as the base speed from the tractive system voltage was lower than 6500 and could never cause damage. This was a very disappointing and expensive error for the team. Whilst it occurred directly after turning on field-weakening, it was also a result of other setup errors and if it hadn’t occurred at that time it was likely to happen at some point in the future with increased tractive system voltage or further field weakening experimentation.

From this knowledge, and a review of track day data, the motor speed did also surpass 6500 during the acceleration runs with field weakening enabled as the speed graph had a small but distinctively flat top. Disappointingly, when this data was originally viewed at the track it was thought to be an indicator that the speed limiter had engaged correctly, in fact providing a sense of achievement with the vehicle reaching such high speeds. In hindsight this was not the case, and unfortunately provided a hidden clue as to what was to occur.

This event signified a significant change in direction for the team and particularly tractive system changes. Field-weakening was no longer considered an option based on these events and the team no longer had a spare motor meaning damage to the other motor caused by changes to controller settings would have dire consequences for the competition. From this point forward, focus was placed on maximising the performance of the powertrain with gearing and leaving it at that. This left gearing at 11:60 as 10:60 provided an almost identical acceleration time however placed the motor at higher RPM.

Controller Change

Whilst powertrain electrical setup seemed locked in after the events described above, controller issues continued to affect vehicle performance in the months leading to comp. Explained more thoroughly in Jacob Bush's report, issues progressively worsened until the month leading up to competition where the team had two motor controllers which would fault under small amounts of throttle rendering the car almost undrivable.

One motor controller began having issues immediately after being reassembled following repackaging investigations early in the year. This was the 'spare' controller and faulted in error mode 9 after minimal throttle was applied under load (i.e. with the car driving, this would not occur on the stands). Upon investigation a loose current shunt was tightened which temporarily fixed the issue however this only lasted one charge of driving before the error came back. This is the only time this controller was successfully driven with in 2023. Bolts were checked again however this did not fix the issue. It was installed to see if the random controller locks at full throttle were isolated to the other controller or not. Whilst it never occurred with this controller, the minimal runtime did not provide any valuable information.

The 'main' controller was used throughout the year and begun faulting in the same error state, starting under high loads, and progressively worsening until it also could barely drive the car (this occurred following months of testing where it had on rare occasion randomly locked on full motor output forcing several emergency stops). In its two last track days of use, this controller could complete an entire endurance event, however after charging it faulted in autocross style driving. Once the first fault occurred, it would worsen with less and less throttle required for subsequent faults. It is unknown if this was related to heat but at carpark testing after these two SMSP track days errors would appear within minutes of driving.

It is not known exactly what damaged the two inverters throughout the year. It is likely incorrect assembly damaged the first controller, while high motor speeds paired with opening the tractive system under load possibly caused damage to the main controller. There were also some instances where DC power cutout due to the DCDC shutting down and the LV batteries overcurrent protection removing LV power with the motor still running at high speed. This also may have damaged the controller. Overall, the 2023 experience with Bamocar Inverters was poor and entering December it appeared that driving at competition might not be possible with these controllers. There is potential that these Inverters could be repaired (potential being used with the 188 in a future autonomous vehicle). It is suggested that the IGBT's are tested to see if any are damaged.

A last-minute decision was made to purchase a Cascadia CM200DZ inverter for use at the 2023 competition. Whilst it required complete integration with the vehicle and came at a significant cost, the benefits of improved documentation & support, better quality, and more durable components favoured this option over trying to source another Bamocar.

Delays in shipping saw the controller not arrive at the University until the Tuesday morning exactly one week prior to the team's scheduled departure for the competition. The last remaining SMSP test day was booked for the Wednesday, and so an incredibly optimistic timeline of implementing the controller prior to the aimed departure to Sydney at 5:30 AM the next morning was set. The majority of the required coding, fabrication for mounts, and cooling system changes to suit the new controller were done in advance leaving only wiring and parameter setup. Incredibly this timeline was achieved by the team, with the wheels spinning at approximately 11 PM on the Tuesday night ready for a full track day the following day.

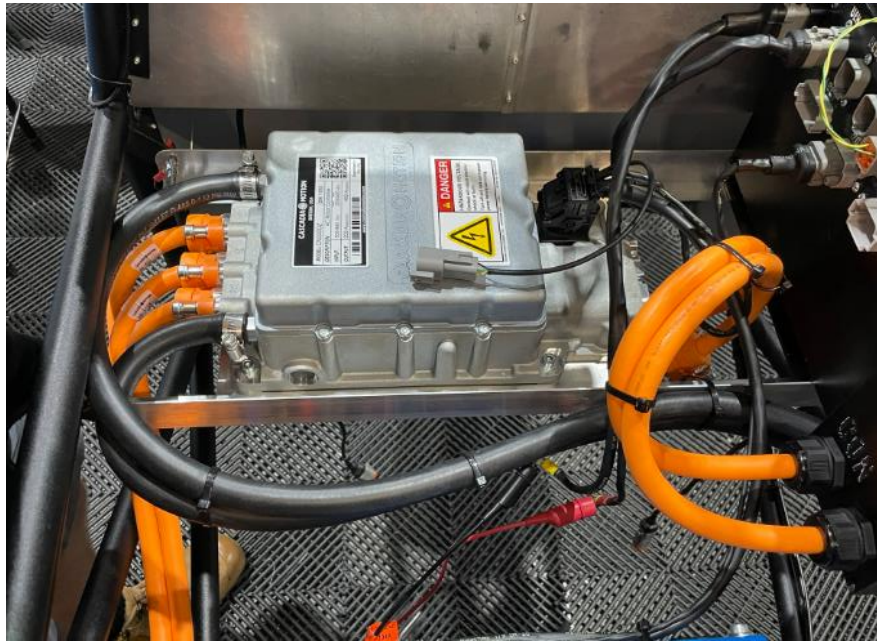


Figure 32 - Installed Cascadia Motor Controller

This track day was a great success with a complete comp simulation completed. Due to the incredibly short turn-around of the controller all default Cascadia settings for the 188 motor were untouched. This resulted in field-weakening being activated once again (with a correct speed limit enforced) resulting in the best acceleration performance ever observed in the acceleration and autocross practise events of the whole year. Controller and motor temperatures were stable despite a complete cooling system overhaul marking an incredible day and achievement for the team. This is a significant point for consideration when selecting hardware – paying more for a premium controller led to better performance immediately compared to almost a year of development with a cheaper product.

Competition Performance

Following on from the last testing day, competition performance was excellent considering the limitations of a Single Emrax 188 powering the car. Whilst not placing particularly well in the acceleration even, 0.8 seconds of time was dropped compared to EV.Three. This was a very significant achievement considering that half of the powertrain was removed. The powertrain also delivered excellent output in the Autocross event and was reliable for the Skidpad and Endurance Events.

It should be noted that a slight oscillation in output was observed at low speeds and torque outputs. There was no time to investigate this prior to competition however the Cascadia software includes parameters which can be adjusted to rectify this.

2024 Design

Due to the limiting performance of a single 188 motor a purchase was made for 2 Emrax 228 Medium Voltage (MV) motors in late 2023. Whilst a 208 could potentially provide the torque and RPM required in a lighter package, the 228 was the recommended motor with headroom in performance, reducing the criticality of gearing in powertrain design. With further information from Unitek and recommendations from Emrax, a 228 will provide a much higher torque output, allowing gearing to be lower without compromising wheel torque. Lower motor speeds place it in a more stable region reducing the need for field weakening or pushing motor capabilities. Choosing a MV winding instead of HV also improves performance at lower voltages by increasing the base speed and further lessening the need for field weakening.

Gearing for 2024 was selected such that with the motor at full torque output the rear grip limit would be slightly exceeded. With fine tuning of the maximum motor torque output the wheel torque output can be optimised for the acceleration launch. A required wheel torque of 750 Nm was derived through simulation, leading to an ideal gear ratio of 3.26 as an Emrax 228 outputs 230 Nm of peak torque. Considering available sprocket sizes, gearing of 14:46 was selected, resulting in a peak wheel torque of 755 Nm. Should this amount of wheel torque not brake traction, a 13T drive sprocket could be implemented.

2024 Recommendations

For 2024, the 228 motor must be implemented, and all motor controller settings reviewed. Should oscillation issues arise again with the 228 they should be rectified, while battery current limiting strategies as discussed earlier will need to be implemented due to the increased potential motor output. If the Cascadia controller does not directly limit supply current, the following concept should be explored for the PEN code. Note that the NU23/NU24 cell arrangement of 108 Energus Modules can realistically only output a maximum of 72kW and so provided accumulator maximum current is never exceeded competition power limits should never be approached.

```
(Assuming 90% system efficiency)

For every pen iteration:

Take instantaneous motor rpm and pack voltage (may need to be smoothed)
maxCurrent = 180 (amps; set for 6p VTC6 Energus Module)
n = 0.9 (efficiency factor for system)
maxAccPower = maxCurrent*packVoltage
maxMotorPower = maxAccPower*n
motorRadians = motorRpm*pi/30
maxTorque = maxMotorPower/motorRadians

If torqueRequest greater than maxTorque
    torqueRequest = maxTorque
```

Figure 33 - Potential Solution for Correct Power Limiting

This will provide a robust and high performing electrical system for the 2024 powertrain. The 228 will be a far less-stressed motor allowing the team to improve performance without pushing component limits like the 2023 team did. Whilst it is necessary to take risks in racing to get the most out of a vehicle, the practices of the 2023 team are not sustainable for an FSAE team. Overall, the change to a 228, paired with improved launch control and power limiting strategies, should allow the team to become far more competitive in the acceleration event for the 2024 competition. Continuing the overarching NU Racing goals, the resulting powertrain will retain its simplicity with improved performance.

Longer term goals for the team and concepts to always keep in mind may include the following:

- Developing a new accumulator which can deliver at least the competition power limit of 80 kW when factoring in the voltage sag which will occur at this load.
- The cells should be selected such that they can be cooled effectively with the current demands required for complete depletion of the cell within a competitive endurance time.
- Motors should be selected such that they can deliver the required amount of power at minimum accumulator voltages.
- The nominal battery voltage should reflect the motors design voltage.
- Motor selection and gearing should be such that the wheel torque must be limited by traction and competition power (or accumulator supply) for an entire acceleration event (i.e. wheels would slip, or power limit be exceeded, if more wheel torque applied at any point through a 75 m acceleration)

Powertrain – Structure

The drivetrain is responsible for taking mechanically taking the torque output of the electric motor and translating this to the drive wheels. This generally involves some sort of gear reduction paired with a diff/spool and axles. The drivetrain is also responsible for mounting the motor(s) and allowing for drive throughout complete suspension travel.

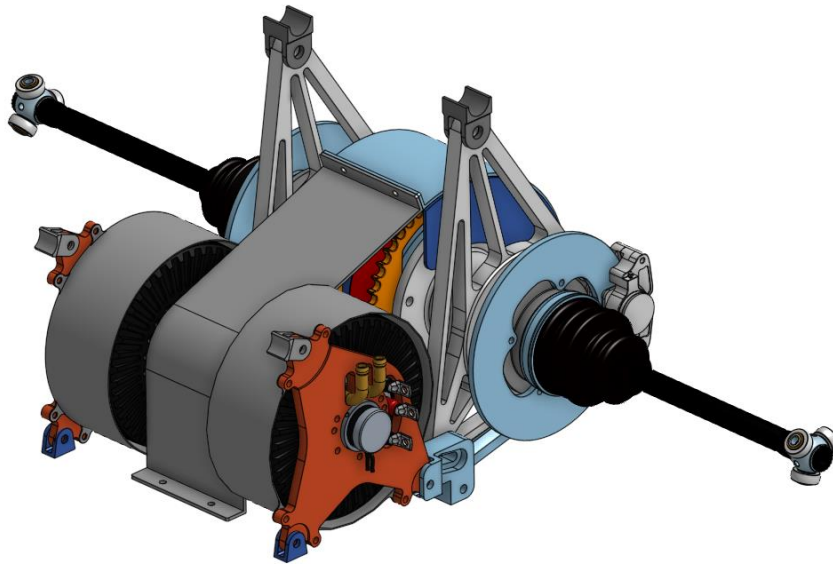


Figure 34 - EV.Three Drivetrain

The EV.Three drivetrain comprised of two Emrax 188 motors mounted with brackets to the chassis. These independently drove one rear wheel each, with separate chain drives for reduction. A coupling was used in the final drive to allow for independent rotation of each axle and wheel. Drive went from each side of the coupling to CV housings which translated torque to drive axles connected to the outer CV's/hubs. The final drive assembly (comprising of the coupling, cv housings and sprockets) was mounted with bearings to two final drive brackets. These were pivoted from their upper chassis mounts, with adjusters on the bottom allowing for chain tension.

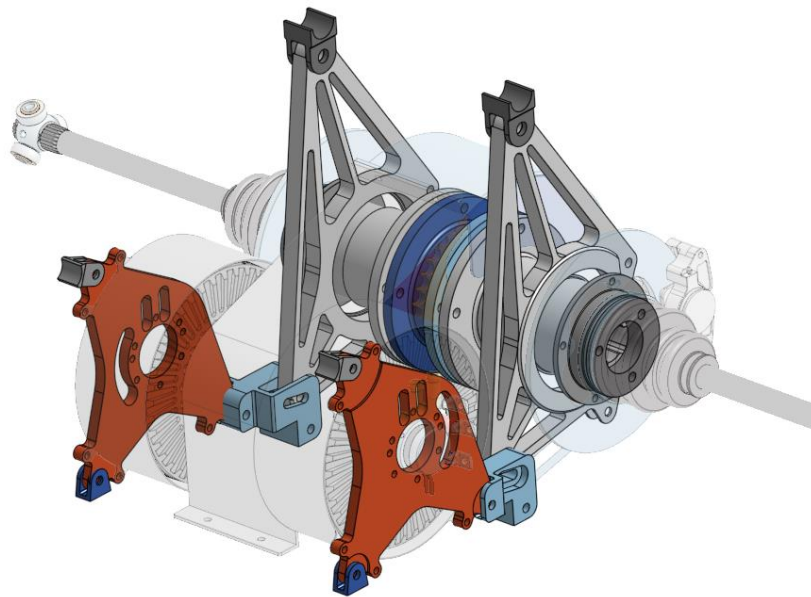


Figure 35 - Machined Components of EV.Three Drivetrain

This design was mechanically impressive with an independent drive allowing for torque vectoring by vary motor torque outputs to each chain drive. It also allowed for high power output and worked without fault at the 2022 competition. There were, however, several areas in the design which could be improved. These included:

- Weight. All brackets welded to the chassis were made from machined solid mild steel while the coupling and dual drive also added significant weight. This was in addition to the second motor and inverter weight penalty.
- High part count and significant number of parts requiring machining. Pictured above, over twenty components required some level of machining
- Poor usage of available torque/power. The motors were only ever used at approximately 65% of their peak torque output, and only ever output approximately one third of the peak power potential.
- Difficult serviceability. The entire powertrain was very difficult to remove from the vehicle, and chain adjustment was time consuming with many fasteners requiring adjustment.
- Final drive bearing alignment relied on adjustment by eye.
- Chain tension was not equal due to relying on drive sprockets being aligned. This was not the case after manufacturing resulting in one chain being tighter than the other with no way to adjust this (as the spool adjustment tensioned both chains at the same time).

Overall, any revisions to the drivetrain for 2023 would mainly focus on reducing part count and manufacturing requirements, with the by-product of weight reduction. Ideally the drivetrain would be simplified with improved utilisation of hardware.

2023 Design V1

With the decision made to only use a single 188 motor for the 2023 competition year, and changes made to the rear of the chassis for compliance, many mechanical components of the powertrain would need revision to suit this new arrangement. The main changes required were:

- Removal of the coupling to be replaced by a spool with an additional 50mm of length (to make up for the increase in rear track)
- Ability to house a larger 60T rear sprocket despite decreasing packaging length by 30mm (in order to create the required gear reduction)
- Improved chain tensioning (to stop final drive bearings walking due to incorrect adjustment and simplify process)

In addition to these requirements, an emphasis was placed on reducing part count and the complexity of components to reduce manufacturing loads and align with top level goals for the year. Ideally many parts of the existing drivetrain would be reused.

Initial design options

There were two design concepts considered initially for the 2023 powertrain. The first was to recreate a similar design to EV3, with two adjustable final drive hangars and a fixed motor bracket to develop chain tension, with revisions to suit the new chassis. This would utilise additional frame members which were added as optional in the chassis design.

The second option considered was to incorporate the motor in a final drive bracket as one single machined part. This had the potential to remove complexity in manufacturing by using the parts machined geometry to align the motor and drive sprocket, as opposed to relying on accurate positioning of brackets welded to the chassis. This concept also offered a way for the powertrain to be separated from the vehicle as a complete unit, improving serviceability and allowing for bench testing. This design concept was called the 'Powerbox'.

Whilst a refined design like EV.Three presented less risk with known manufacturing requirements, the potential of reducing part count and machining loads with a Powerbox was very appealing. As a result, initial design work focused around seeing if this was possible. As this would be a proof-of-concept first time prototype, the design focused around ensuring it could physically fit, while reusing as many parts as possible. Designs for new parts would focus on easy manufacturing with little emphasis on weight or refinement provided the whole package would not see an increase in weight compared to the EV3 drivetrain.

Should the initial prototype not be viable, a more traditional powertrain could be implemented as the first prototype design would be done relatively quickly. As this design work was being completed in parallel with the chassis manufacturing, if the outcome of the initial design suggested this concept was not feasible, the chassis could have optional members welded in and a more traditional powertrain design manufactured prior to the first drive of NU23, predicted for late July. This did present some risk however the overall concept was sound and removing weight saving from design goals greatly reduced the chance of issues occurring.

Powerbox initial design

The first Powerbox prototype focused on reducing machining to get the minimum functional product that could be used at competition. This would be done relatively quickly and modularly, such that if it was successful a lighter second-revision could be implemented before comp.

The key points for the first prototype were:

- Reuse existing inner tripod housings and axles (using spacers to make a spool)
- Mount the motor to one final drive bracket
- Some form of chain adjustment for testing different sprockets
- Cater for mounting of single brake rotor/caliper
- Ability to fit a 60t driven sprocket

The overarching design would be a box with three aluminium plates (one for the base and each side) and 3mm steel plates for the front, top and back. The steel would serve as the chain guard hence requiring the 3mm thickness and steel material for compliance. The base plate would provide mounting to the chassis whilst also forming structure for the chain tensioner to bolt to. By using the inherent strength of the chain guard as a structural element, weight could be saved by reducing the amount of additional bracing required.

Packaging was very tight from front to rear with the much larger 60t sprocket. Just over 406 mm of space was allocated in the chassis from the front surface of the rear accumulator mount RHS to the front surface of the RIS. When removing the 6mm for the front and rear chain guard plates, 400 mm was left for the motor and final drive. Measuring the 60t sprocket with chain used in early testing on EV.Three.5, this was just under 260 mm in diameter, requiring a minimum clearance of 130 mm to any surface from the centre.

As the existing tripod housing flanges had an overall diameter of 150 mm, this left only 195 mm of space for the motor. The existing motor scatter shields had a diameter of 200 mm and so would not fit in this space. Instead, the outer steel pieces would also be utilised as the scatter shield. This did come with a weight penalty as the shields only require a thickness of 1 mm however would be 3 mm. This compromise was accepted as it would reduce part count and complexity for a first prototype. Overall, this left approximately 4 mm of clearance around the motor, between the spool and front cover. The motor was also placed 4 mm off the bottom to lower its centre of mass and slightly increase room.

The side plates were designed around 18 mm aluminium plate, mirroring the EV3 final drive brackets and allowing the same 16 mm deep bearings to be used. With the motor mounted on the inner surface and reusing the existing spline sprocket adapter from EV.Three, the driven sprocket would need to be spaced 37 mm off the motor side tripod housing to keep chain alignment.

The outer surfaces of the side plates were spaced 292 mm apart, 50 mm more than in EV3. This moved the tripod housings 25 mm out on each side, accounting for the 50mm increase in wheel track allowing for the existing drive shafts from EV.Three to be reused. At this spacing, a second 75 mm spacer was required to fill the gap between the sprocket and the brake side tripod housing.

The brake disc was positioned on the outside of the tripod housing flange. This would require a spacer to move the brake caliper 22.5 mm in from the side plate for correct alignment, however, was deemed the simplest solution instead of spacing the brake rotor off the tripod housing flange. Due to the diameter of the housing, a larger rear rotor would be required. Mounting on the inner

side of the side plate, as opposed to on the outside like EV.Three, was chosen to remove components from the outside of the Powerbox and utilised the empty space inside it.

A top-down view of the components discussed above can be seen below.

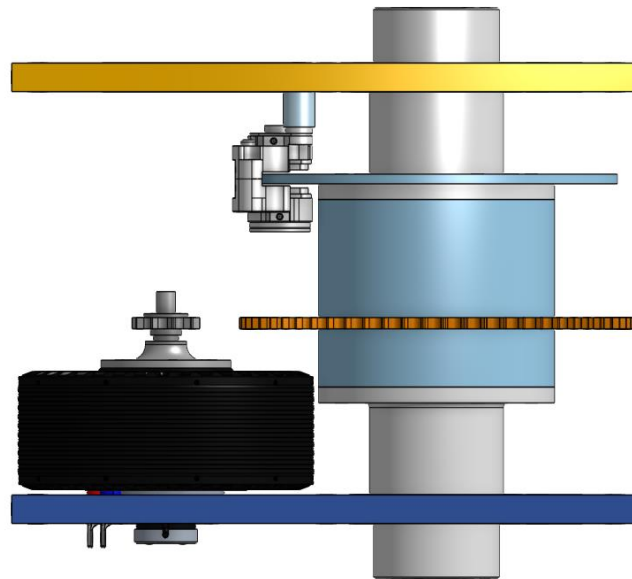


Figure 36 - Powerbox V1 Top-down View of Packaging

Initially, the base was also modelled from 18 mm plate. This sat in between the side plates allowing them to be bolted together with no head protruding from the bottom surface as this would interfere with chassis members. Holes would allow for access to internals when assembled which would be required, particularly for bleeding brakes and changing drive sprockets. A slot was also added into the base for chain clearance. The initial assembly can be seen below. Square cutouts were added for captive nyloc nuts to sit removing the need for a second holding tool during assembly and meeting critical fastening requirements.

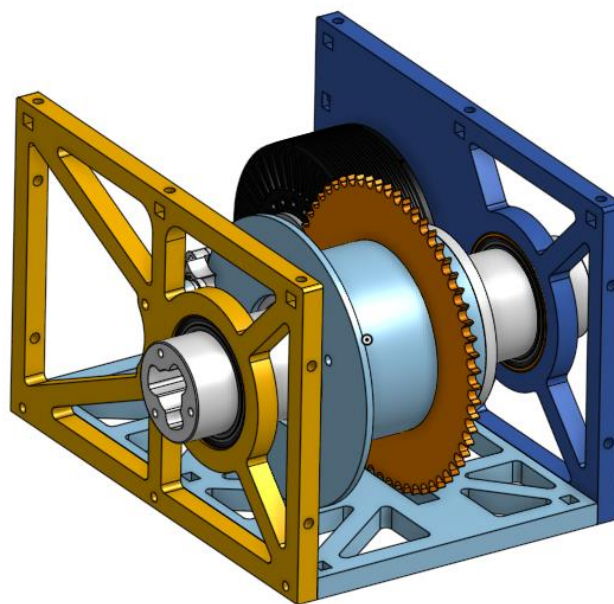


Figure 37 - Initial V1 Assembly

After some consideration of chassis mounting, the base was altered to extend out the sides of the box with tabs aligned with chassis members for bolting to holes through the chassis. Two tabs would be added at the base of the RIS for the rear mounts. The base thickness was also reduced to 12 mm as it would no longer house nuts. Instead, the side plates were to house nuts with counterbored M8 bolts coming up into the plates from the base plate. This did add additional machining to the base plate, however, was deemed the simplest solution and separated the assembly of the drivetrain from its installation in the vehicle as different bolts were used.

The three steel surfaces were combined into one folded piece to reduce part count and increase rigidity. Some material was also removed from this on the brake side to improve access (this could not be done on the motor/chain side as it needed to act as a chain guard and scatter shield. The final design can be seen below.

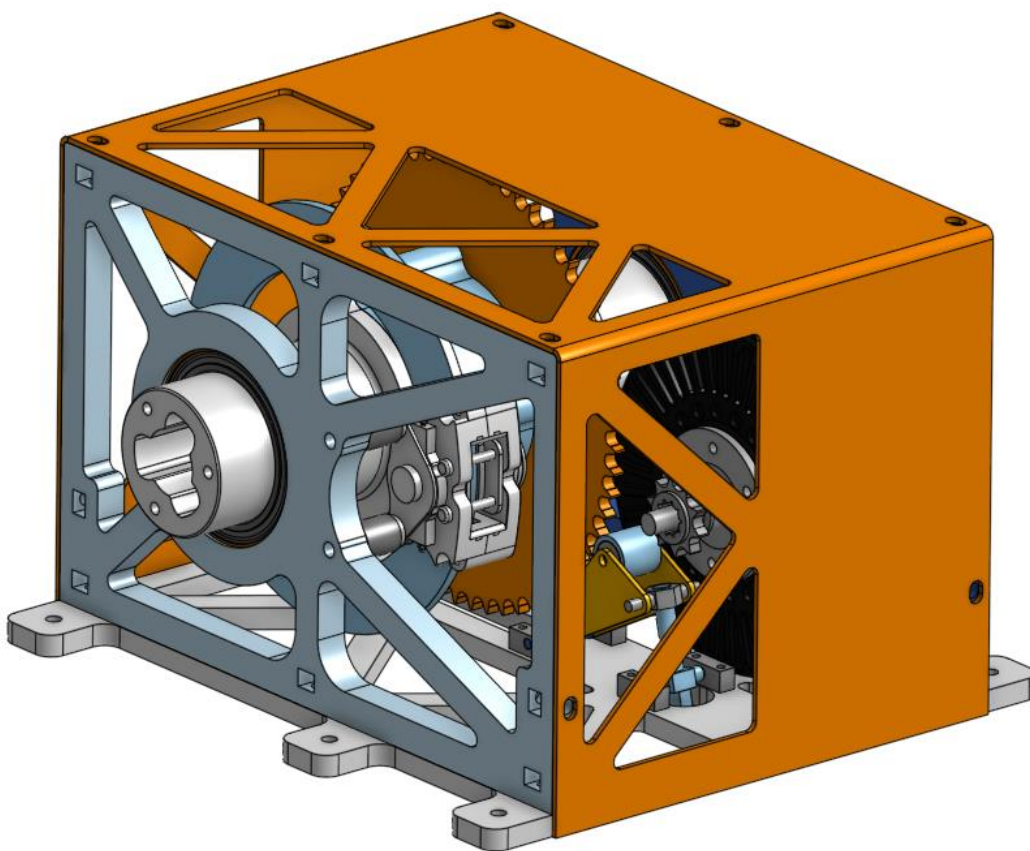


Figure 38 - Final Powerbox V1 Design

The chain tensioner comprised of a roller and two side plates. It was to be hinged from under the driven sprocket producing an arc for the roller to follow between the sprockets. The first revision used a modified bolt which when rotated would move the tensioner up/down. A hole added into the base plate provided access to this bolt head. A half nut would lock the bolt in position when adjusted. This required only two tools to perform a chain adjustment and the loosening of the nut before adjusting the single bolt. This was far easier than the multiple bolts and adjusters which required loosening, adjusting, and tightening on the EV. Three drivetrain.

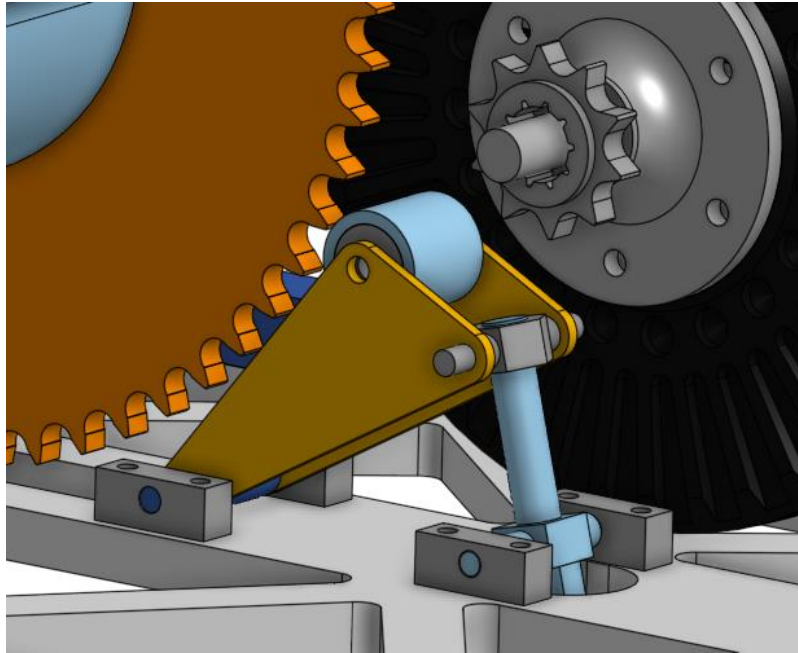


Figure 39 - Chain Tensioner Preliminary Design

There were concerns surrounding the life of the roller as it is somewhat loaded in this arrangement, particularly if a lot of slack needs to be taken up increasing the chain wrap and total load on the roller. Due to the short design life of FSAE components it was deemed acceptable pending testing.

With a complete viable model of the first Powerbox prototype and proof that this concept could fit within the confines of the 2023 chassis design, it was decided to move forward with manufacturing of the first prototype and not progress with a more traditional design. This required manufacturing of the chain tensioner, base and side plates, chain guard, final drive spacers and the brake caliper spacer.

Manufacturing V1

A key improvement in the Powerbox design was a significant reduction in the number of machined parts required for the system. Whilst EV.Three required 12 machined components for mounting the motors and final drive brackets, the Powerbox only required minor machining for the motor plate, base plate, and chain tensioner. The following describes how all components were manufactured for the first prototype.

Base and Side plates

The base and side plates were machined from 19.05 mm and 12.5 mm aluminium plates. These were supplied by Calm Aluminium as sponsorship, with pieces ordered approximately 20 mm larger than required in length and width to account for machining. These were first machined down to thickness before all non-critical 2D features were cut using the on-campus water cutter. For the bearing surfaces on the side plates these were cut at the diameter of the lip as this is not particularly critical, leaving 2 mm to remove for the bearing surface.

Through experimentation with test cuts, it was determined that through using the machines built-in cut-offset functionality no adjustments would be required in the DXF files prior to cutting. This was particularly a concern for the square holes housing the nuts as these required a good fit to not spin. With a nominal M8 nut width of 13 mm, a 13 mm dimensioned square cut provided excellent fitment with slight clearance. With this information, all water cut holes were done at size providing a small amount of clearance. Water cutting was only used for 8 mm holes in the base plate as 4 mm holes were considered too small for water cutting.

With the water-cutting completed, all tight tolerance features were completed in the MEWS workshop with the CNC Milling machine. The base plate had counterbores completed, while the brake and motor plates had bearing surfaces machined for a press fit. The motor plate also had features for the motor cut out and a 6 mm pocket machined out on the outer surface where the motor mounted. This was required to reduce the wall thickness to 12 mm to allow the electrical and coolant connections to pass through.

Chain Guard

The chain guard was very simple to manufacture. Modelled using OnShape's built in sheet metal tool, it was exported as a flat pack for laser cutting and folding at Hancock Speedway. Due to the slight errors associated with one-off part folding, holes were elongated to reduce the chance of them not lining up with the machined holes in the side plates when installed. This part had great fit with no modifications required.

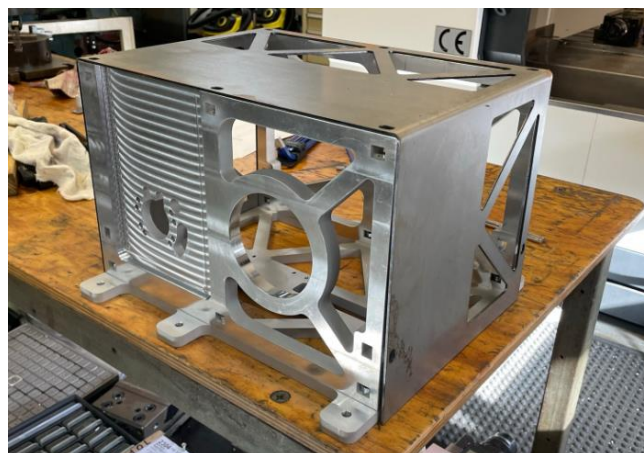


Figure 40 - Completed Side & Base Plates and Chain Guard

Final Drive Spacers

The final drive spacers were originally modelled as simple circular pieces for quick manufacturing on a lathe. These were not weight optimised with the intention was that additional material could be removed later in the year. After discussions with MEWS, the design was altered to reduce weight. This did increase manufacturing complexity somewhat significantly leading to delays in completion. In hindsight, this was not required as these parts did not end up being used in the competition vehicle. Mock spacers were 3d printed to check fitment with the assembly as seen below.



Figure 41 - Powerbox V1 with Motor and Chain Installed

Chain Tensioner

The tensioner was manufactured from two laser cut steel plates and two components turned on a lathe. A bolt was modified with a washer welded on for tensioning. This is detailed further in Rene Sternly's report.

Brake Caliper Spacer

Due to the 2D nature of this part, it was water cut from aluminium plate before being machined down to thickness.

2023 Design V2

The first Powerbox prototype was successful at providing a functional and comp compliant system. Initially, through some level of coincidence, with a 13T drive sprocket and 60T driven sprocket no additional chain tension was required negating the need for a tensioner. Through testing there was also not a significant amount of chain stretch observed, another reason for the addition of a chain tensioner. Whilst not at the ideal drive ratio for acceleration performance, this showed that if the distance between the motor and final drive centrelines was fine-tuned, the tensioner could potentially be removed. This would remove significant complexity and part count, however, would prevent gearing changes.

This Powerbox allowed for testing of 13, 12, 11 and 10 tooth sprockets to determine the best final drive ratio for the acceleration event with the chain tensioner taking up the increased slack created by smaller sprockets. This was successful with no major failures; however, it was noted that as the slack taken up and therefore chain wrap of the tensioner roller increased, wear and noise became quite high.

Weight, of course, was another issue for the prototype with the solid chain guard adding significant weight to the package of 4.7 kg. The final drive was also far heavier than required, formed with multiple parts without clear design intention. As a result, due to the quick turnaround of the first prototype, it was decided that a second weight optimised Powerbox would be commissioned prior to competition. This second iteration would focus on the following improvements:

- Development of a new spool, with less parts and reduced overall diameter allowing for additional motor clearance
- Additional of scatter shield to remove need for chain guard material over motor utilising the room provided by a new spool
- Significant material removal from the chain guard
- Weight reduction in base and side plates.
- Design around 11:60 gearing (as by this time testing had shown this to be the most appropriate gearing).

For the side and base plates, material was removed from most areas. General widths were reduced from 18 mm to 12 mm, while the front upper corners were trimmed on both sides. The motor plate was also changed to have a motor recess machined into the inner face for the motor. This cleaned up the outside and reduced the material required for the scatter shield.

The motor plate was also modified with the spacing between the final drive and motor centres changed to suit 11:60 gearing with no tensioner. This distance was found through 3D printing hubs at different centre-to-centre distances to check chain tension. The distance between the sprockets was adjusted until it was sufficiently tight enough that no chain tensioner would be required. Whilst a 10:60 ratio would have provided the most wheel torque, the plate was sized for an 11T sprocket as performance was very similar. This still provided the option for a 10T sprocket to be installed with the use of the tensioner.

The chain guard had essentially all material outside the required 60 mm zone (3 times chain width of 19.5 plus leeway) removed aside from where folds occurred, and tabs required. The folds added significant strength to the chain guard, and it was intended that these would provide bracing for the lightened structure.

A scatter shield from EV3 was repurposed for the new Powerbox. Whilst potential weight saving could have been achieved by fabricating a new one from aluminium, the reduced workload of reusing the existing steel shield was favoured. This was modified with the removal of one tab and shortening of the length by 6 mm at MEWS.

The additional folding in the chain guard did increase manufacture difficulty slightly however this was still easily outsourced. It should be noted that this part did require slight revision after the first part was made due to a fold line being too close to an edge preventing the fold from being completed correctly. This was rectified and the second attempt fit well.

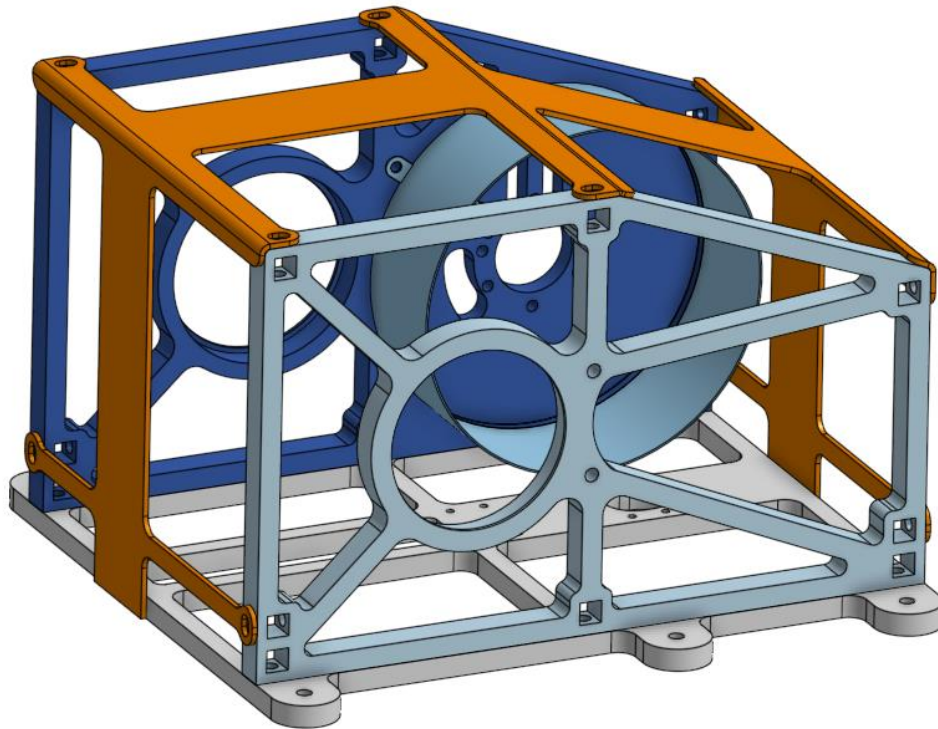


Figure 42 - Powerbox V2 Structure Design

A slight modification was also made to the tensioner roller by adding material along the centre line. This aimed to provide a surface for the chain rollers to run along reducing wear from the side plates which was very evident on the first prototype.

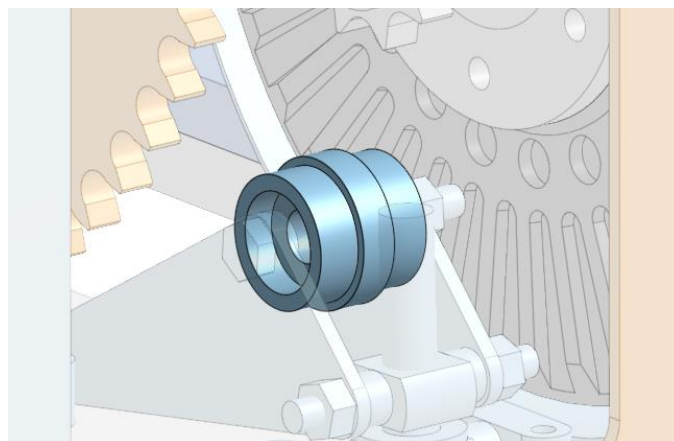


Figure 43 - Modified Roller Design

The spool was redesigned out of two pieces (a single piece was considered however could not be easily hollowed on the inside) with geometry to suit the existing bearings but with sprocket alignment adjusted to suit the 6 mm change in motor position. It also relocated the brake disc removing the need for the caliper spacer. The smaller diameter of the new spool allowed for a smaller EV3 rear brake disc to be reused, further reducing weight.

Component	Weight (kg)		
	Powerbox V1	Powerbox V2	Reduction
Motor Plate	2.9	2.1	0.8
Brake Plate	2.1	1.3	0.8
Base Plate	2.2	1.5	0.7
Chain Guard	4.7	1.9	2.8
Spool	4.5	2.1	2.4
Scatter Shield	-	0.4	-0.4
Brake Rotor	1.1	0.9	0.2
Total Reduction (kg)			7.3

Table 4 - Powerbox V2 Weight Reduction

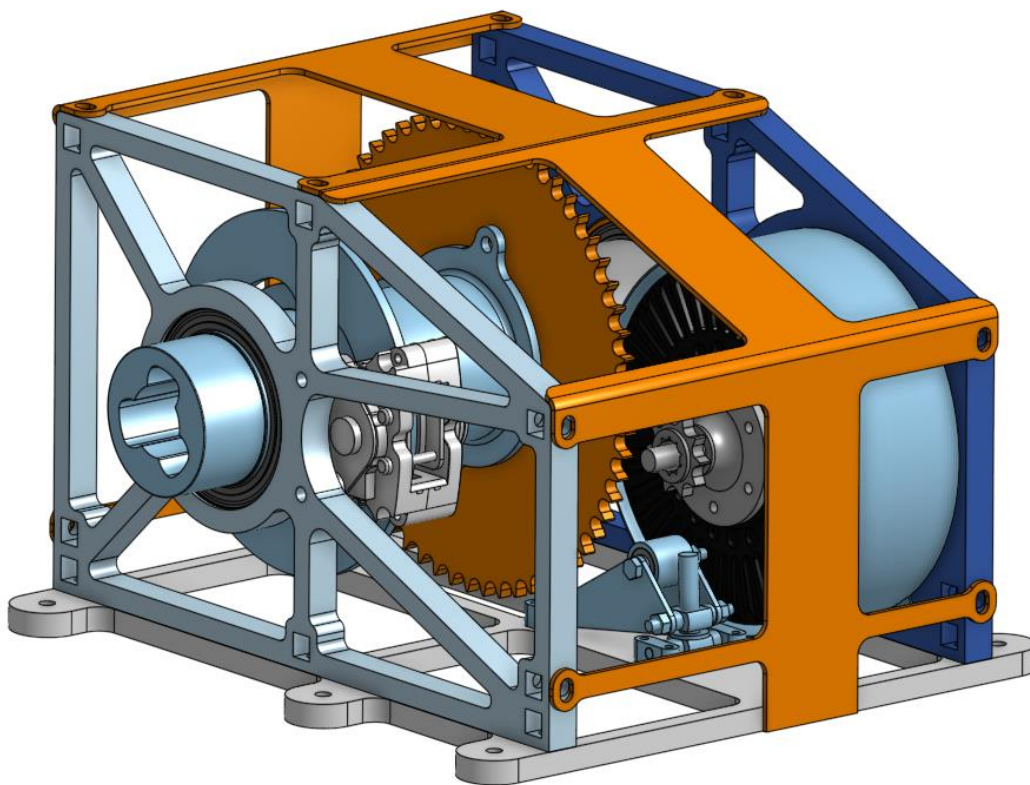


Figure 44 - Completed Powerbox V2 Design

Manufacturing V2

Most manufacturing was completed the same as the original Powerbox however some improvements were made. These are as follows:

- The spigot fit for the motor was machined instead of water cut
- Placing the motor cutout on the inside allowed this to be completed in the same operation as the bearing pocket for the motor plate
- All steel components were powder coated for competition

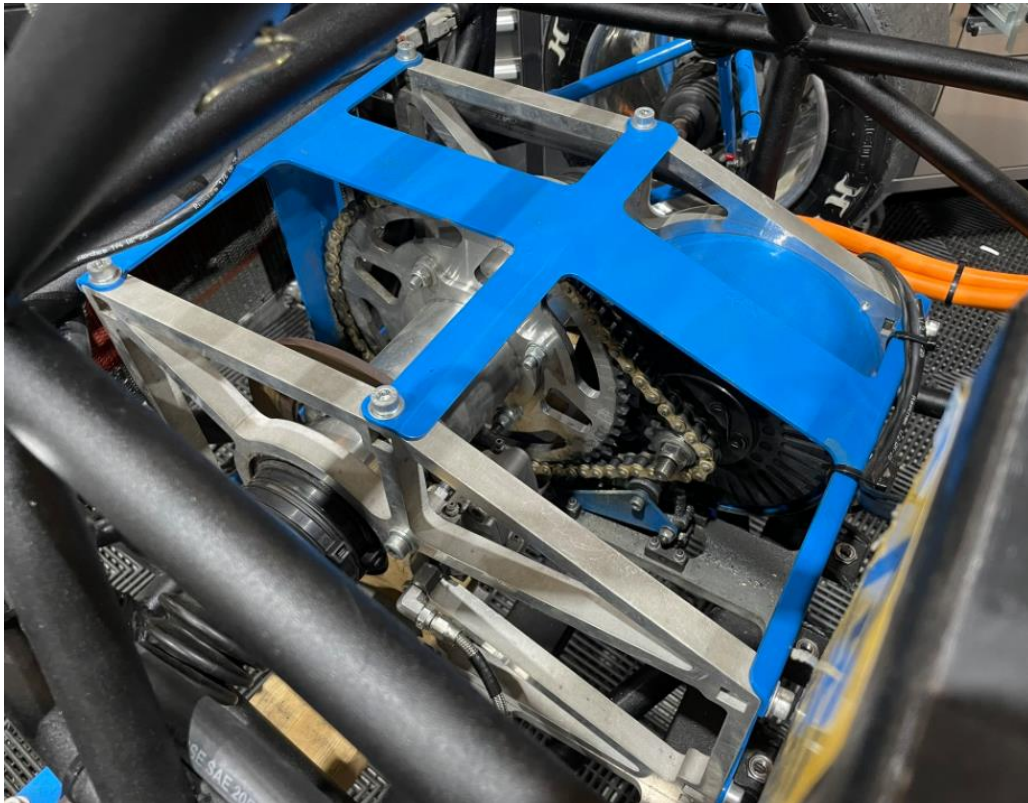


Figure 45 - Powerbox V2 Installed in NU23 Chassis

Issues

Whilst the Powerbox V2 manufacturing and implementation was overall a success, there were some issues particularly around the final drive bearings. Whilst this did not seem to be an issue with the original Powerbox, the slight increase in compliance of the lightened Powerbox saw issues during acceleration loads. The noise was not always consistent however was an issue leading to spool wear which increased the bearing clearance worsening the issue.

When investigated, there were two issues which are believed to have caused this to occur. Firstly, it seemed there was an error in the machining of the motor plate. Water cut and machined features did not align correctly in this plate, unlike the brake plate. This may have been by:

- Incorrect zeroing of part for machining
- Inaccuracies in the water cutting process
- Using a different origin when machining compared to the brake side

Whilst the root cause is unknown, the affect was that the motor plate final drive bearing sat 2-3 mm further forward compared to the brake side bearing leading to shaft alignment issues.

Secondly, mirroring the EV.Three design, the inner bearing race to spool clearance was made a sliding fit, allowing for easy assembly and disassembly. Paired with the bearing misalignment, this allowed the inner race to move easily causing the bearing to walk a concerning amount.

Whilst there were extensive discussions with other students and machinists around the assembly order, tightening sequence and torque of the bolts when assembling the Powerbox, none of these adjustments made fixed the issue.

Instead, the prior to comp the bottom holes in the side plates were elongated allowing the plates to be offset forward and back to counteract the difference in bearing machining locations. This stopped the bearing from walking on the bench. In addition, bearing retaining compound was used between the inner race and spool. This was successful with no noises presenting themselves at competition.

To remove this issue occurring again, it is recommended that an interference fit also be used between the spool and bearing inner races, while machining of the side plates should use the same origin when considered as an assembly (i.e. the front lower corner of each plate when positioning on the car is considered) instead of using say 'bottom left' for the origin in drawings for both pieces. Using origins this way prevents water cutting inaccuracies leading to bearing misalignment (although it is unclear if this actually cause the issue).

Aside from these manufacturing issues, there were also other issues with the packaging of this design. Whilst it worked well in the NU23 chassis, the powertrain did contribute to the vehicles increased wheelbase length. Whilst this couldn't change for the 2023 year, it was believed that improvements to packaging could be made for 2024 by changing the placement of the motor to bring the spool, axles, and overall wheels closer to the accumulator.

2024 Design

For NU24, a Powerbox was designed in preparation for changes in the chassis and implementation of an Emrax 228 motor. This design focused on small changes to improve issues with the second Powerbox while also reducing weight and part count. It would consolidate all learnings from the previous two designs while refining design and manufacturability.

The key objectives of this Powerbox design were as follows:

- Reduced length to shorten wheelbase
- Package 228 motor
- Suit reduction in rear track of 53 mm
- Improve bearing tolerances
- Remove need for chain tensioner at 14:46 gearing

Whilst reusing parts would be ideal to reduce manufacturing loads, unfortunately many components would not be transferable for the 2024 Powerbox. The spool would not suit the change in track without requiring shorter axles, while the scatter shield, side plates, chain guard and base plate were would all require adjustments in geometry. The spline adaptor would also not suit the new motor.

To reduce packaging length the decision was made to move the motor from in front of the spool to on top. Whilst this did move mass up and require a taller RIS, it would allow for the package to decrease in length by a significant amount reducing vehicle wheelbase.

Due to the significantly increased peak torque output of an Emrax 228, less gearing was required despite increasing wheel torque. As a result, the driven sprocket could be reduced in size allowing for surrounding components to be smaller. A 46T sprocket was chosen for the driven sprocket as it was the smallest size available off the shelf. The team also already had one of these. A rear sprocket as small as 42T could be used without the chain fouling on the spool, however this would increase chain loads with no reduction in Powerbox size as the brake disc still requires clearance. Incorporating a scatter shield around the motor resulted in a diameter of 240 mm. For some clearance a length of 248 mm was chosen for the side plates.

With the concept of no chain tensioner proven in the 2023 Powerbox designs, the intention for 2024 was to remove this assembly completely. With the significant increase in power and torque available from the 228 motor, mechanical gearing has a larger margin for error allowing for the gear ratio to be chosen prior to testing (with some calculations required). With the chosen gear ratio of 14:46, the distance between the final drive and motor was calculated using the formula:

$$c = \frac{25.4P}{8} (2L - N - n + \sqrt{(2L - N - n)^2 - 0.81(n - n)^2})$$

Where:

C = Centre -to-centre distance in mm

L = Chain length in pitches (must be even)

P = Pitch of chain in inches

N = Number of teeth in large sprocket

n = Number of teeth in small sprocket

This provided a value of 178.79 mm for 60 links with a 0.5-inch pitch (for 428 chain). This was the smallest distance which could be used whilst leaving approximately 4 mm between motor plate final drive bearing pocket and motor pocket. Should a 13T drive sprocket be used, the separation centre-to-centre distance would become 181.4 mm.

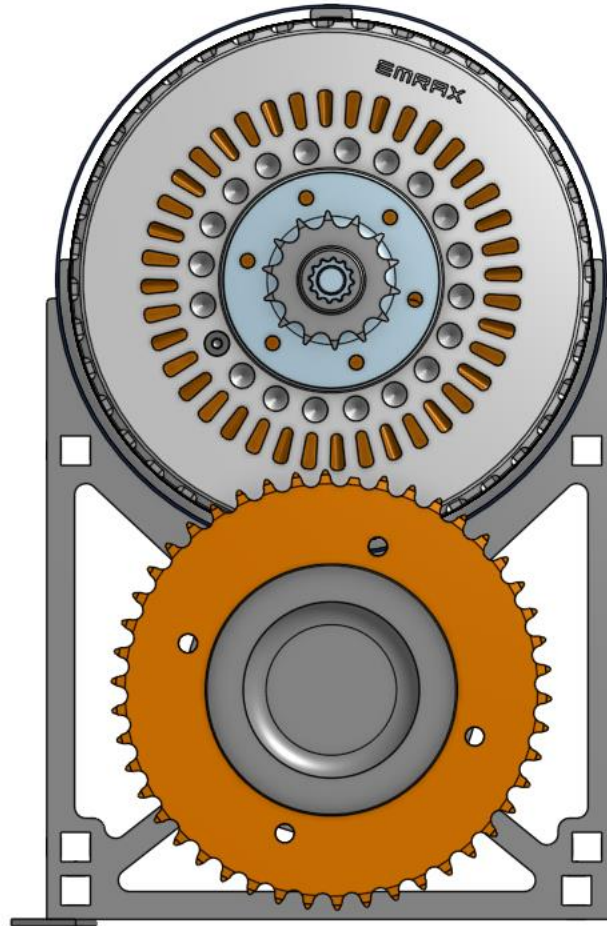


Figure 46 - Vertical Arrangement and Sprocket Arrangement

A formula had never previously been used for the NU23 Powerbox designs. To confirm formula accuracy, the values for both NU23 Powerboxes were entered for the ratio which didn't require a tensioner (13:60 for V1 and 11:60 for V2). The numbers calculated were 0.31 mm higher for V1 and 0.23 mm higher for V2. This seemed consistent with the small amount of chain slack remaining in these designs. Therefore, for the NU24 design 178.9 was used. Being only 0.09 under the calculated length should further remove slack whilst still keeping some margin of error (albeit very small) for machining tolerances.

The use of 428 chain was kept from the NU23 Powerbox despite concerns over additional load from the increase in motor torque output. The chain never broke on NU23 even with a 10T sprocket however at full motor output the new arrangement would see an increase in chain load of 64%. The calculated peak load under full motor torque output is 8070 N which provides a factor of safety of 3.71 to the chain's tensile strength of 30 kN. Whilst not the recommendation of 6 for chain drives, this was deemed acceptable for an FSAE vehicle with a short design life. It is also intended that the motor will never run at full torque output due to traction limitations.

With no chain tensioner this removes the need for structure for it to mount to. Therefore, no base plate equivalent was necessary in the 2024 Powerbox. Instead, the side plates could bolt directly to the chassis. To reduce height, and due to packaging around the motor, the chain guard was altered to only mount to the front and back surfaces of the side plates. This removed susceptibility to errors in folding and so elongated holes were changed to tight tolerance circular holes. This change would help the chain guard become more integral in the alignment of the structure, required as there would not be a base to do this.

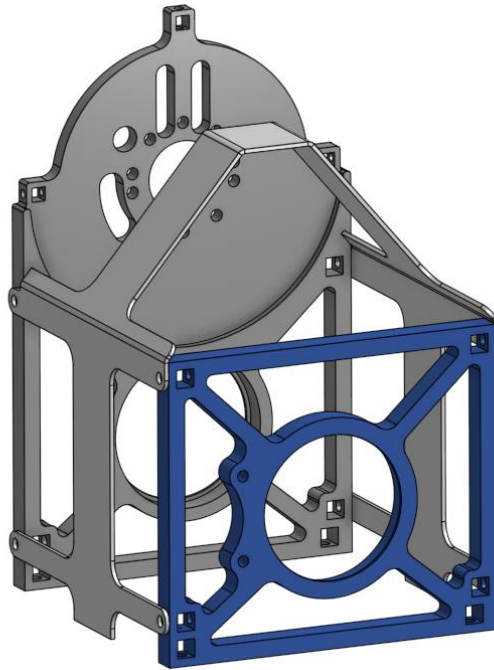


Figure 47 - Side Plates and Chain Guard

A new scatter shield was designed to suit the larger 228 motor diameter with a different style of mounting tab removing the need for these to be welded on. A new spline adaptor and spool were also modelled heavily based on the Powerbox V2 components with adjustments to suit new geometry where required. It should be noted that FEA has not been completed on these components and should be done prior to manufacturing. The spool has been shortened, with bearings moved slightly closer to the ends reducing protrusion from the sides of the Powerbox. A lip has also been added to the outside at each end to assist in retaining CV boots.

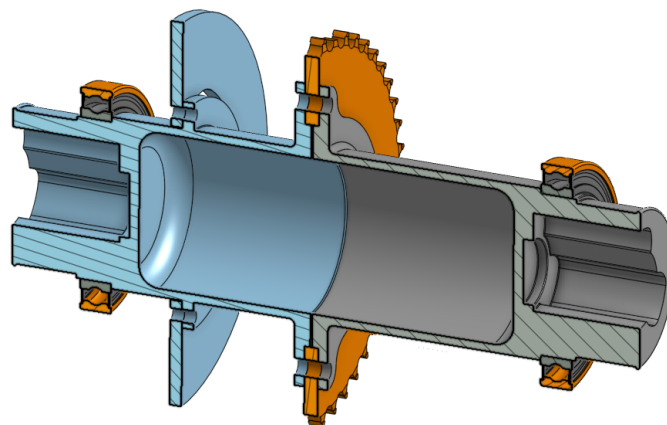


Figure 48 - NU24 Spool Assembly

The brake caliper has been placed for easier access at the rear. With no guard above it, bleed nipples should be far easier to open compared with the NU23 Powerboxes. The below table summarises the complete design in terms of weight reduction at the time of writing.

Component	Weight (kg)		
	Powerbox Light	Powerbox NU24	Reduction
Motor Plate	2.1	1.9	0.2
Brake Plate	1.3	0.9	0.4
Base Plate	1.5	-	1.5
Chain Guard	1.9	1.6	0.3
Spool	2.1	1.9	0.2
Scatter Shield	0.4	0.5	-0.1
Tensioner	0.4	-	0.4
Total Reduction (kg)			2.9

Table 5 - Weight Reduction of NU24 Powerbox

While this does suggest a reduction in weight of 2.9 kg, it should be noted that the change to an Emrax 228 brings a 5.6 kg weight penalty resulting in an overall increase in weight of 2.7 kg. This should be acceptable given the increase in drivetrain output it will provide.

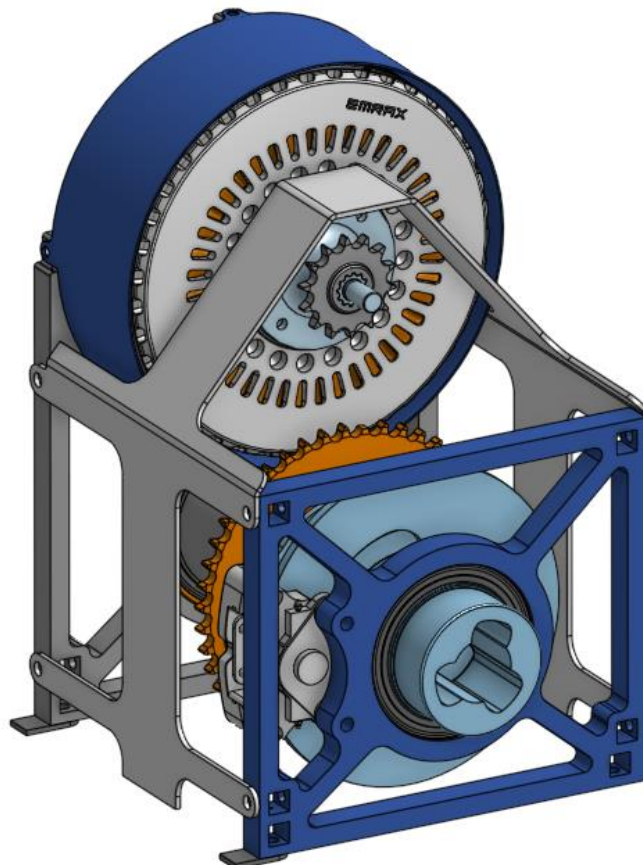


Figure 49 - NU24 Preliminary Powerbox Design

2024 Recommendations

The following provides recommendations for manufacturing the 2024 Powerbox to ensure it is done easily without issues:

- Side plates – The brake and motor plates are to be made from 19.05 mm 6061 aluminium plate supplied from Calm Aluminium (with at least 15 mm additional border for water cutting clearance). These will need to be skimmed down to 18 mm thickness before water cutting basic shape including square holes and brake caliper holes (motor features and bearing pockets should be suppressed to create surface for DXF generation). An additional 2mm of material can also be added to the base edges for machining a proper square surface if desired (water cutting does not cut perfectly square). The motor plate will then require machining of the 6mm motor pocket and all motor features, while both plates will also require the external holes drilled.
- Spool – The spool will require machining from a 6061 rod supplied by Calm Aluminium (again ensure additional length supplied for machining ~ 50 mm). Geometry in OnShape as of 30/01/2024 should be correct however it is recommended this is checked over.
- Chain guard and scatter shield will require laser cutting at Hancock speedway from 1 mm aluminium/steel and 3 mm steel respectively. The chain guard will require folding at Hancock due to its thickness (MEWS cannot bend 3mm steel plate) while the scatter shield is most likely best done in MEWS due to its round shape. Both have been modelled using the sheet metal tool so flat views are available for immediate DXF export.
- Sprocket spline adaptor – this will require machining from steel (grade of material should be checked against existing 188 adaptor). The spline will need to be done at Newcastle gears (NOTE: Due to the manual process of making these splines supplying a sprocket to check fit will be far better than trying to dimension this detail in a drawing). This part has been designed to reuse the existing sprocket spacer however a new one has been modelled to remove the loose fitment from this piece if desired.
- Brake disc – this has been designed with reusing the existing brake rotor from NU23 in mind. These were made from 6 mm steel plate which was surface ground to remove mill scale.

Two tabs have been modelled to be welded to the RIS for rear mounting to the chassis similar to NU23. It may be beneficial to change these from plate to 25 mm SHS for increased strength as these make up half of the mounts in NU24 unlike NU23 where a second RHS member was utilised for 2 additional mounts.

The motor plate has bolt holes reflecting those used for the 188 motor with dimensions altered to suit the 228 model. It should be noted that while bolt hole locations are detailed in these models not all locations are typically threaded (as is the case with the 188 motor). Prior to these holes being machined the physical motor should be checked to ensure these are located correctly. The locations of the temp sensor and resolver wiring should also be checked (note though that the resolver can be rotated if the position is recalibrated in the controller).

The spool geometry has been modified to include a 0.02 interference fit on the inner races of the final drive bearings. This should be calculated correctly for a press fit to double check the interference is appropriate. It should be ensured a press-fit is achieved to prevent bearing walking issues from occurring again.

A plate has also been modelled to use as an assembly tool to ensure the side plates are assembled squarely. To use this the side plates will require additional material on the bottom surfaces when water cut to allow a perfectly square edge to be machined. This can also be used as a jig to weld the rear Powerbox mounts to the chassis. The steps for assembly should be:

1. Assembly spool (tripod housings, driven sprocket, and brake disc)
2. Assemble side plates (press final drive bearings into plates before pressing plates onto spool)
3. Install motor on motor plate (with spline adaptor and drive sprocket) and scatter shield.
4. Install chain guard (use assembly tool here if made and tighten side plates to this before tightening chain guard bolts).
5. Install chain and brake caliper.

If the bottom surfaces are not machined the assembly tool should not be used. Instead, the chainguard alignment will be provided by the spool and side plates already being pressed together into an assembly.

The most critical note for all drivetrain components is that these designs have not yet been validated through FEA. Whilst a conservative approach has been taken, these new parts (particularly the motor plate) should be analysed to ensure loads will not exceed the capabilities of 6061 aluminium.

If FEA analysis suggests the motor plate may require additional reinforcement, the chain guard could potentially be extended vertically with a tab across to the upper scatter shield bolt to support this area and prevent chain tension bending the motor plate.

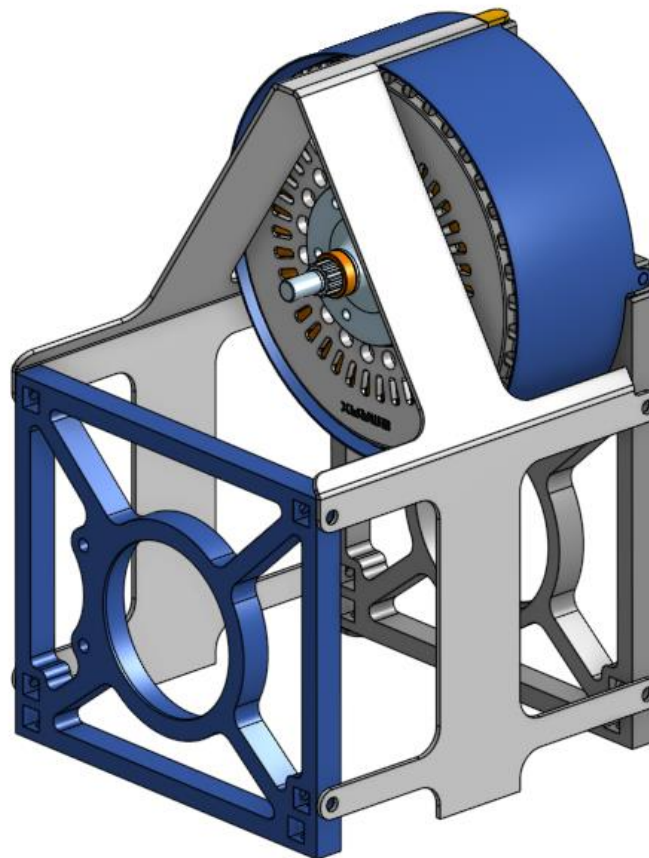


Figure 50 - Additional Motor Plate Bracing Concept if Required

Braking / Pedal Box

The braking system of a race car is as fundamental for performance as the powertrain and suspension – for the best lap times and overall pace a vehicle must be able to slow down for corners as quickly as possible to enable the driver to hold higher speeds longer and deeper into a corner. Whilst improvements in this system will not directly gain points in the skid pad or acceleration events, it is vital for autocross, endurance, and safe practise throughout the year at test days.

The fundamental purpose of the braking system is to take pressure applied by the driver's foot at the brake pedal and use this to slow the car down. For race vehicles this is achieved mechanically through hydraulic circuits which apply pressure to brake pads which act directly onto disc rotors. There are cable and drum type brake systems on some cars however these are not suited to high performance applications. An effective braking system allows the driver to easily place as much braking force as possible on all wheels without exceeding the grip they have at any one time providing as much braking overall as possible without locking up and losing control of the vehicle.

Components

A typical hydraulic braking circuit has 3 main components. These are:

- Master cylinder(s) (with Brake pedal)
- Slave cylinder(s) (with Brake caliper)
- Disc Rotor

Master cylinders



Figure 51 - Tilton 78 Series Master Cylinder as Used by NU Racing

The master cylinder takes an applied force and converts this into hydraulic pressure. The resulting pressure can be calculated by the simple equation:

$$P = F/A$$

Where P is the fluid pressure, F is the linear force applied to the master cylinder, and A is the master cylinder piston area.

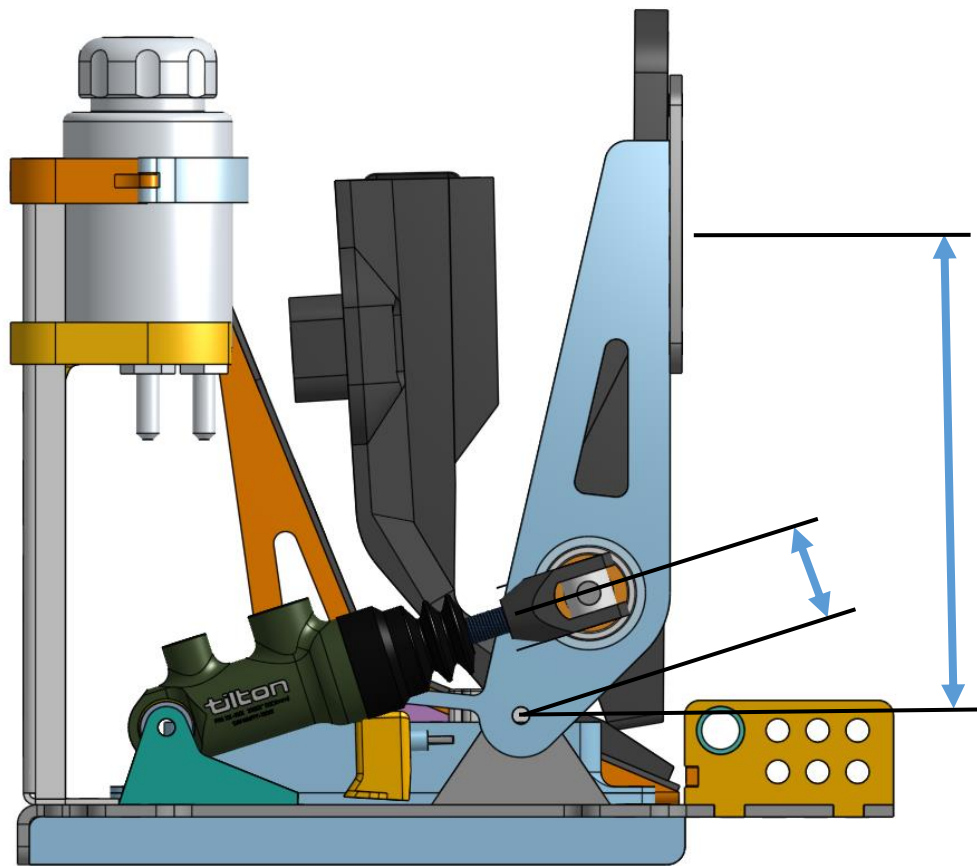


Figure 52 - True Moment Arms of Brake Pedal & Master Cylinders

The force F is determined by the pedal ratio of the brake pedal. This corresponds to the mechanical advantage of the brake pedal which can be calculated by the ratio of the lengths of the true moment arms for the force applied by the driver's foot and the linear force acting through the master cylinder. The above figure demonstrates where these measurements are taken.

It should be noted that both master cylinder surface area (generally specified instead by bore diameter) and the pedal ratio work together to provide the relationship between the applied force on the pedal and the fluid pressure. A master cylinder of larger diameter will require a higher pedal ratio to maintain the same applied force to brake pressure relationship. These relationships can be seen in the below table.

Property	Effect on brake pressure (for constant pedal force)	
	Increase	Decrease
Pedal ratio	Increase pressure	Decrease pressure
Master cylinder diameter (area)	Decrease pressure	Increase pressure

Table 6 - Effects of Pedal Ratio and Master Cylinder Diameter on Brake Pressure

It is also important to remember that in a dual circuit system 2 master cylinders are used and so the areas must be added, or force divided between both cylinders. When using a balance bar (explained below) the force acting on each master cylinder must be calculated first before assessing each circuit independently.

Slave cylinders

Slave cylinders take the hydraulic pressure provided by the master cylinders and convert this back into a force to apply to brake pads to push against the rotor. These are housed within a brake caliper and can vary in the number of cylinders (generally referred to as 'pots') and cylinder diameter.

Like master cylinders, these still follow the $P = F/A$ relationship (more usefully rearranged to $F = PA$). As they are reversed converting pressure into force, changes to the total area (calculated by piston area * no of pistons) has the opposite effect on braking performance as shown in the table below:

Property	Effect on pad force (for constant brake pressure)	
	Increase	Decrease
Slave cylinder diameter	Increase	Decrease
Number of cylinders	Increase	Decrease

Table 7 - Effects of Slave Cylinder Diameter and Number on Pad Force

Disc rotors

Disc rotors provide a moving surface for brake pads to apply a friction force to. As they are a disc and the brake pads provides a force to the outside of this disc, a moment is made opposing the rotation of the disc. This moment is transferred to the wheels as they are mechanically connected to the rotors. Combined with wheel/tyre size, disc diameter determines the relationship between pad force and the braking force being applied to the vehicle. Larger discs provide a higher braking force and improved heat dissipation however come with the downside of increased weight. They are also physically limited in size when positioned inside the wheel to ensure adequate clearance for the caliper. The relationships to braking force can be seen below:

Property	Effect on braking force (for constant pad force)	
	Increase	Decrease
Disc diameter	Increase	Increase
Wheel Diameter	Decrease	Decrease

Table 8 - Effects of Caliper and Wheel Diameters on Braking

Other components for a brake system include are:

- Balance bar
- Brake fluid
- Brake pads
- Proportioning valve

Balance Bar



Figure 53 - Balance Bar (Tilton Brake Balance Bar Assembly, n.d.)

A balance bar provides a mechanical mechanism in the brake pedal to adjust the ratio of forces applied to the master cylinders. This is achieved by attaching the master cylinders to a bar which has a spherical connection to the pedal. When adjusting the balance bar, the spherical point moves closer to one master cylinder increasing the force on this cylinder. This is due to the sum of moments about the spherical always being zero and changing the lengths of the true moment arms of each master cylinder.

Another helpful aspect of the balance bar is that it provides freedom for discrepancies in master cylinder positions providing a constant force as opposed to a constant displacement when the cylinders are locked together. This is important as it removes the criticality from master cylinder positioning and lengths.

Brake fluid

Brake fluid is the essentially incompressible fluid transferring hydraulic force through the system. It is specifically formulated to have a high boiling point. This is essential as if the fluid in a hydraulic system does boil and become gaseous, this gas be compressed and prevents pressure from building when the master cylinders are displaced, removing the ability for the system to apply force to slave cylinders.

Brake fluid is classed by its boiling temperature with common fluids being dot 3, 4, and 5.1 (dot 5 is a silicon-based fluid not typically compatible with most brake systems). Higher dot numbers refer to higher boiling points however this generally comes with the downside of increased hygroscopic properties. Generally, for race vehicles which see high brake temperatures and regular fluid changes (removing concerns for water absorption over time) a higher dot rating is preferred provided all components specifically are compatible.

Brake pads

Brake pads provide a wearable high friction surface to apply force on the calipers. Available in many different formulas these are generally optimised for performance, life, feel or noise. They are also usually specific for different rotor materials. Generally, for FSAE applications options will be limited and based on the caliper choice. For instance, the calipers used on NU23 only have one pad type and compound available.

Proportioning valve



Figure 54 - Brake Proportioning Valve (AeroFlow Brake Proportioning Valve, n.d.)

A proportion valve is another means of providing adjustable front-rear bias. Instead of mechanically adjusting the force on each master cylinder and therefore resulting fluid pressure of each system, proportioning valve change the response of a circuit to master cylinder displacement by working as a regulator at higher fluid pressures. These work well when the master cylinders are mechanically connected and constrained to have the same displacement. These are generally placed on the rear circuit to reduce pressure compared to the front.

Rules

For FSAE the braking system must follow many rules, the main ones of concern for system design being:

1. T.3.1.2 The braking system must:
 - a. Act on all four wheels
 - b. Be operated by a single control
 - c. Be capable of locking all four wheels
2. T.3.1.3 The braking system must have two independent hydraulic circuits. A leak or failure at any point in the system will maintain effective braking power on minimum two wheels.

From these the braking system must be comprised of two separate hydraulic circuits which act on all four wheels. For easy tuning of front-rear bias this is achieved by having independent front and rear circuits.

2022 System review

System

With independent drive to each rear wheel, the 2022 braking system comprised of two rear inboard rotors and 2 outboard front rotors. These were made from 8 mm laser cut steel plate ground down to approximately 6 mm. Two Tilton 78 series master cylinders provided pressure to four Willwood Dual piston calipers.

A proportioning valve was incorporated into the rear circuit for break bias adjustment accessible to the driver in the cockpit.

Pedal Box

The pedal box was a simple but effective design for 2022. Utilising a single laser cut baseplate and three machined clevises for the brake pedal and master cylinders, the design was very sound with inherent strength. The Accelerator pedal was used from a Volkswagen golf supported by a bracket made of laser cut steel sheet. The baseplate also housed the Pedal Box Electronics Node and brake pressure sensors.

The brake pedal had a tube welded in as shown below for master cylinder mounting. This locked the cylinders together restraining them to equal displacement when braking to suit proportioning valve based bias adjustment.

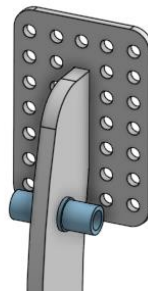


Figure 55 - EV.Three Brake Pedal Master Cylinder Mounting

Issues

Whilst an effective braking system, there were several minor issues which affected brake performance. These generally stemmed around the mechanical design of the brake lever and placement of the master cylinders.

The main issues presented by drivers and those working on the car were:

- Poor pedal feel – there was insufficient pedal travel to allow for small adjustments to brake pressure coming into corners
- Difficulty bleeding master cylinders due to inclined position preventing air from escaping
- Unable to remove pedal box without opening brake lines
- Proportioning valve appeared to make no discernible difference to braking feel
- Susceptible to incorrect adjustment of the threaded master cylinder ends – if these were not exactly the same length both cylinders would not be able to fully release when pedal pressure was removed.

Potential solutions

In order to rectify these issues, solutions which were to be explored for the 2023 system included reposition the master cylinders to be more horizontal, improving the pedal ratio and incorporating a balance bar to improve pedal feel and bias adjustability.

2023 Design

System

As there were no significant functional issues with the majority of the braking system, and rotors, calipers and master cylinders were all relatively new and in good condition, all of these components were utilised again in the 2023 system. The majority of brake lines were also reused aside from the rear line which required replacement.

A new master cylinder arrangement was to be developed to house a balance bar in the brake pedal. This would provide true mechanical brake bias adjustment and was to replace the prop valve implemented in EV.Three.

Changing to a spool rear end negated the need for second rear caliper and rotor simplifying the system at the rear. It should be noted that by doing this rear brake force was halved however this was offsets somewhat by the larger rear rotor in the Powerbox V1. Reduced braking force in the rear is also generally the preferred bias as weight transfers to the front wheels providing more grip here under heavy braking. The Powerbox V2 saw an original EV.Three rear rotor reintroduced for weight savings and so the intention was that balance bar tuning could be used to increase rear bias if required.

Pedal Box

Pedal Box design was initially completed by Brady McNeil. Whilst conceptually sound, modelling was not to a manufacturable standard and so it was redone in order for manufacturing to commence. The base plate was changed from 6 to 4 mm aluminium sheet with ribs incorporated into the underside for stiffening.

As master cylinder and calliper specifications were not changing, a new brake pedal was also design in multiple versions to adjust the pedal ratio in order to find the best brake feel. This incorporated a tube for the balance bar assembly to be installed into. The accelerator and bracket were reused while pressure sensors also carried over.

A new aluminium reservoir was also incorporated to replace the plastic one used in EV.Three as this was prone to leaks and had been damaged. An existing mater cylinder clevis was used for the pedal pivot with a shoulder bolt, while two new clevises were designed to be cut from RHS greatly simplifying manufacturing compared with a machined aluminium piece.

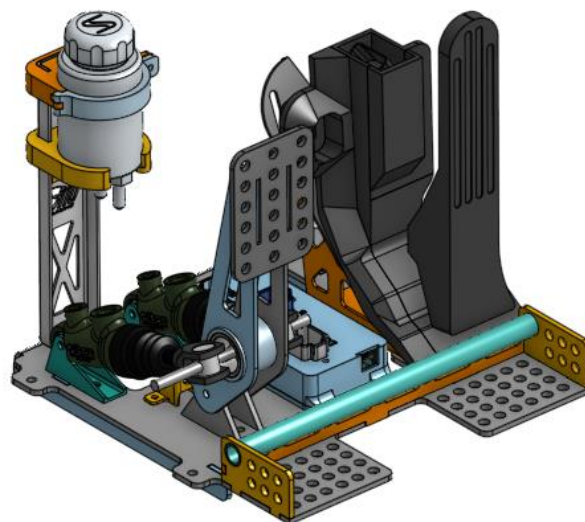


Figure 56 - NU23 Pedal Box Design

Manufacturing

The base plate was laser cut and folded at Mits Alloy before being tig welded at MEWS. The pedal was also welded at MEWS from 3 mm and 4 mm aluminium sheet which was water cut in EC. The tube was also welded in and machined to final size after being turned to rough size on a lathe. The clevises and reservoir bracket were also made in MEWS, cutout from RHS and folded/welded from a laser cut net respectively. These were also powder coated prior to competition.

New Brake lines were made up with Aeroflow quick release fittings allowing for easy removal and installation of the pedal box with no need to bleed the system after.

A knurled adjuster handle and lock nut were also added to the balance bar to lock the adjuster to avoid movement resulting in unintended bias changes. Whilst the system cannot physically come apart in a way which could affect braking safety, A nyloc nut was used in case required in the technical inspection.

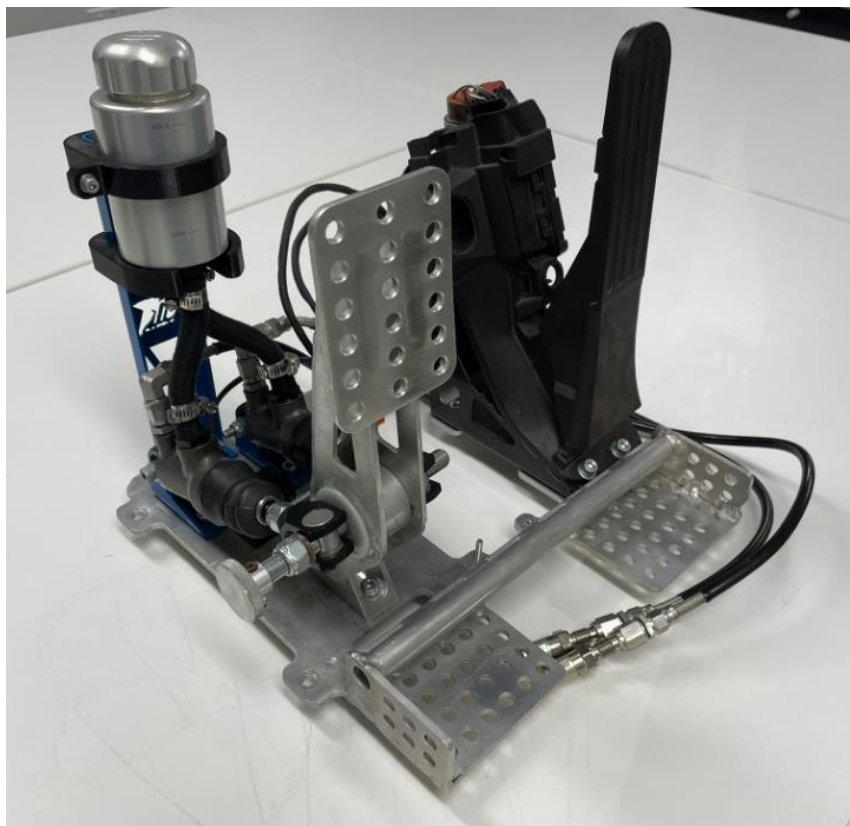


Figure 57 - Complete NU23 Pedal Box

2024 Recommendations

Retaining parts from the existing pedal box and designing components around materials available (such as RHS) greatly reduced the manufacturing load for making a new pedal box. This allowed all parts to be completed quickly resulting in a complete upgraded pedal box with improved brake feel and adjustment ready for competition. Unfortunately, due to other events in the months leading up to the competition significantly reducing testing time preventing bias tuning from occurring.

For 2024 it is recommended that a similar approach is made for the pedal box. An overall simplistic designed should be retained with components reused where possible. In particular, it is believed the pedal assemble should be retained at least initially with a new base plate to suit the chassis changes. With additional testing brake bias can be tuned further improving braking performance.

Manufacturing & Coordination

Overview

As Mechanical Lead in semester 1, significant time was spent liaising with team members producing designs and MEWS and other third parties to have components made. A strong focus early in the year was ensuring that designs were 'made for manufacture' with designs reflecting the geometry of common profiles or with ability to be formed from laser components.

Laser cutting and folding was a very common forming method used extensively throughout the car. Simple or thick folding was completed and Hancock and Mits Alloy while more complex but thin folds were completed in MEWS. The majority of laser cut parts were ordered through Hancock as outlined in Justin Li's Report.

Parts which utilised sheet metal included:

- Accumulator container
- Pedal box base plate
- Suspension pickups
- Firewalls
- Radiator supports
- Cockpit floor
- Closeout panels
- Headrest and support
- Chain guard
- Chain tensioner
- Steering brackets

Parts utilising common material profiles included:

- Accumulator chassis mounts
- Accumulator lid mounts
- Master cylinder clevises

Extensive communication with MEWS was essential throughout the year to ensure components were produced on time in accordance with key vehicle milestones. Fortunately, there were minimal issues requiring fixes with reduced delays throughout the year.

Spreadsheets were typically used as a tool to keep track of part progression through from design into manufacturing.

Changes to design cycle

Whilst difficult to implement due to the minimal design work carried out in 2022 for the 2023 vehicle, a strong focus of the team in 2023 was to shift the design cycle back 6 months. While traditionally NU Racing would design, build, and compete with the same car in one year, this would see the team build a car designed by the last year's team in the first half of the year before designing next year's car alongside testing in the second half of the year.

This change is detail in Marisa McLean's report and has many advantages including:

- Design work done by members who have experienced manufacturing of components and therefore have a better understanding of requirements
- Brings forward the build stage to earlier in the year allowing for far more testing in the second half of the year

Whilst a complete car design was not achieved in 2023 for the 2024 competition year, significant subsystems including the chassis, accumulator, drivetrain, and steering were all at a manufacturable stage prior to passing on to the new team. This is a far more sustainable approach for the team moving forward into future years and it should be a focus of the team to maintain this rhythm.

Alongside this change of timelines, it is important to also bring back design and manufacturing deadlines to allow for extended testing. Ideally no changes to vehicle design should be occurring in the second half of a year leading up to a competition unless well understood and low in risk. This is particularly important for low-voltage systems.

2024 Recommendations

With simplified design of subsystems NU23 parts were generally easy to manufacture with limited complex geometry. It is intended that through a continued design ethos of simplified and manufacturable designs this should continue into 2024 and beyond. Initial NU24 designs have kept this in mind, and many parts on the vehicle should be able to utilise the same manufacturing methods seen in the development of NU23.

The 2024 team should at least be able to match the schedule of the 2023 team and have a driving car by July. With learnings from NU23 adopted into the NU24 designs, these ideally will not require major revision during the testing phase allowing for less downtime and increased track time in the second half of the year.

Manufacturing of new suspension required for the 2024 chassis will result in additional manufacturing loads however if this begins early in the year it should not cause any significant delays.

2023 Competition Performance

Introduction

Despite an unfortunate end to the competition weekend, overall performance for NU Racing and NU23 at the 2023 Formula-SAE Australasian comp exceeded the majority of expectations and reflected the significant amount of work put into improved vehicle performance throughout the year. A focus was put on simplifying the weekend and minimising vehicle setup changes to reduce stress and the potential for team conflict during this intense period. The following outlines the tasks involved with the competition weekend from a vehicle performance perspective.

Competition Car Schedule

Just prior to the car being packed for driving to the competition it had one last drive to scrub the new tyres in. This left in a complete state for comp, fully compliant with no known changes required for technical inspections. The following lists the tasks required to be done on the vehicle throughout the week. Note that for the 2023 competition the acceleration and skid pad events were reversed for this year with Skidpad the first event on the Saturday.

Arriving at comp:

- Acc 50% SOC
- Car fully assembled
- Car on wet tyres
- 4 x new scrubbed slick tyres on rims + 2 x used slick tyres on rims

After receiving energy meter (prior to tech inspections):

- Acc loom De-pinned at CEN end on Negative side
- Current sensor installed – Ensure correct orientation
- Harness bootied following sensor installation
- Voltage sensor pinned
- Energy meter mounted

Before Accumulator inspection:

- Accumulator removed from car
- ACCUMULATOR INSPECTION

After accumulator inspection:

- Accumulator installed in car
- HV switched on to check functionality
- TECH INSPECTIONS: EV STATIC -> MECH -> EV FUNCTIONAL -> RAIN TEST

Between Rain and Brake tests:

- New slicks on rear (or used if 4 available)
- Used slicks on front
- Higher tyre pressures (25 psi)
- 25km/h speed for brake test conveyed to driver
- BRAKE TEST

After brake test (All tech inspections finished):

- Remove accumulator
- Charge to 95% SOC (445V) (Doesn't have to be fully charged but will reduce charging time on Saturday between events)
- Swap over to new fronts

- Accumulator installed and HV switched on to check functionality

Before Skidpad:

- Tyre pressures set
- Clean tyres when staging for runs
- SKIDPAD EVENT

After Skidpad:

- Remove Accumulator
- Charge to 448V Important this is fully charged for Accel!
- Install Accumulator and check HV functionality

Before Accel:

- Tyre pressures set
- Clean tyres when staging for runs
- ACCEL EVENT

After Acceleration:

- Remove Accumulator
- Charge to 448V Important this is fully charged for Autocross!
- Install Accumulator and check HV functionality

Sunday before Autocross:

- Tyre Pressures set
- AUTOCROSS EVENT

After Autocross:

- Charge to 448V (Important this is fully charged for Endurance)
- Cool charger as much as possible whilst charging

Before Endurance:

- Tyre pressures set
- ENDURANCE EVENT

After Endurance:

- Parce Ferme (requires snips and ASHD de-pin tool)
 - Cut boot from Acc Loom
 - De-pin ASHD and remove current sensor
 - Reboot for use next year
- Charge to 50% SOC (Important for battery storage)

Notes for all events:

- Wets always on hand ready to go
- No suspension changes made to car unless well tested with planned methodology for changes prior to comp
- BMS should only have settings adjusted for endurance based on profile already developed
- Motor controller settings should only be changed if necessary for improved performance and tested/planned prior to comp

Technical Inspection Overview

Technical inspections for 2023 were well practised and executed in a timely matter with minimal issues. Many mock technical inspections were completed in the months leading to competition. This was reflected in NU Racing being the first team in the competition to complete all inspections. For most inspections only 2-3 people are required in the inspection bay with one of these people responsible for providing samples or datasheets as required. The following provides an overview of any issues which came up in these inspections.

Accumulator

The accumulator inspection was the first and most concerning inspection for the team. Having never been formerly completed at an FSAE event, efforts were made to tidy the wiring of the top plate as much as possible to reduce the chance of issues being flagged in the inspection. Whilst the majority of this inspection went well, there were several major issues brought up in this inspection:

- 225 A Main fuse current rating exceeded the nominal current rating of the 16 mm² wiring used for the tractive system
- The main accumulator ASHD HV connector did not have a datasheet displaying its voltage and current rating
- Some fasteners were not positively locked
- Some wiring had the potential for chaffing on some sharp edges of the top plate

The wiring and fasteners were easily fixed at competition, and whilst the exact data sheet could not be found for the ASHD connector a family connector data sheet was accepted.

The fuse required more substantial work. Luckily a lower rated value fuse was sourced however it was smaller in size. This required slight modifications to the fuse ends to allow it to fit the existing studs on the accumulator top plate.

There was concern as a 'fast-blow' fuse was the only one available. It was rated to 125 A and peak accumulator currents of up to 180 A had been observed. Whilst this current exceeded the fuses named rating, it was not an issue. Like the 130 A rated 16 mm² cable which it was to protect, brief spikes of current would not cause damage to the fuse. Even at 200 A, this fuse would not blow for over one minute according to its datasheet. This is an important concept for future fuse selection – the current rating must not exceed the cables continuous rating and does not need to be above the peak current expected to prevent blowing. This fuse was installed and worked without issue across all events throughout the competition weekend.

EV Static

EV Static was mostly passed first go with only one minor issue. This required two ground wires to be added to the HVD bolts to ground them and was checked after the mechanical inspection. There was slight concern over the booting of the DCDC connector however this was not failed in the inspection. A 12V Post LVMS test point would have been beneficial for this inspection.

Mech

The mechanical inspection passed with no issues at all. This was the result of many mock inspections and an overall neat build quality. It is important to consider that a neat and well-built vehicle provides a great first impression to scrutineers and is likely to make scrutineering more relaxed provided all inspection points appear to have been addressed.

EV Functional

EV functional also passed first time. Again, this was the result of many mock inspections providing a well-practised team. A note to consider here is that the functional display of the trail functionality is difficult to achieve when done by hand as significant pedal pressure is required before it is triggered. Holding the throttle at the threshold for this to trigger will also spin the wheels incredibly fast.

The method used for the trail test was as follows:

- Apply a very small amount of throttle to make the wheels spin at a relatively slow rate
- Begin applying a significant amount of pedal pressure (ensure this is enough for the brake light to be activated)
- Whilst in the process of applying pedal pressure increase throttle in order to pass the threshold for trail
- Once both thresholds are passed at the same time the motor output will stop (there will be a small period where the motor is loaded up. At this point the inspector is shown that the throttle is not responding and must be reset by allowing the throttle back to its zero position)

A reduced speed limit could be considered for this inspection in 2024 to remove the need of minimising throttle for this test. Alternative the torque request from the PEN could be scaled down.

Rain

The rain test passed without issue. The body kit and upper firewall provided excellent protection to internal components during this test which is believed to have helped significantly. All electronic nodes were also sealed with silicon by Nick Lyall and Brock Symons prior to the competition.

Brake

The brake test also passed without issue. This was practised at the last track days prior to competition and a method developed as follows:

- Driver to accelerate to 25-30 km/h
- Dash E-stop to be pressed once this occurs
- Driver to coast until at braking zone
- Brake pedal pressed as hard as possible and held until vehicle comes to a stop

It is important to consider that there is no minimum speed for this test and so was done at the lowest speed possible. This not only reduced tyre wear and flat spots, but also reduced the chance of the car rotating whilst locked up which can result in a failure. Used front tyres were installed for this test to prevent wear to the new front tyres for competition. Tyre pressure was also checked for this inspection and so it is not recommended that tyre pressures are increased significantly to aid in locking up as this will raise suspicion.

With all technical inspections completed the car was ready for dynamic events to occur on the weekend.

Event performance and Results

Acceleration

In the acceleration NU23 achieved a time of 4.45 seconds in its first run. Due to the motor not being able to break traction even at full output, output could not be increased and so no improvement was made from this time. This was a significant improvement from the 2022 time of 5.2 seconds. Points scored were 50 compared to 15 in 2022 with a position of 7th.

Skidpad

Skidpad saw NU Racing achieve 2nd position in the EV class with an averaged time of 5.29 seconds. This was an improvement on the time of 5.44 from 2022 and an increase in points from 56 to 65.

This time was achieved by the primary Skidpad driver on their second attempt who was sent out in the second stint on a cleaner track with warmer tyres. It should be noted that a change in tyre compound was also made in 2023 to Hoosier LC0 compared to R25B in 2022. This appeared to be an improvement.

Autocross

NU Racing also achieved 2nd in the Autocross event for the EV class with a score 120 compared to 93 in 2022. Times are not comparable due to track changes.

This time was achieved with the primary Autocross driver on their second attempt. Like Skidpad they were sent out as the second driver for a cleaner track surface and warmer tyres.

Endurance

Endurance unfortunately saw an issue with the car causing it to retire in the second driver sprint. This was an incredibly disappointing end to the weekend for the team. Based on the pace seen approximately 200 points was predicted for this event, however only 9 were scored. This was a significant reduction compared to the 159 scored in 2022.

Efficiency

Despite not finishing as over half of the endurance event was completed the team was still eligible for efficiency points. Due to the increased pace and resulting power usage efficiency scored 52 down on the 85 from 2022. Based on the temperatures observed it is believed this pace and power usage would have continued for the remainder of this event meaning this scaled score would likely be very similar had the endurance event been completed.

Endurance issue

Explained more thoroughly in Jacob Bush's and Jedd Reeve's reports, unfortunately towards the end of the 11th lap in the endurance event an AMS hard fault caused the car to shut down. This was very unexpected based on the reliability of the team developed electronics seen in the testing leading up to the competition.

From data pulled it appeared that the LVD teensy had stopped working with its heartbeat stopping. With this teensy no longer sending a digital Temp OKHS over CAN the BMS's CAN timeout functionality kicked in, pulling down the analogue AMS OKHS to the CEN triggering the fault. This was a huge disappointment for the team and its failure mode surprised many. Unfortunately, as there is only one attempt permitted in the Endurance event, minimum points were scored. This immediately removed any chance of any overall podium position predicted for the team prior to this event.

Whilst similar faults had been seen during the initial implementation of the CAN-based temperature monitoring, this fault was not seen in the months leading to competition. The same issue did occur once during the last SMSP track day during an endurance simulation with the new Cascadia controller, however unfortunately coincided with the cells exceeding 60 degrees which suggested the hard fault was acting as intended. The issue came good after several power cycles and did not show up again when driving the car through the last carpark driving session nor any events leading to the endurance event at competition. This issue may have been occurring while charging in the days leading up to and at the competition, but paired with other charger issues this was not considered at the time.

2024 Recommendations

It is recommended that a similar approach is taken to the 2024 competition with focus throughout the year on more driver training and development of a balanced car setup for all events. The additional work and risks surrounding suspension changes across the competition weekend is likely not necessary.

With the implementation of a larger motor in 2024, increasing wheel torque substantially, more work will be required to maximise the acceleration performance potential. It is recommended that grip limits are tested extensively throughout the year with adjustments made to maximum motor torque output. At competition the first run can be completed at a lower wheel torque to minimise slip before being increased to find surface limitations if desired.

Mock technical inspections should continue to be routine in the leadup to competition with those responsible for talking to inspectors well practised and confident with answers for all relevant aspects of the car. From experience, talking to an inspector with confidence and clarity with clear preparation will minimise questioning and streamline the process.

Whilst technical inspections were successful in 2023, it is believed that to reduce intensity leading to competition mock inspections and compliance deadlines should be pushed earlier into the year. Relevant tech inspection requirements should also be provided to those responsible for systems earlier in the year to ideally have systems completed to compliance from the beginning. A compliance cutoff in August would be ideal.

The AMS fault in endurance was a substantial blow to the 2023 competition for NU Racing. It is of the opinion of the author that whilst there were some events which if more closely examined may have provided data suggesting intermittent faults with the LVD, ultimately in the context of the two weeks leading to competition there was simply no way that any team member could have foreseen this issue occurring in the endurance event. It is no small feat that in the seven-day leadup to competition the team took an undriveable vehicle and completed the full integration of a new controller in less than 24 hours, which was followed by a full comp simulation the next day with no major issues. Whilst it can't simply be put down to bad luck, there is no regret over the decisions made leading into competition weekend nor the endurance event.

Whilst there are obvious ways to alter or restrict the abilities of key safety systems on the car required for compliance and to meet the rules, this is never an appropriate way to improve performance in any event and definitely not in the spirit of the competition. Whilst never a guarantee of success, the only way to improve on system performance is to stick to the basics of better unit testing, improved vehicle testing, and to avoid changes to critical electrical systems in the months leading to competition. Ideally, there should be no changes in the second half of the year for a competition vehicle, with LV systems work focussing on the following years design.

Despite this, there are aspects of NU Racing's LV systems design process which could be improved. Failure mode analysis is critical in any system and should be taken seriously for all items on the car, particularly electronics which have an effect on the shutdown circuit. This frustratingly had been somewhat considered for CANAMON and LVD code with open or short circuit temperature readings and even complete failure of a segment Teensy being accounted for. Very unfortunately, this analysis did not consider the effect of the LVD teensy itself failing. Subject to rules, and safety at general track days, it may be beneficial to consider changes to circuit and topology design to either better manage component failures or include redundancy.

While the competition finished on a very low note, when considering the teams position at the start of 2023, the undersized motor hardware utilised and the very simple mechanical design of systems compared with other high achieving teams, dynamic performance was exceptional. Whilst the competition results were not fantastic overall, besides a very minor reliability issue in the accumulator, the 2024 team has been set up with fundamental vehicle design and performance worthy of a podium finish at a Formula-SAE Australasia event.

With small, iterative development of subsystems NU Racing can expect to be a highly competitive team in future years. Overall, it is expected that with a substantial improvement in acceleration, and maintained performance in Skidpad and Autocross, paired with improved reliability and continued improvements in cell thermal management, NU24 should be a very high performing vehicle.

2024 Initial Design

As somewhat already discussed, initial 2024 design work for NU24 was completed towards the end of 2023 and in the early stages of 2024. With the impressive dynamic performance observed at the 2023 competition, focus was to revolve around minimising changes and implementing the larger 228 motor to improve acceleration. An iterative design change for all subsystems would also provide sustainable weight reduction, in line with a 25 kg reduction goal.

Key changes for NU24 would be:

- Reduced accumulator weight through a change to an Aluminium container
- Increased power and torque output through implementation of an Emrax 228
- Repackaged powertrain to reduce vehicle length
- Slight reductions in vehicle track and wheelbase
- Simplification of chassis with consolidation of some members
- New suspension to suit the new chassis

Aside from the powertrain, these initial designs are spread across the reports of Joseph Barker, Jacob Searle, Jusitn Li and Drew Bender. The extent of design at the end of January in 2023 can be seen below.

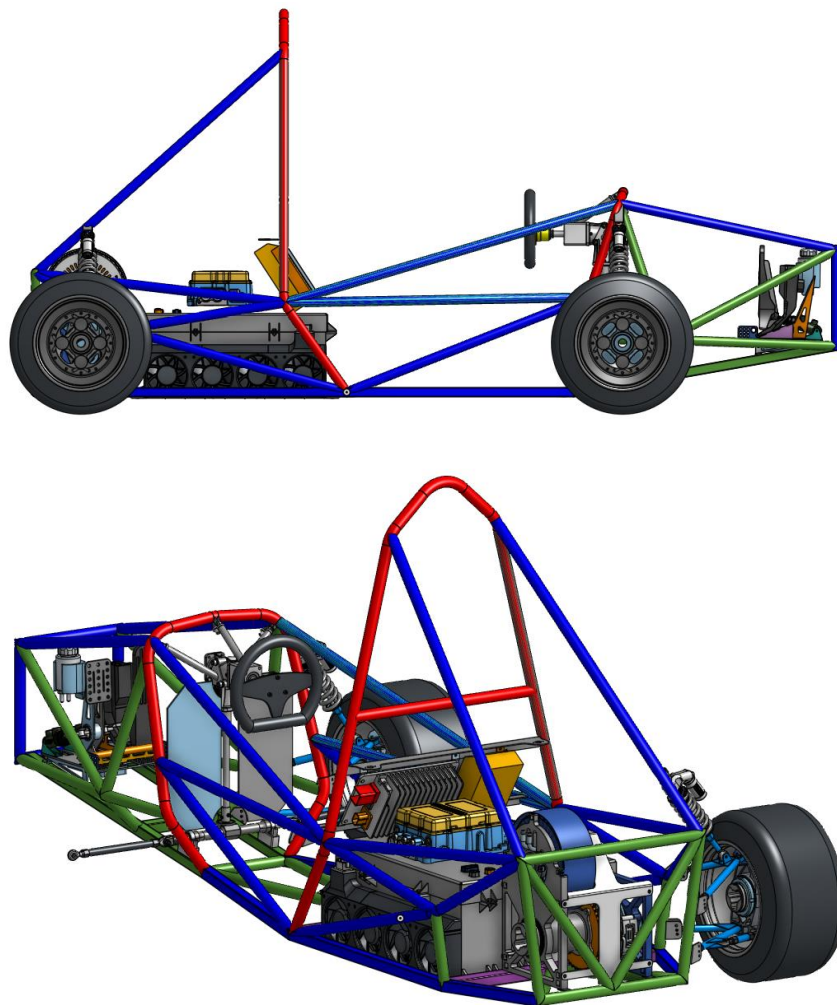


Figure 58 - NU24 Design as of Early February 2024

Conclusions and Recommendations

2023 was a very successful year for NU Racing with significant growth in vehicle performance, simplified design, and improved subsystem cohesion. NU23 and its designs provide an excellent base for continued growth in 2024 and beyond.

It is believed that the major changes made in 2023, particularly to the powertrain, accumulator, and general vehicle design, have set a sustainable future development path for what is a small team in the FSAE competition. It is essential to keep iterative growth at the core of NU Racings designs, with the 2023 competition clearly showing that excellent event results do not require highly complicated systems.

The design ethos of NU23 was set very intentionally for long-term team growth and understanding, with initial the 2024 design a clear demonstration of the degree of changes teams should aim for when designing a new vehicle.

Ideally, NU Racing should build towards a design process where major subsystem changes are developed and tested in parallel with iterative car designs. Changes should only be made if they provide a measurable gain in performance, cost, or manufacturability. This must be validated with clear and conclusive experimentation. Systems should never be used at competition if they do not have data to justify being on the car.

It should be highlighted that many of the problems, concepts, decisions, and designs discussed throughout this report are simple in nature when broken down. Whilst technical knowledge is required for the role of Chief Engineer, the most important aspect of this position is having a general understanding of the entire vehicle with particular focus on how subsystems relate and influence each other. Simple engineering concepts are generally all that is required to inform decisions on an FSAE vehicle, with the key being the understanding of how these concepts relate to vehicle functionality. The flow on effects of all decisions must also be well considered and understood.

One particular intent of this report was to try and provide a general overview to as many systems as possible, with focus on the tractive system and its components as this is generally the least understood aspect of an electric FSAE vehicle. In a competition where dynamic performance constitutes a significant number of points in the competition, maximising output from the tractive system is essential.

Overall, it is hoped the 2024 team is successful and benefits from the work of the 2023 team. With a shift in timelines and design ethos, team performance should improve while reducing peak workloads and improving preparation for competition.

One last consideration which must always be kept in mind is that whilst vehicle and team performance is always a key goal, and makes up the majority of this report, at its roots the FSAE competition is truly about improving undergraduate Engineers and providing key experience before moving into industry. It is strongly believed that all team members of the 2023 NU Racing team grew significantly throughout the year, developing experience and knowledge that can be drawn upon for years to come.



2023 NU Racing Team

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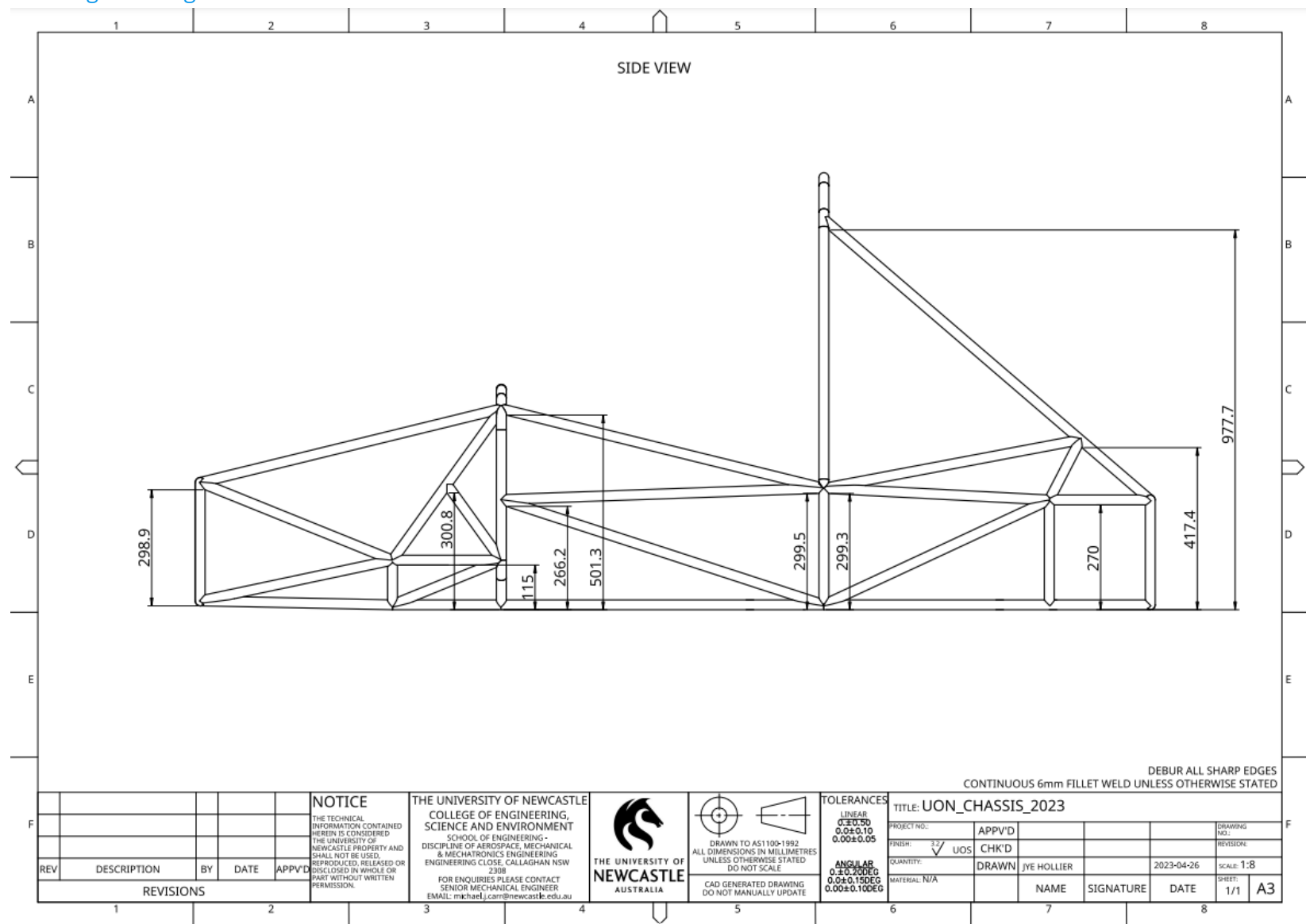
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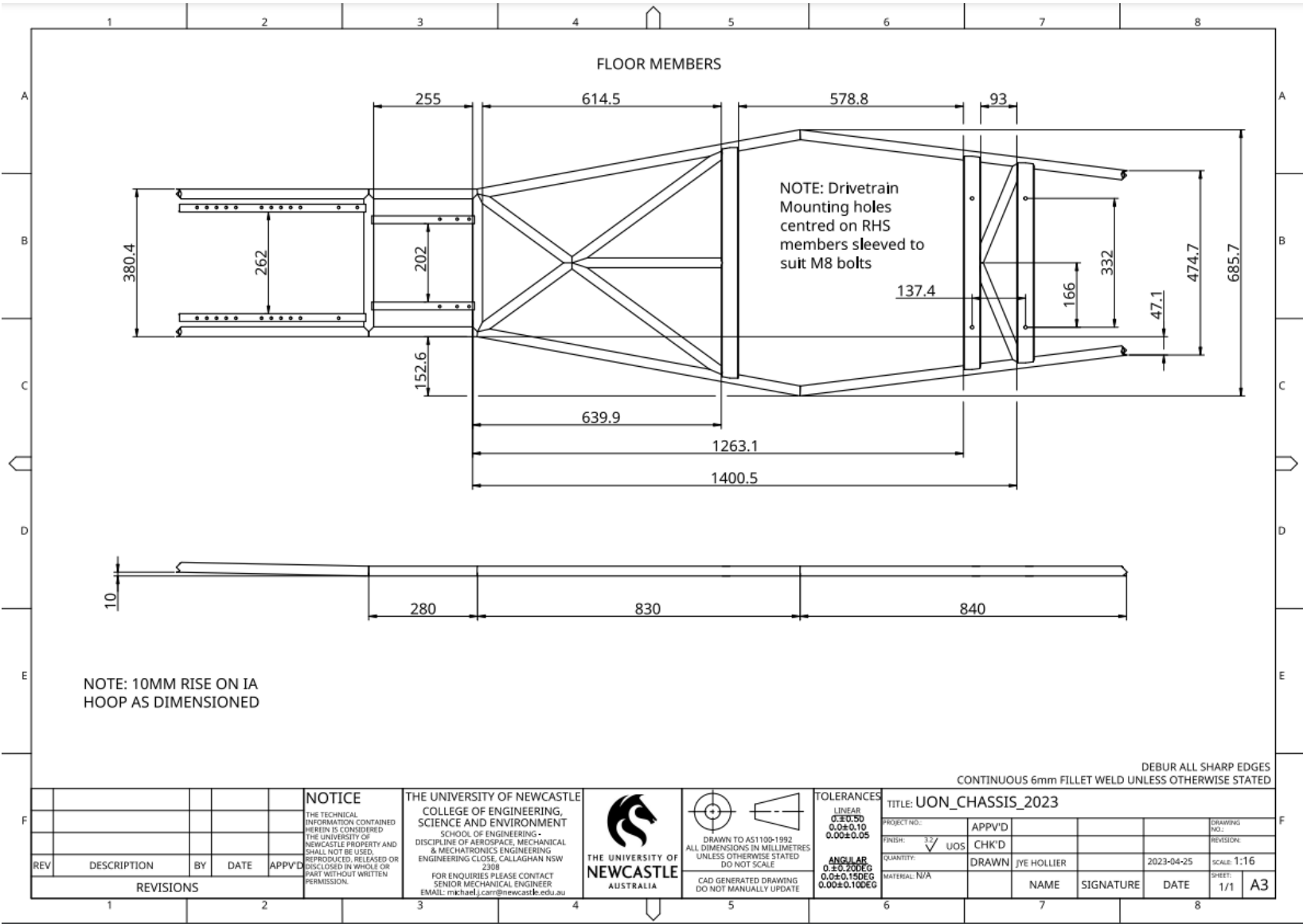
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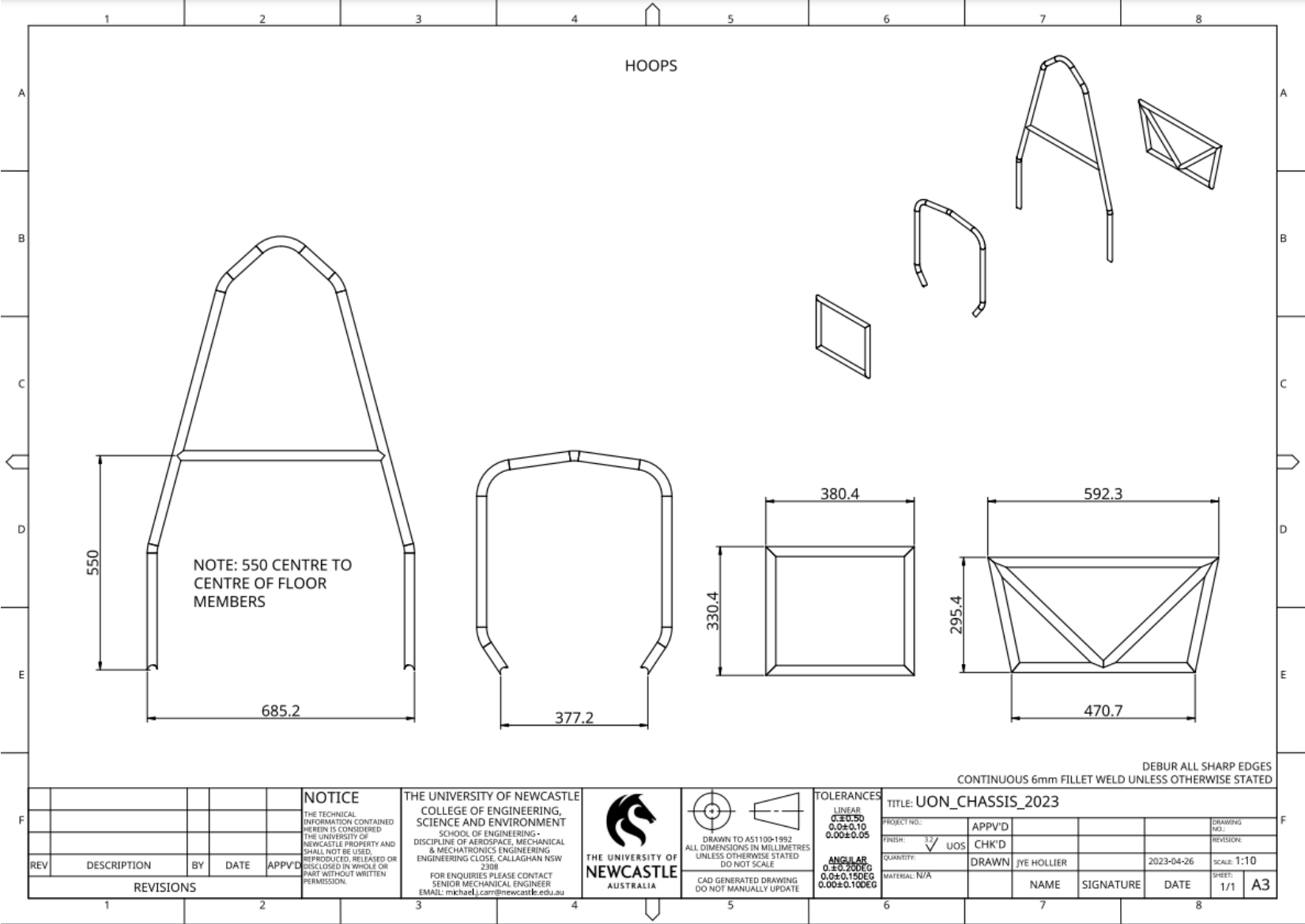


Appendix

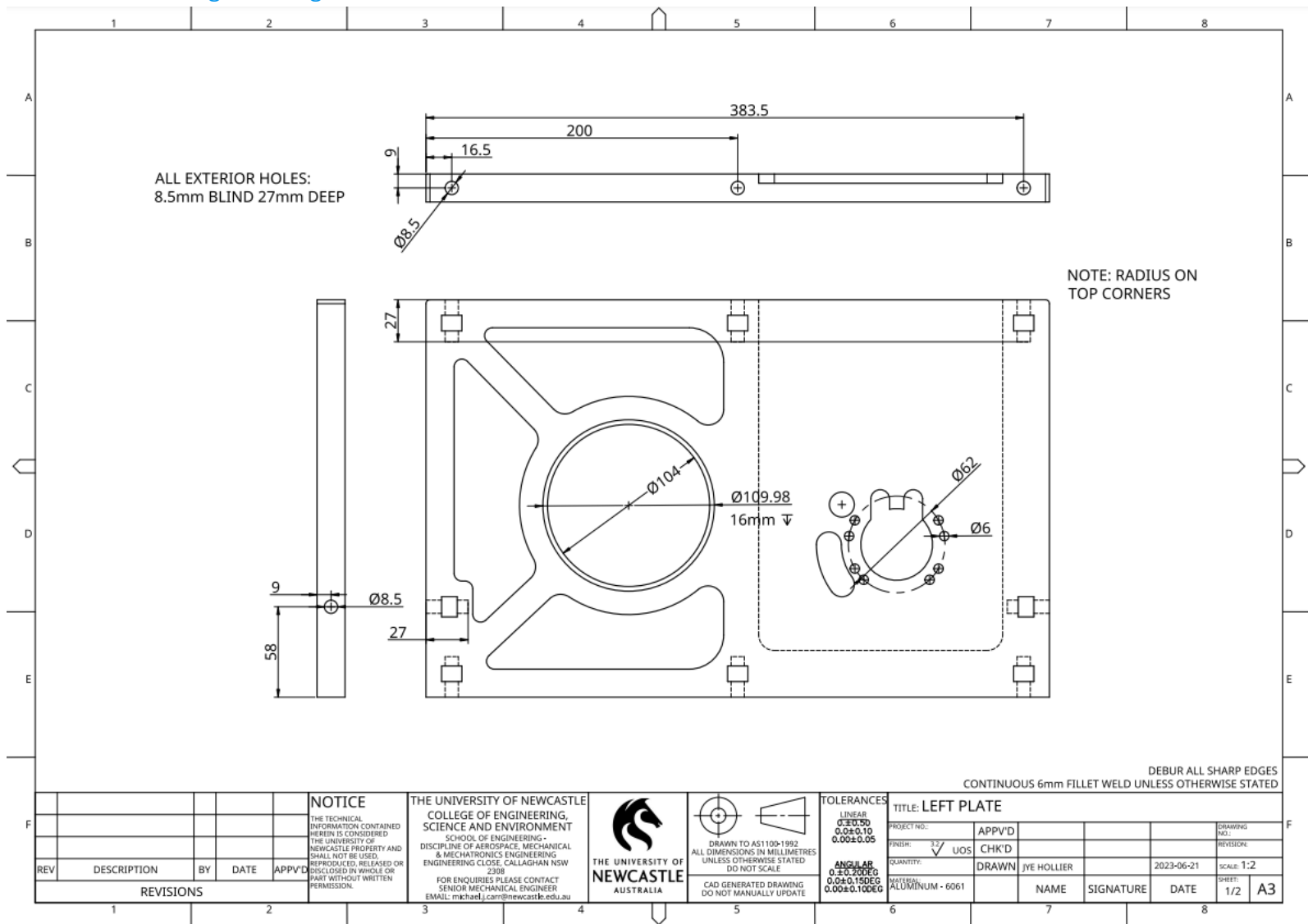
Chassis Welding Drawings

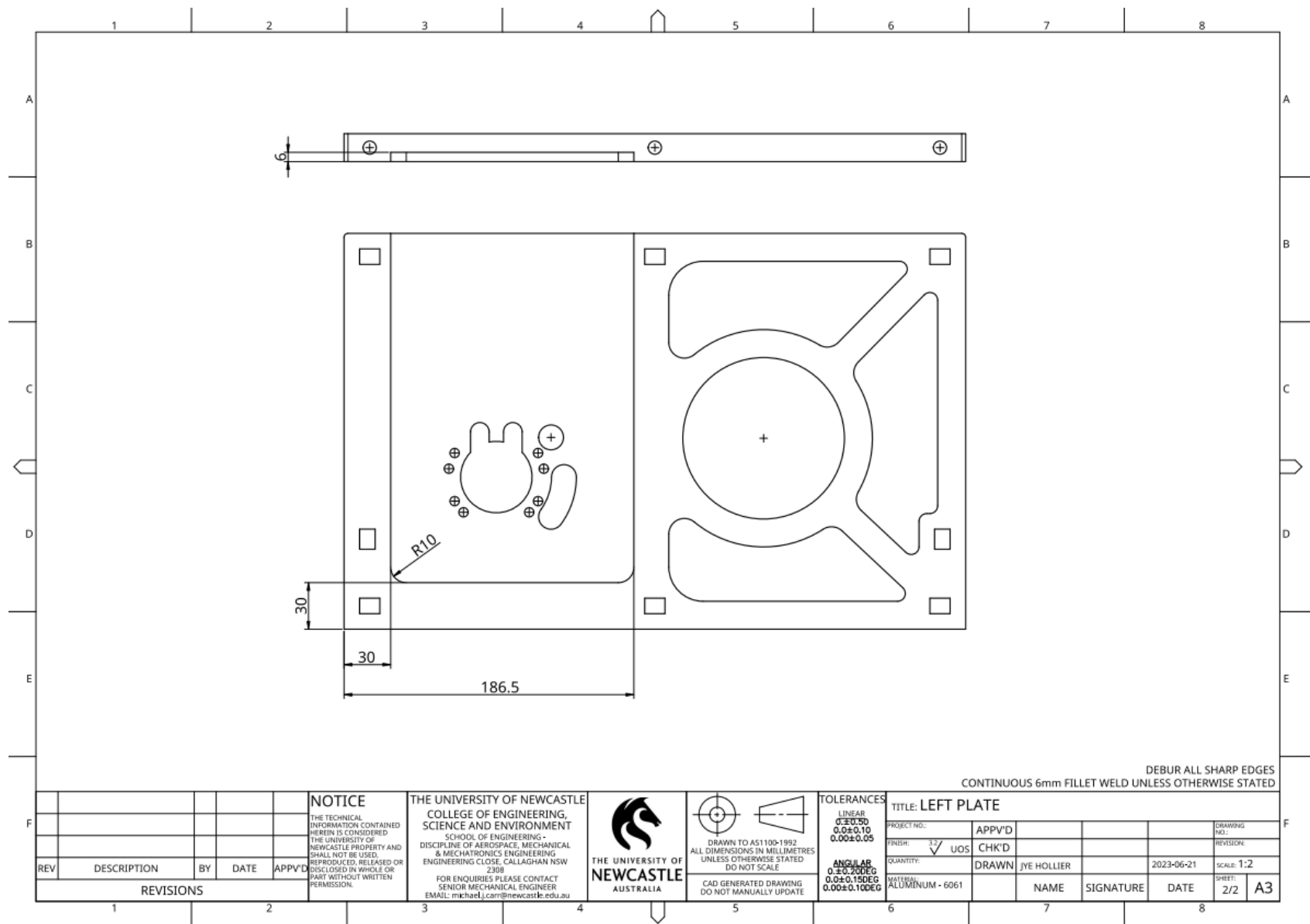


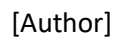


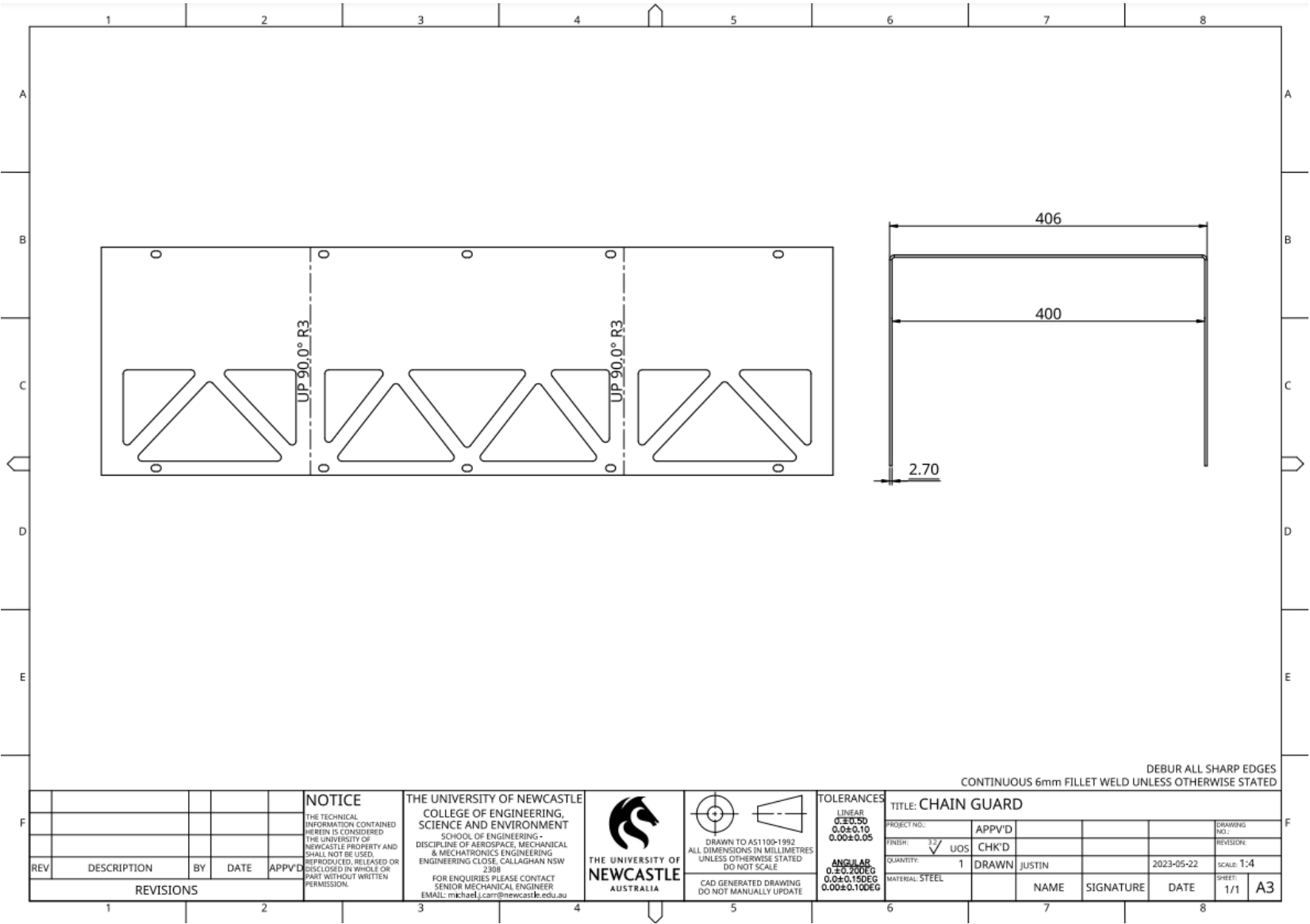


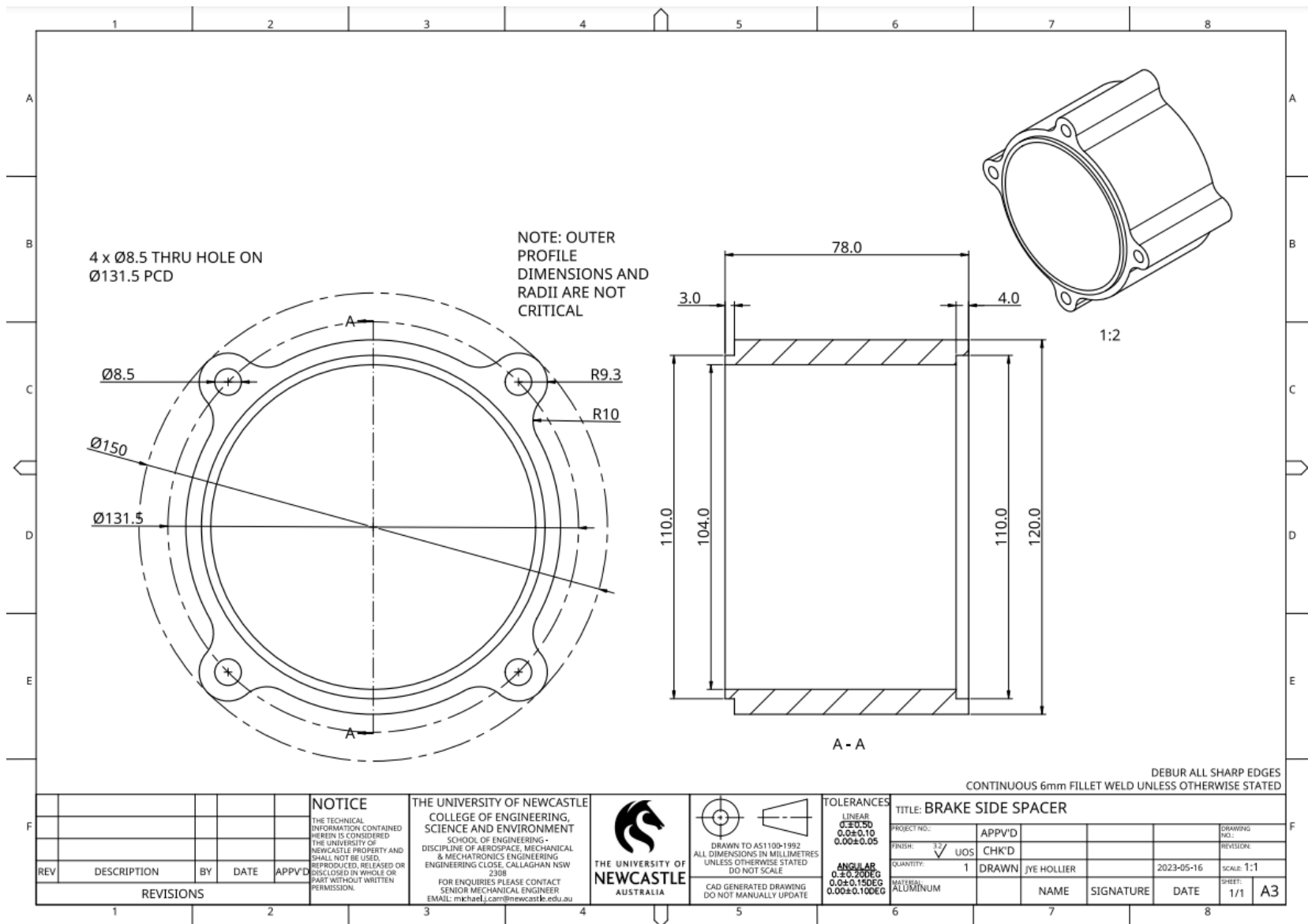
Powerbox V1 Manufacturing Drawings

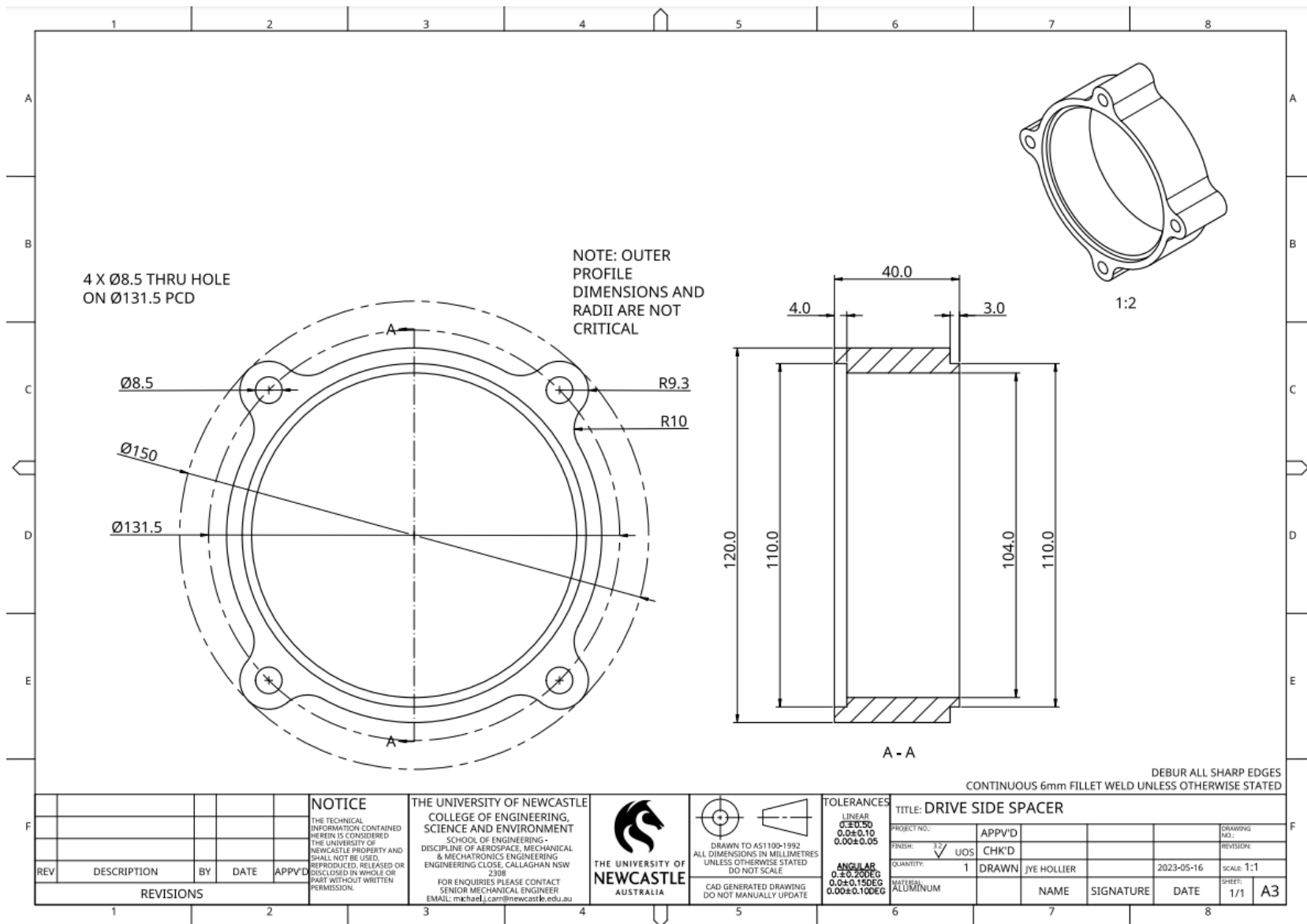




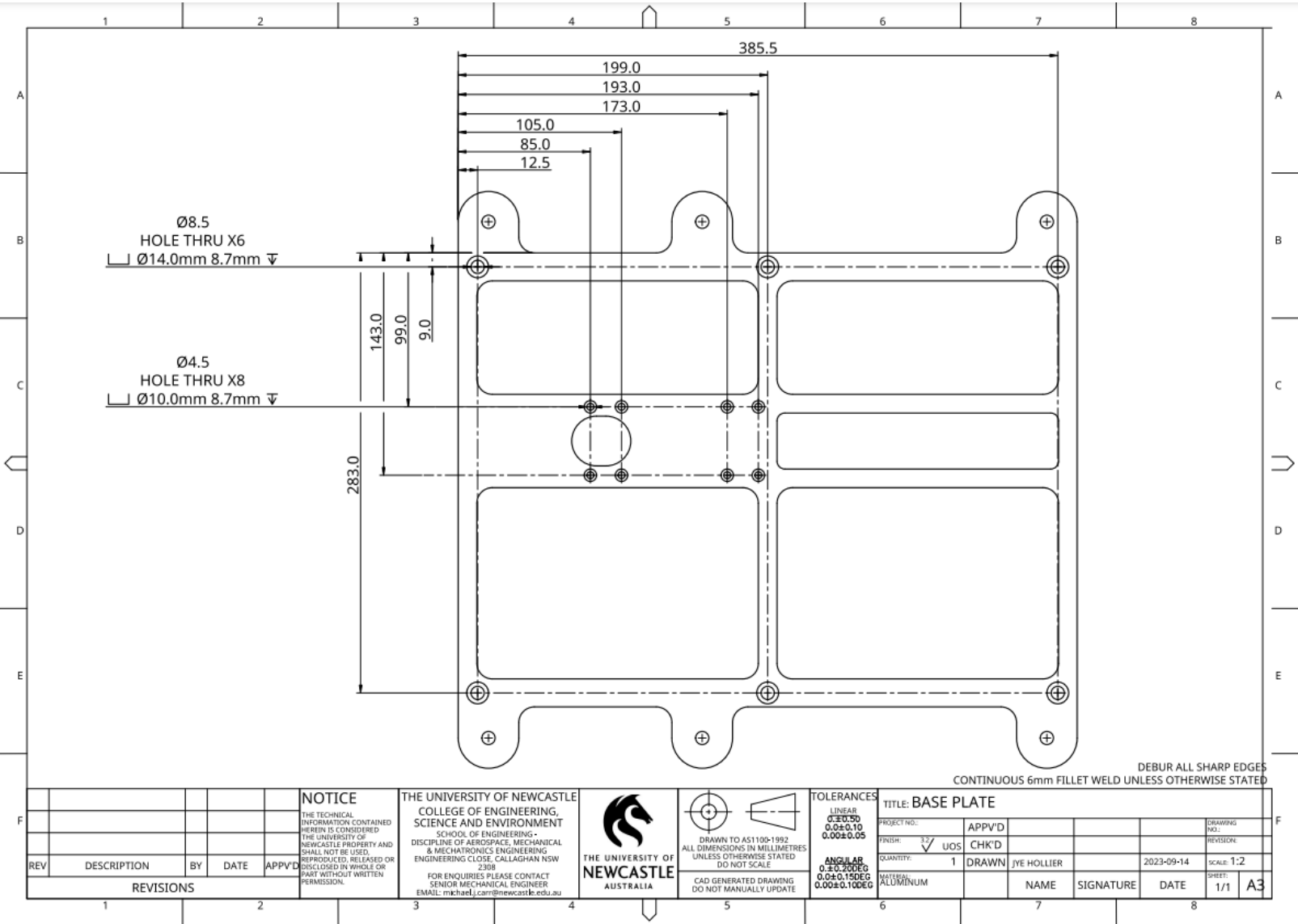


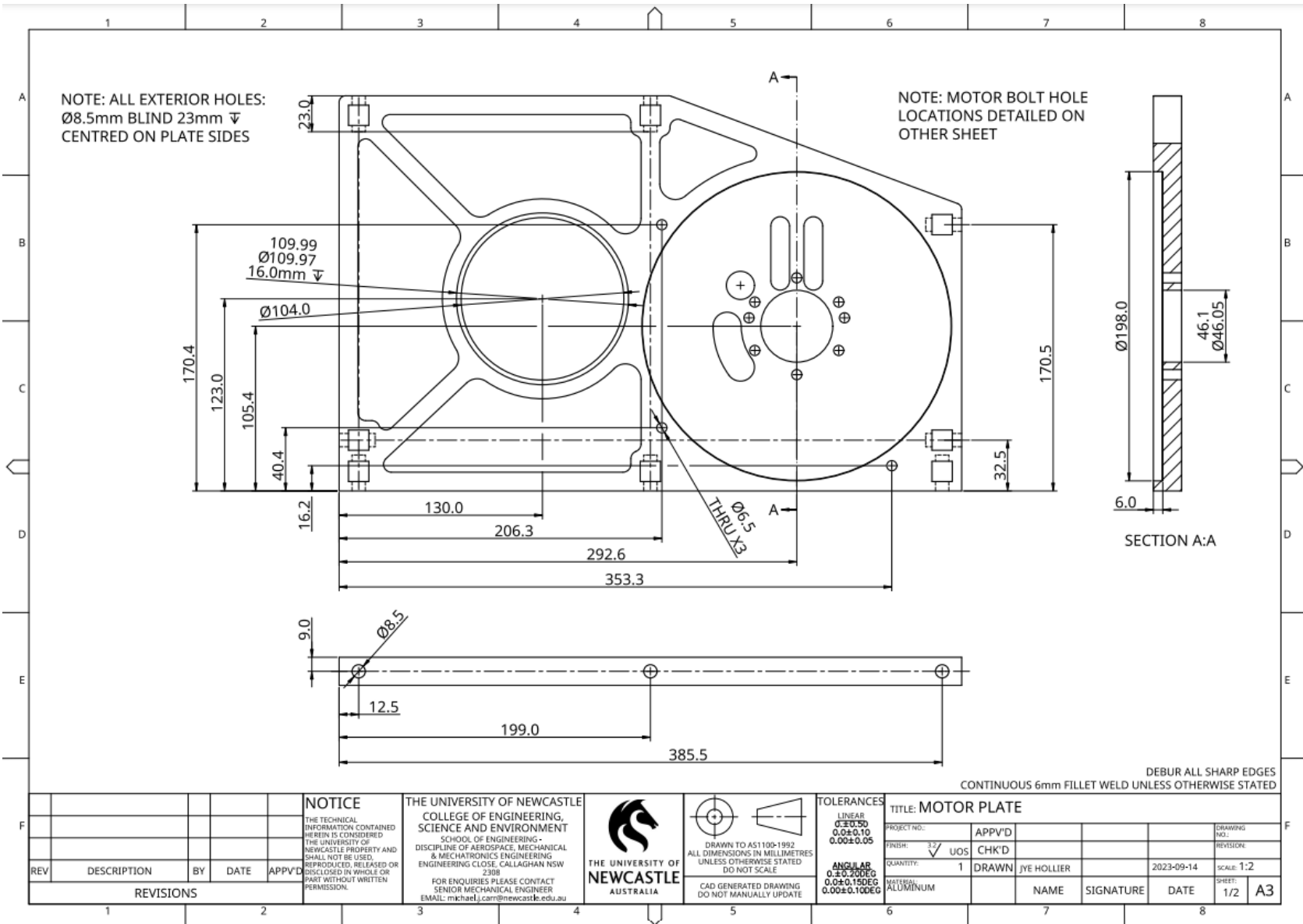


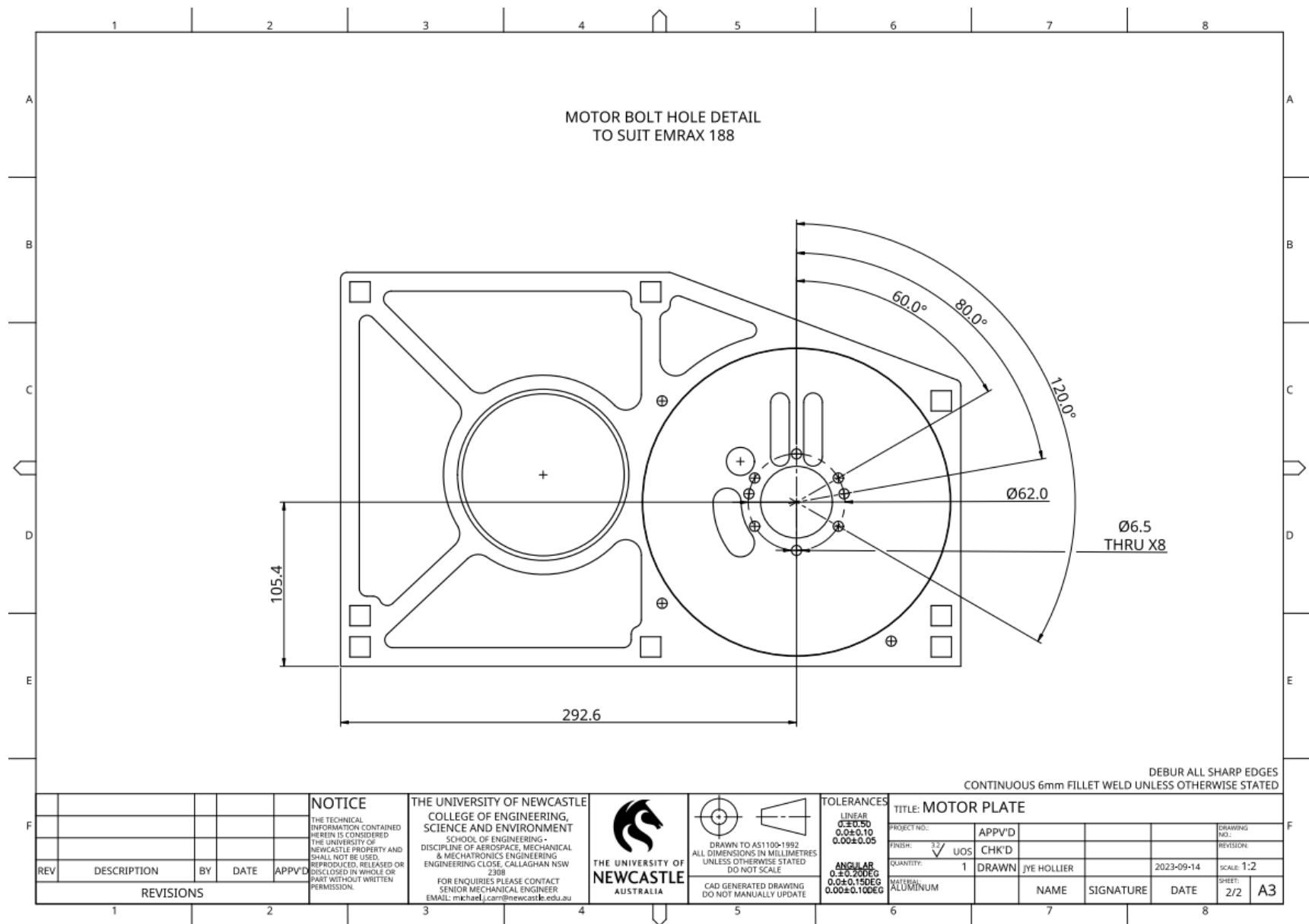


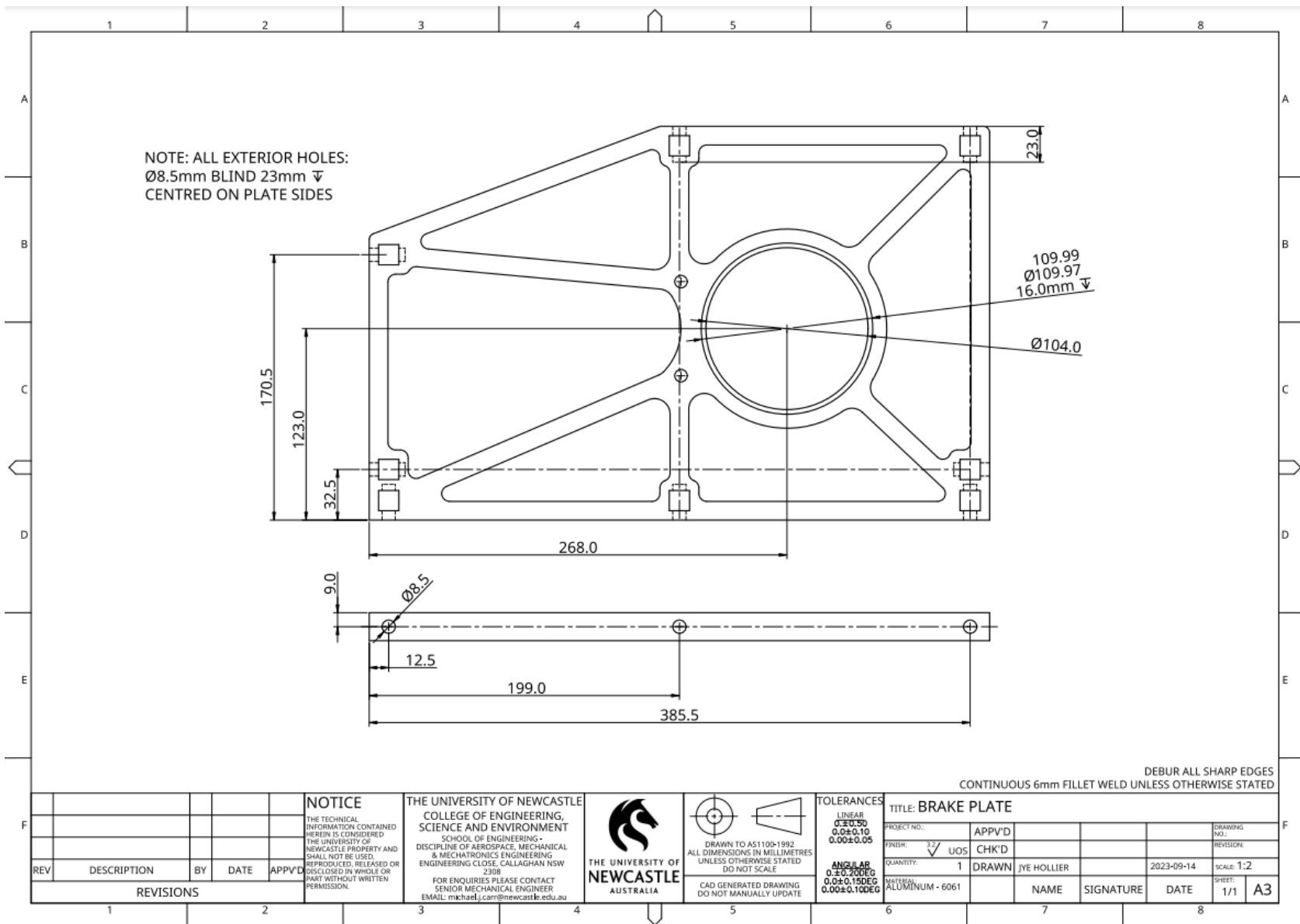


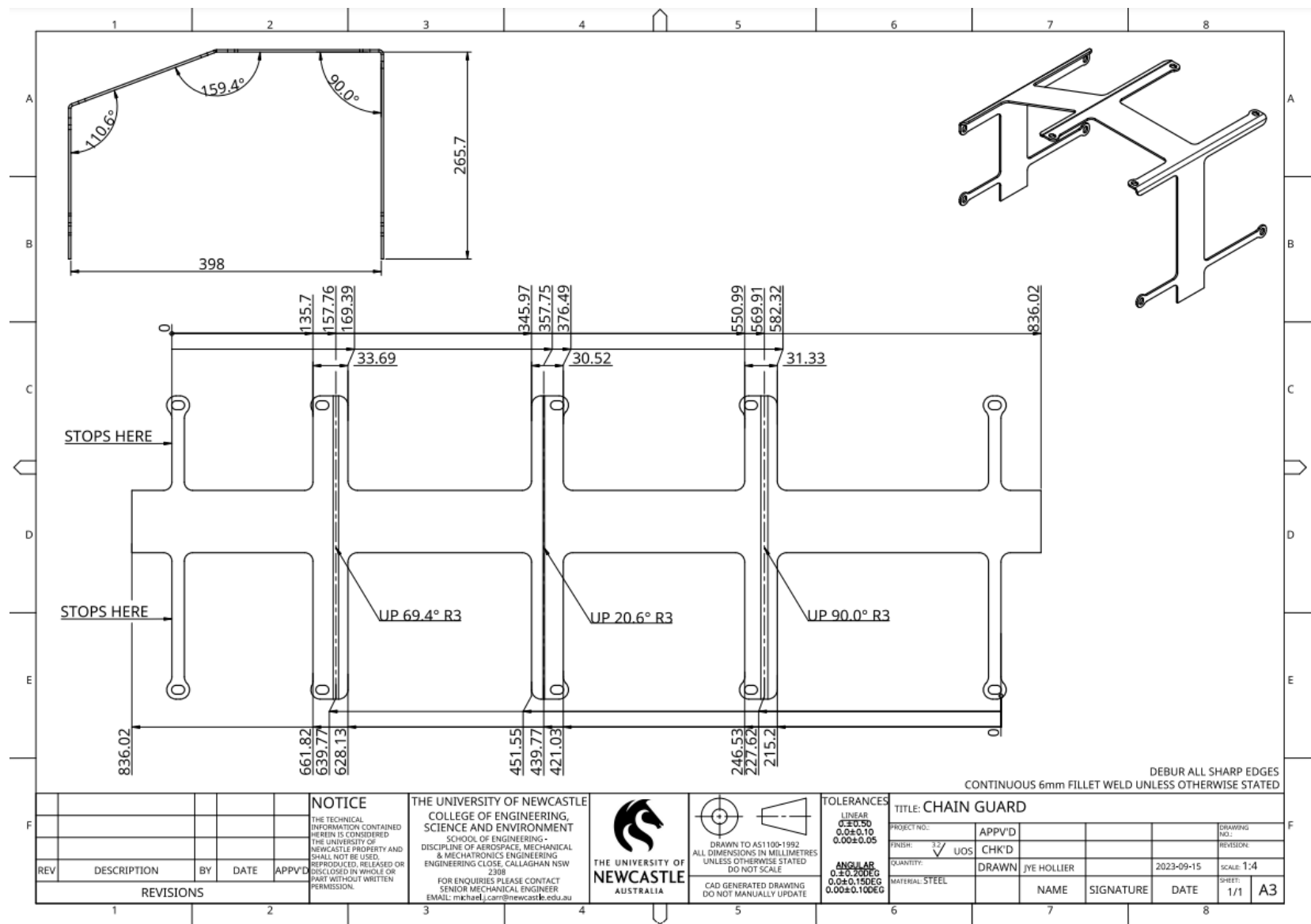
Powerbox V2 Manufacturing Drawings



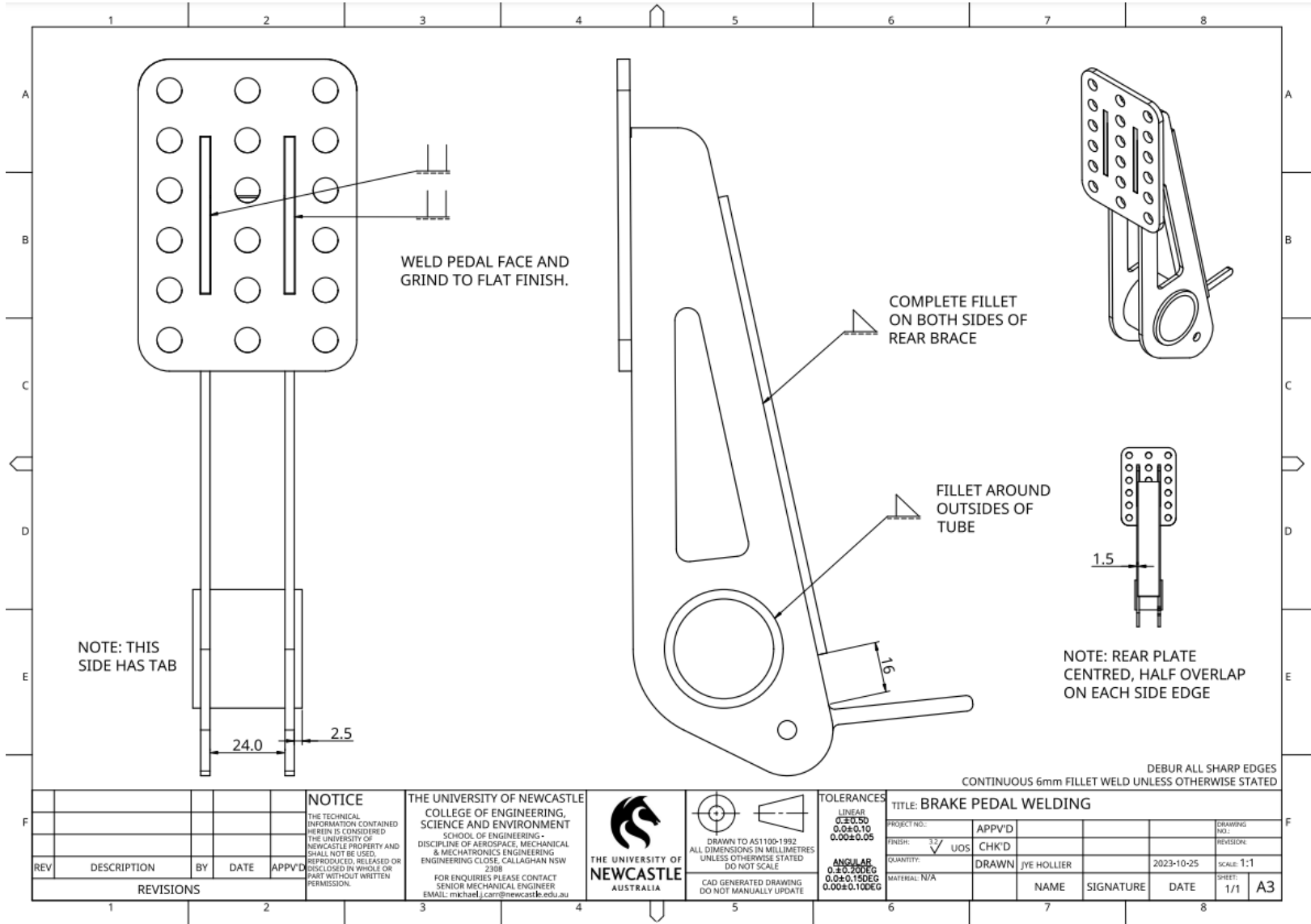






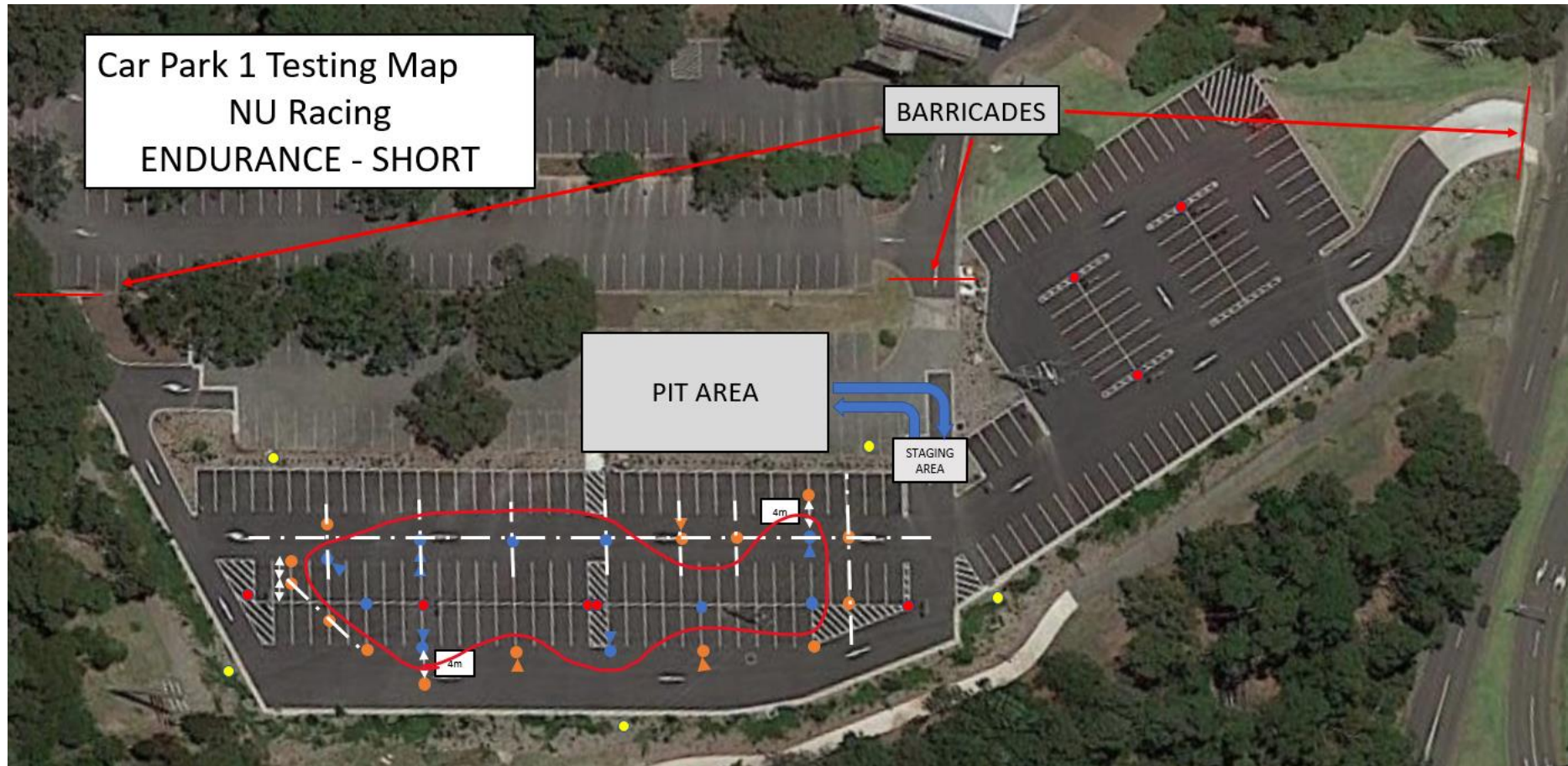


Brake Pedal Welding Drawing

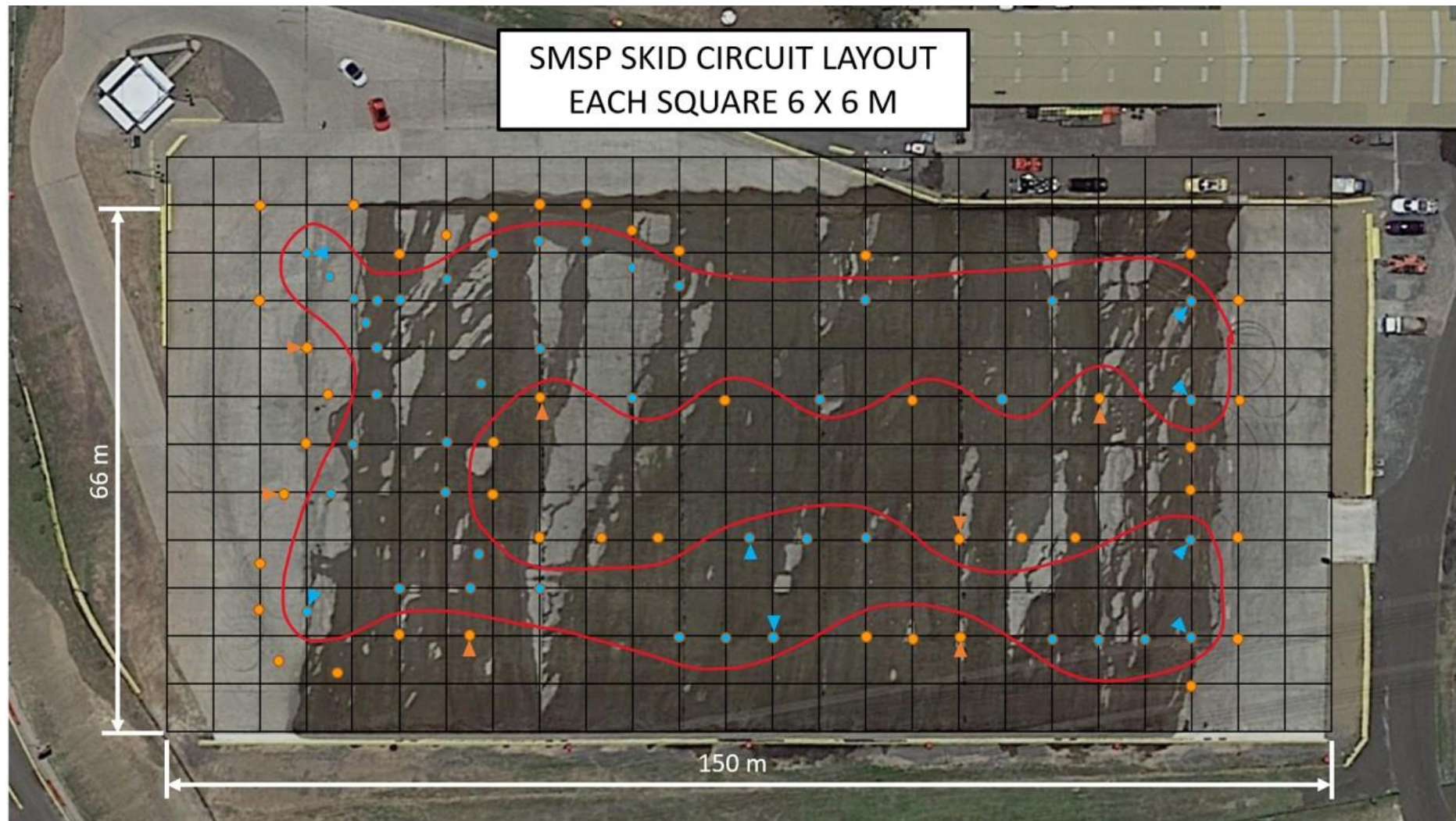


Track Designs

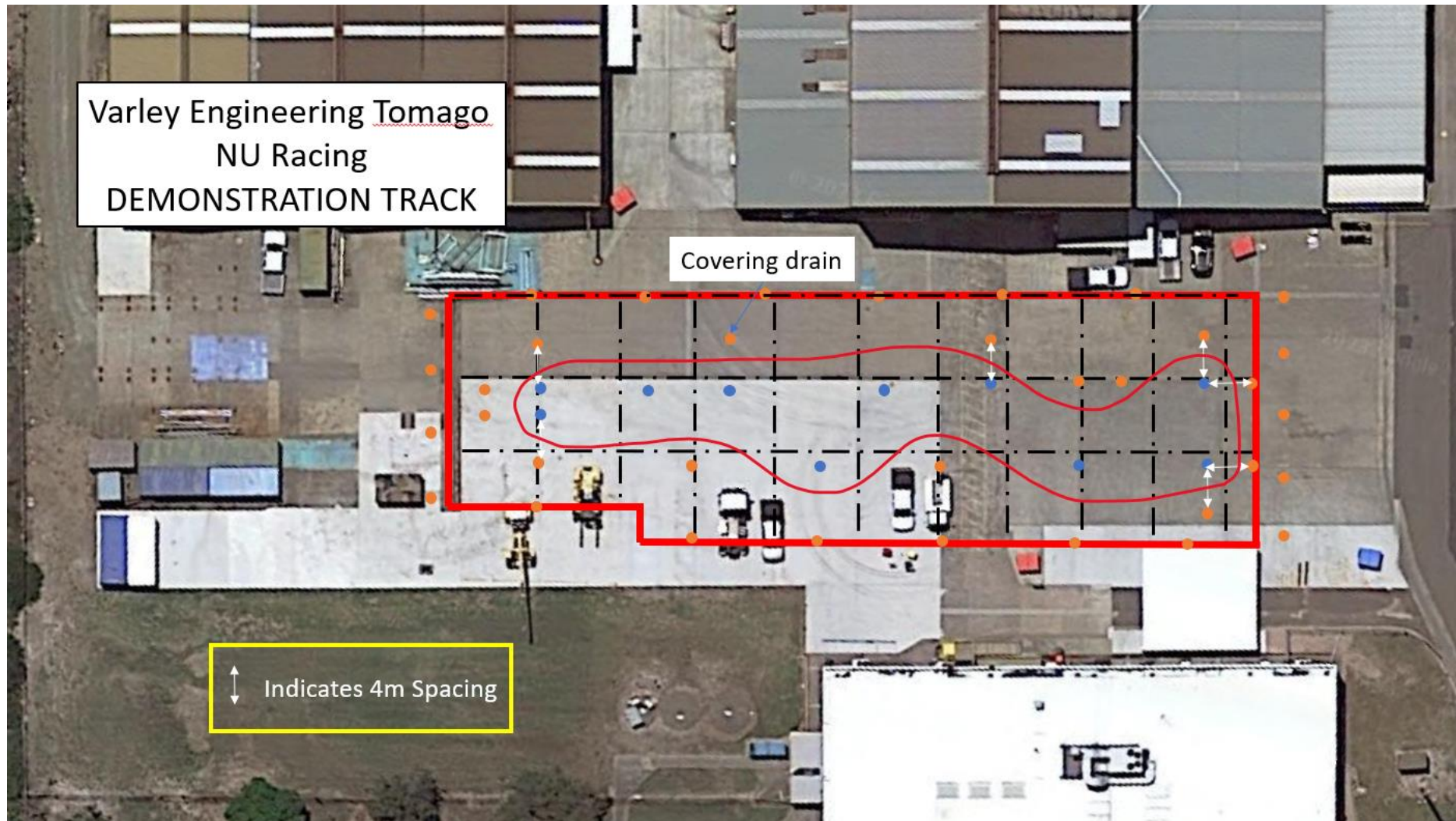
Car Park 1 Testing



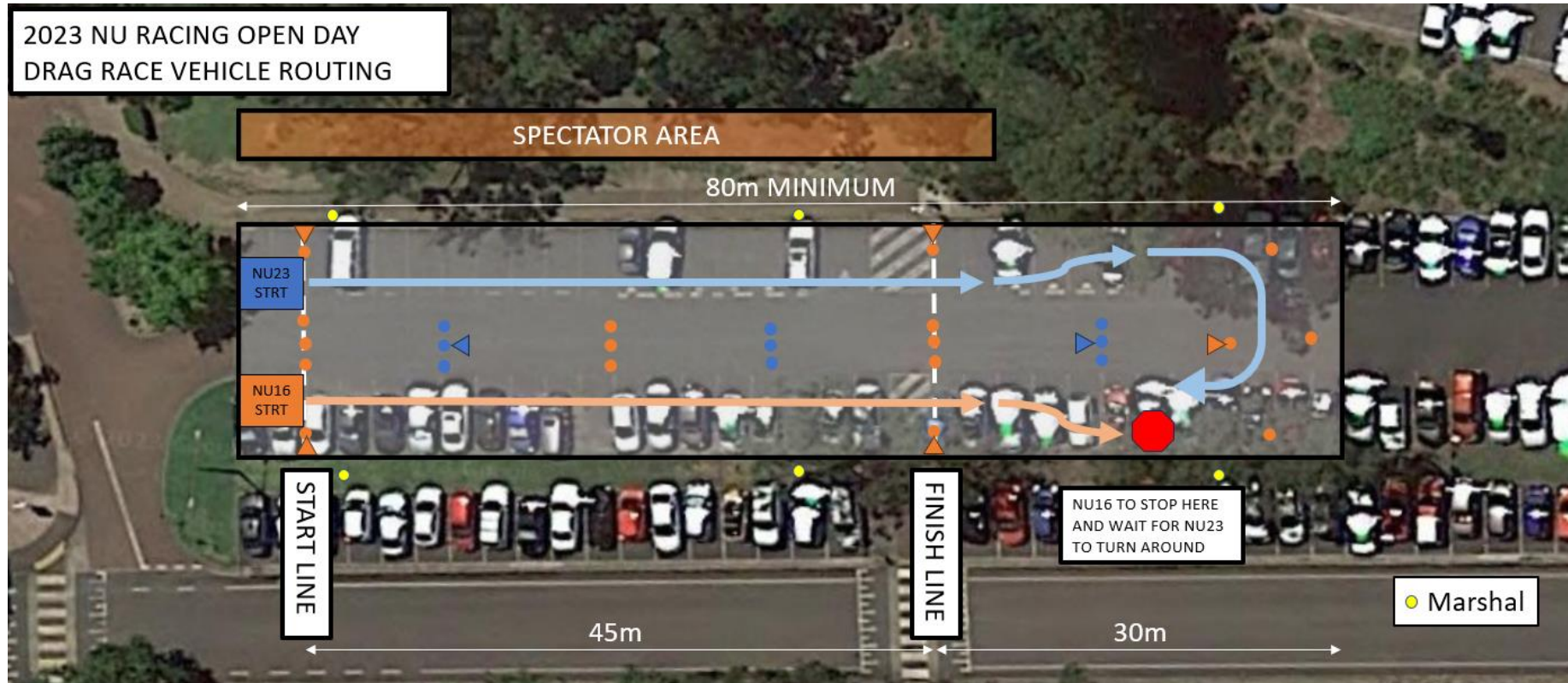
SMSP Testing



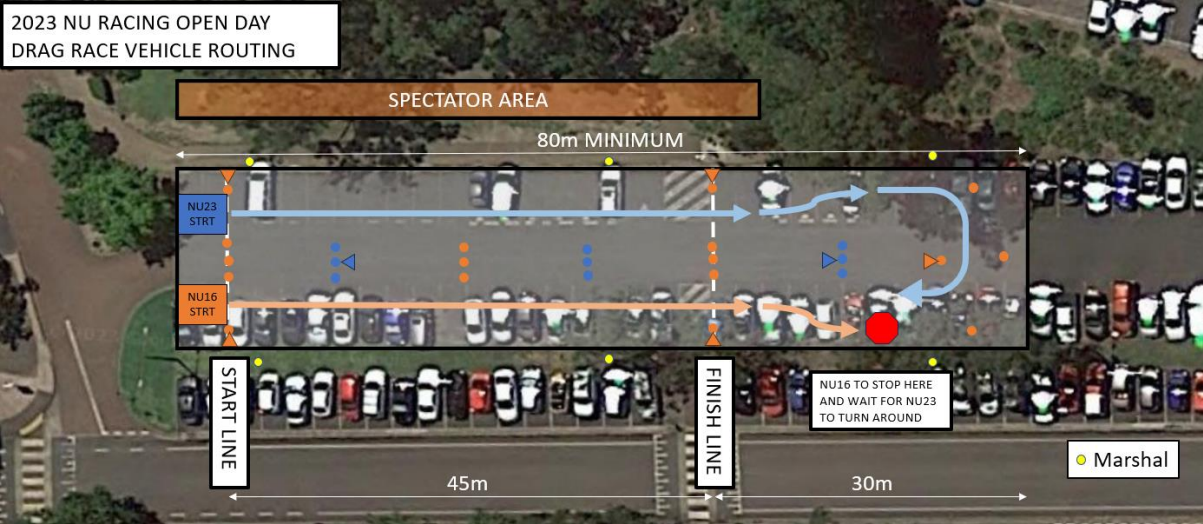
Varley Demonstration



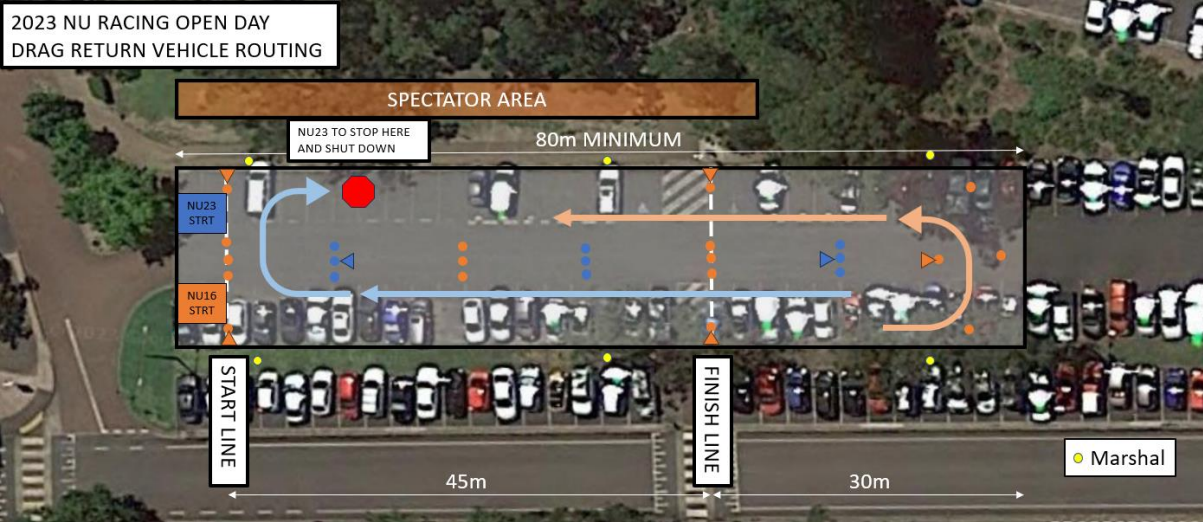
Open Day Demonstration



2023 NU RACING OPEN DAY
DRAG RACE VEHICLE ROUTING



2023 NU RACING OPEN DAY
DRAG RETURN VEHICLE ROUTING



2023 NU RACING OPEN DAY
DRAG RETURN VEHICLE ROUTING

